MAT 2040 Lectures Notes

Course taught by Prof. Tom Luo and Prof. Dongdong He

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Preface

This book contains the lecture notes I took from the course MAT 2040 taught by Prof. Tom Luo and Prof. Dongdong He in The Chinese University of Hong Kong, Shenzhen in the summer of 2019. These notes were compiled in LaTeX.

The purpose of these notes is to record the materials on whiteboards during lectures and my reflections upon the materials. Thus this book is by no means a dictionary or an encyclopedia. Still, I aim to build a self-contained and complete structure. Therefore I have made several modifications for the sake of mathematical rigor and the completeness of the structure of this book. For example, I have modified several definitions for mathematical rigor, the order of the materials in this book does not completely follow the original lectures, and several theorems have been added. Prof. Tom Luo and Prof. Dongdong He have not edited this book.

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Chapter 1

Week One

1.1 Monday (Prof. Tom Luo)

1.1.1 Introduction: What is Linear Algebra?

Linear Algebra is a foundational subject in mathematics. There are three foundational subjects in math: calculus, linear algebra, and probability and statistics. More advanced topics include differential equations, differential geometry, topology, abstract algebra, etc. In general, calculus deals with continuous phenomena while linear algebra deals with discrete data. Linear algebra is used in a wide range of areas such as data mining, bioinformatics, speech recognition, etc. You can even say linear algebra is more useful than calculus!

Solving linear systems

An important topic in linear algebra is linear systems. A *linear system* of m equations in n unknowns takes the form

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases} \tag{1.1.1}$$

We can pack the a_{ij}'s and b_i's in $(1.1.1)$ in a more compact form:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}. \tag{1.1.2}$$

The $m(n+1)$ numbers in $(1.1.2)$ *completely* determine a linear system.

In an $m \times n$ linear system, if there are more unknowns than equations ($m < n$), we say that the system is *underdetermined*. If there are more equations than unknowns ($m > n$), we say that the system is *overdetermined*. If there are as many equations as unknowns ($m = n$), we say that the system is *square*.

We are interested in the following questions:

1. *Does a solution exist?* If not, then the system is called *infeasible*.
2. *How many solutions does it have?* At first glance, we may think that the number of solutions to a linear system can be 0, 1, 2, 3, and so on. As it turns out, there are only three possibilities: a linear system must have no solution, exactly one solution, or infinitely many solutions.

3. *Finding methods to solve systems of linear equations.* In practice, the methods should be efficient. In other words, the complexity of a method should be low. The complexity depends on m and n, and we hope that it can be polynomial in m and n. Also, the methods should be numerically stable. That is, we do not want numbers that are either too large or too small to appear.

There are Abstract Notions

Apart from its wide applications in real life, linear algebra also involves abstract concepts. Take vector space as an example. The Euclidean vector spaces $\mathbb{R}^n$ are perhaps the most elementary vector spaces. For example, $\mathbb{R}^2$ consists of all Euclidean vectors of size 2:

$$\mathbb{R}^2 = \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \middle| \, x_1, x_2 \in \mathbb{R} \right\}.$$

However, the concept of vector space can be extended to more abstract spaces. For example,

$$V = \left\{ f \middle| \, f : \mathbb{R} \mapsto \mathbb{R},\ f(t) = \sum_{i=0}^{N-1} a_i \cos\left(\frac{2\pi i}{N} t\right),\ a_i \in \mathbb{R},\ i = 0, 1, ..., N-1 \right\} \tag{1.1.3}$$

also defines a vector space whose elements are functions.

Here we see an example of the real-life applications of linear algebra. In signal processing, a signal is usually represented as the sum of sine and cosine functions (as in (1.1.3)). By doing so, we are able to separate different signals and denoise a signal. These topics are discussed in a subject called *Fourier analysis*.

Further, in vector spaces, we introduce and discuss abstract concepts such as dimension, rank, etc.

Linear Algebra is Visulizable

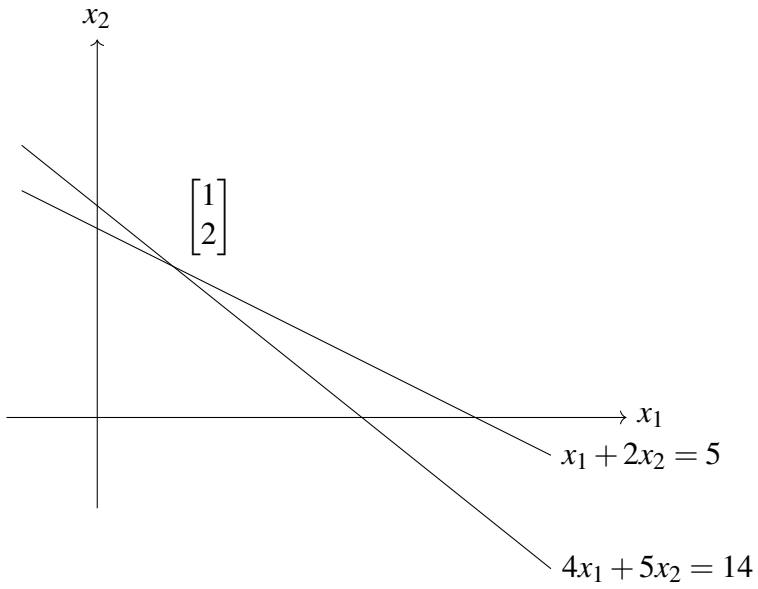
Linear algebra is visualizable. As an example, consider the following 2×2 linear system:

$$\begin{cases} x_1 + 2x_2 = 5 \\ 4x_1 + 5x_2 = 14 \end{cases}. \tag{1.1.4}$$

The solution to $(1.1.4)$ is

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

We can visualize $(1.1.4)$ with a row picture:

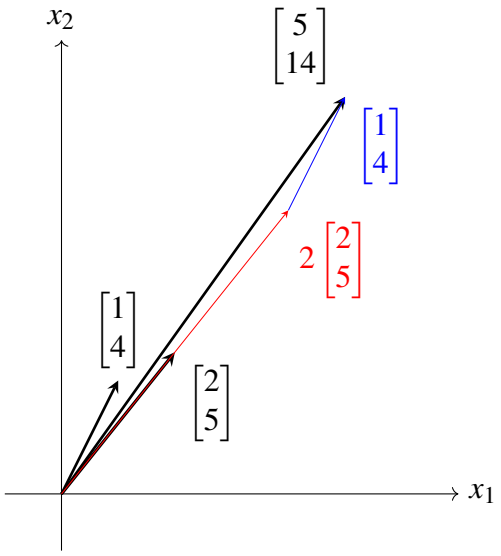


In the row picture, each equation represents a straight line on the graph and the intersection of the two lines is the solution.

We can also visualize (1.1.4) with a column picture. To do so, note that (1.1.4) is equivalent to

$$x_1 \begin{bmatrix} 1 \\ 4 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 5 \\ 14 \end{bmatrix}.$$

The column picture is as follows.



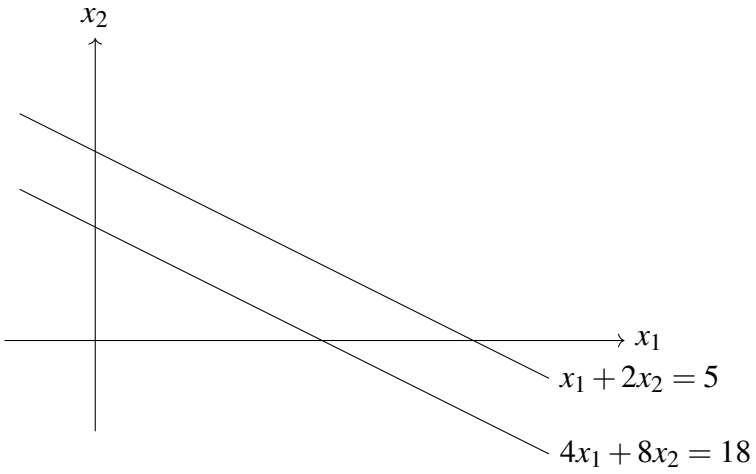
The column picture can be interpreted as finding a linear combination of the vectors $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$ that gives $\begin{bmatrix} 5 \\ 14 \end{bmatrix}$. That is to say, we need to find x_1 and x_2 such that

$$x_1 \begin{bmatrix} 1 \\ 4 \end{bmatrix} + x_2 \begin{bmatrix} 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 5 \\ 14 \end{bmatrix}.$$

Consider another system:

$$\begin{cases} x_1 + 2x_2 = 5 \\ 4x_1 + 8x_2 = 18 \end{cases} \tag{1.1.5}$$

This system does not have a solution because (Eqn.2) $-$ (Eqn.1)$\times 4$ gives $0 = -2$, which is a contradiction. We can also use a row picture to explain why (1.1.5) has no solution:



We see from the row picture that the two lines are parallel and do not intersect. Therefore $(1.1.5)$ has no solution.

Can a 2×2 linear system have infinitely many solutions? The answer is yes, when the two lines in the row picture coincide. In this case, the two lines have infinitely many intersections and the system has infinitely many solutions. For example, the system

$$\begin{cases} x_1 + 2x_2 = 5 \\ 4x_1 + 8x_2 = 20 \end{cases}$$

has infinitely many solutions because $x_1 + 2x_2 = 5$ is equivalent to $4x_1 + 8x_2 = 20$.

The reader may have now noticed that we have listed all the possibilities of the number of solutions to a 2×2 linear system. Indeed, as two lines in the xy-plane must intersect, coincide, or be parallel, then there are only three possibilities of the number of solutions to a 2×2 linear system: 0, 1, and ∞.

1.1.2 Contradiction for Inconsistent Systems

We can extend our previous result in $(1.1.5)$ to a more general case. In fact, we have the following theorem.

Theorem 1.1.1. *The system of linear equations*

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

has no solution if and only if there exist $\alpha_1, \alpha_2, ..., \alpha_m \in \mathbb{R}$ *such that*

$$\begin{cases} a_{11}\alpha_1 + a_{21}\alpha_2 + \cdots + a_{m1}\alpha_m = 0 \\ a_{12}\alpha_1 + a_{22}\alpha_2 + \cdots + a_{m2}\alpha_m = 0 \\ \quad \vdots \\ a_{1n}\alpha_1 + a_{2n}\alpha_2 + \cdots + a_{mn}\alpha_m = 0 \end{cases}$$

and $\alpha_1 b_1 + \alpha_2 b_2 + \cdots + \alpha_m b_m \neq 0$.

We also have an analogous theorem for systems of linear inequalities.

Theorem 1.1.2. *The system of linear inequalities*

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \leq b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n \leq b_2 \\ \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \leq b_m \end{cases}$$

has no solution if and only if there exist $\alpha_1, \alpha_2, ..., \alpha_m \geq 0$ *such that*

$$\begin{cases} a_{11}\alpha_1 + a_{21}\alpha_2 + \cdots + a_{m1}\alpha_m = 0 \\ a_{12}\alpha_1 + a_{22}\alpha_2 + \cdots + a_{m2}\alpha_m = 0 \\ \quad \vdots \\ a_{1n}\alpha_1 + a_{2n}\alpha_2 + \cdots + a_{mn}\alpha_m = 0 \end{cases}$$

and $\alpha_1 b_1 + \alpha_2 b_2 + \cdots + \alpha_m b_m < 0$.

1.1.3 Gaussian Elimination

We have just discussed 2×2 linear systems. Now we turn to the problem of solving a 3×3 linear system. Consider the following system:

$$\begin{cases} 2x_1 + x_2 + x_3 = 5 \\ 4x_1 - 6x_2 = -2 \\ -2x_1 + 7x_2 + 2x_3 = 9 \end{cases} \tag{1.1.6}$$

We represent $(1.1.6)$ with the matrix

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 2 & 9 \end{array}\right]. \tag{1.1.7}$$

The key to solving $(1.1.6)$ is to convert $(1.1.7)$ to an equivalent upper triangular system

$$\left[\begin{array}{ccc|c} * & * & * & * \\ 0 & * & * & * \\ 0 & 0 & * & * \end{array}\right] \tag{1.1.8}$$

and then apply back substitution. To convert a system to an equivalent upper triangular system, we need to perform three types of *elementary row operations*:

(i) Interchange two rows.

(ii) Multiply one row by a nonzero constant.

(iii) Replace a row by its *sum* with a multiple of another row.

We now convert $(1.1.7)$ to $(1.1.8)$:

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 2 & 9 \end{array}\right] \to \left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 3 & 14 \end{array}\right] \to \left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 1 & 2 \end{array}\right]. \tag{1.1.9}$$

We immediately get $x_3 = 2$ from the third equation. Substituting $x_3 = 2$ into the second equation, we obtain $x_2 = 1$. Eventually we solve for x_1 in the first equation and we obtain $x_1 = 1$.

In $(1.1.9)$, we converted a linear system to an equivalent upper triangular system using row operations (i), (ii), and (iii). This process is called *Gaussian elimination*. We now generalize the process to an $n \times n$ linear system.

Suppose we are given an $n \times n$ linear system

$$\left[\begin{array}{cccccc|c} * & * & * & \cdots & * & * \\ * & * & * & \cdots & * & * \\ * & * & * & \cdots & * & * \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ * & * & * & \cdots & * & * \end{array}\right]. \tag{1.1.10}$$

The first step is to zero out the last $(n-1)$ entries in the first column. We do this by adding multiples of the first row to the last $(n-1)$ rows so that the first entries in the last $(n-1)$ rows become 0 (row operation (iii)). However, if the first entry of the first row is 0, we need to first interchange the first

row and another row whose first entry is nonzero (row operation (i)). Then we zero out the last $(n-1)$ entries in the first column. After the first step, (1.1.10) is converted to

$$\left[\begin{array}{ccccc|c} * & * & * & \cdots & * & * \\ 0 & * & * & \cdots & * & * \\ 0 & * & * & \cdots & * & * \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & * & * & \cdots & * & * \end{array}\right]. \tag{1.1.11}$$

In a similar manner, we zero out the last $(n-2)$ entries in the second column by adding multiples of the second row to the last $(n-2)$ rows so that the second entries in the last $(n-2)$ rows become 0 (row operation (iii)). (We may need to interchange rows so that the second entry of the second row is nonzero.) After the second step, (1.1.11) becomes

$$\left[\begin{array}{ccccc|c} * & * & * & \cdots & * & * \\ 0 & * & * & \cdots & * & * \\ 0 & 0 & * & \cdots & * & * \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & * & \cdots & * & * \end{array}\right]. \tag{1.1.12}$$

Note that the zeros in the first column are not ruined.

Repeating the same process for the rest of the columns, we obtain an equivalent upper triangular system

$$\left[\begin{array}{ccccc|c} * & * & * & \cdots & * & * \\ & * & * & \cdots & * & * \\ & & * & \cdots & * & * \\ & \mathbf{0} & & \ddots & \vdots & \vdots \\ & & & & * & * \end{array}\right]. \tag{1.1.13}$$

where all the entries below the main diagonal are 0. Then we apply back substitution and solve for $x_n, x_{n-1}, ..., x_2, x_1$ in turn.

Complexity Analysis

We now calculate the complexities of Gaussian elimination and back substitution.

Complexity of Gaussian Elimination. In the kth step ($k = 1, 2, ..., n-1$), we need to zero out the last $(n-k)$ entries in the kth column. Reducing a row in step k requires $[1+(n-k+1)]$ multiplications and $(n-k+1)$ additions (instead of $[1+(n-k+2)]$ multiplications and $(n-k+2)$ additions because we know that the entries to be zeroed out are going to be 0; the extra multiplication is for the calculation of the multiplier of row k). Thus the overall complexity of Gaussian elimination on an $n \times n$ linear system is equal to

$$\sum_{k=1}^{n-1} (n-k)\left[2(n-k+1)+1\right] = \sum_{k=1}^{n-1} \left[2(n-k)^2 + 3(n-k)\right] \sim \frac{2}{3}n^3 \text{ flops.}$$

Complexity of Back Substitution. Solving for x_n requires only 1 division. For each $k = 1, ..., n-1$, after obtaining $x_n, ..., x_{k+1}$,

$$a'_k x_k + a'_{k+1} x_{k+1} + \cdots + a'_n x_n = b'_n$$

gives

$$x_k = \frac{1}{a'_k}(b'_n - a'_n x_n - \cdots - a'_{k+1}x_{k+1})$$

which involves $(n-k)$ multiplications, $(n-k)$ subtractions, and 1 division, which, in total, are $2(n-k)+1$ flops. Thus the overall complexity of back substitution for an $n \times n$ upper triangular linear system is equal to

$$\sum_{k=1}^{n} [2(n-k)+1] = n^2 \text{ flops.}$$

1.1.4 Possibilities of Number of Solutions to a 3×3 System

Recall that in section 1.1.3, we converted the linear system (1.1.7) to an equivalent upper triangular system:

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 2 & 9 \end{array}\right] \to \left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 3 & 14 \end{array}\right] \to \left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

and obtained a unique solution

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

What if (1.1.7) becomes

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & \mathbf{1} & 9 \end{array}\right]? \tag{1.1.14}$$

We apply Gaussian elimination:

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 1 & 9 \end{array}\right] \to \left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 2 & 14 \end{array}\right] \to \left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 0 & 2 \end{array}\right].$$

As there are no x_1, x_2, x_3 such that $0x_1 + 0x_2 + 0x_3 = 2$, so (1.1.14) has no solution.

If we further change (1.1.14) to

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 1 & \mathbf{7} \end{array}\right], \tag{1.1.15}$$

then we have

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 1 & 7 \end{array}\right] \to \left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 2 & 12 \end{array}\right] \to \left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

We can choose the value of x_3 arbitrarily, and express x_1 and x_2 in terms of x_3. Thus (1.1.15) has infinitely many solutions. We shall return to this example later.

In fact, we have displayed all three possibilities of the number of solutions to a 3×3 linear system: 0, 1, or infinitely many. Later we will see that is true for every $n \times n$ linear system.

■

1.2 Friday (Prof. Dongdong He)

In this lecture we will further our discussion over Gaussian elimination on $n \times n$ systems. We will see cases when Gaussian elimination breaks down. We will also talk about basic matrix operations.

1.2.1 Reduced Row Echelon Form

Before we start our discussion, we shall introduce some matrix notations, which we have already used on Monday.

Matrix Notations

Given a linear system

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases} \tag{1.2.1}$$

of m equations in n unknowns $x_1, ..., x_n$, we can pack the numbers and unknowns in matrices and vectors. The *coefficient matrix*, the *unknown vector*, and the *given vector* are defined as

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \qquad x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad \text{and} \qquad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix},$$

respectively.

Using matrix-vector multiplication, which we will introduce later, we can rewrite (1.2.1) as the matrix equation

$$Ax = b. \tag{1.2.2}$$

The *augmented matrix* is defined as

$$[A|b] = \left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array}\right] \tag{1.2.3}$$

Gaussian Elimination on $n \times n$ Systems

We now continue our discussion on Gaussian elimination. For now we will focus on $n \times n$ systems:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \quad \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n \end{cases} \tag{1.2.4}$$

Ideally, if we perform Gaussian elimination, we will get

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{array}\right] \to \left[\begin{array}{cccc|c} a_{11} & * & \cdots & * & * \\ 0 & a_{22}^{(1)} & \cdots & a_{2n}^{(1)} & b_2^{(1)} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & a_{n2}^{(1)} & \cdots & a_{nn}^{(1)} & b_n^{(1)} \end{array}\right] \to \cdots$$

$$\rightarrow \left[\begin{array}{cccc|c} a_{11} & * & \cdots & * & * \\ & a_{22}^{(1)} & \cdots & * & * \\ & & \ddots & \vdots & \vdots \\ & \mathbf{0} & & a_{nn}^{(n-1)} & * \end{array}\right] \tag{1.2.5}$$

where a_{11} ,$a_{22}^{(1)}$, ..., $a_{nn}^{(n-1)}$ are nonzero. We can then use back substitution to find a unique solution. The entries a_{11} ,$a_{22}^{(1)}$, ..., $a_{nn}^{(n-1)}$ on the main diagonal are called *pivots*. Each of them is used to zero out the entries below.

If a matrix is nice enough, we can get n pivots after elimination. However, this is not always the case. If some entry on the diagonal becomes zero during elimination, things *may* go wrong.

Example 1.2.1. Consider the matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 2 & 2 & 5 & 2 \\ 4 & 6 & 8 & 3 \end{array}\right]. \tag{1.2.6}$$

We perform Gaussian elimination on the first column:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 2 & 4 & -1 \end{array}\right]. \tag{1.2.7}$$

Here, the second entry of the second row of (1.2.7) is 0 and cannot be used to do the elimination. However, this is curable, as the second entry of row 3 is nonzero. So we can exchange row 2 and row 3 using row operation (i) and we obtain:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & -1 \\ 0 & 0 & 3 & 0 \end{array}\right]. \tag{1.2.8}$$

Then there *is* a pivot for the second column. Now we can use back substitution to obtain the unique solution

$$x_1 = \frac{3}{2}, \qquad x_2 = -\frac{1}{2}, \qquad x_3 = 0.$$

In this example, (1.2.8), the so-called *row echelon form* of (1.2.6), has full pivots. In this case, the matrix (1.2.6) is said to be *nonsingular*.

In example 1.2.1, when we are doing elimination, the second entry of the second row of (1.2.7) becomes zero. But this is curable, as we can find another row below whose second entry is nonzero. However, this is not always the case, as we will illustrate in the following example.

Example 1.2.2. If we modify (1.2.6) to be

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 2 & 2 & 5 & 2 \\ 4 & \mathbf{4} & 8 & 3 \end{array}\right] \tag{1.2.9}$$

and perform Gaussian elimination, we obtain in the first step

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 4 & -1 \end{array}\right].$$

Here we fail to find a pivot for the second column as the second entries of row 2 and row 3 are both zero. However, we can continue the process of elimination on the third column. We treat the third entry of the second row as a pivot. Then we get

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & -1 \end{array}\right]. \tag{1.2.10}$$

As no x_1, x_2, x_3 satisfy the third equation, (1.2.10) has no solution.

In example 1.2.2, (1.2.10) has only two pivots and the system (1.2.9) is called *singular*. The system in this example has no solution. However, that is not the only possibility for singular systems. We show the other one in the next example.

Example 1.2.3. We further modify (1.2.9) to be

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 2 & 2 & 5 & 2 \\ 4 & 4 & 8 & \mathbf{4} \end{array}\right].$$

The first step of Gaussian elimination yields

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 4 & 0 \end{array}\right].$$

Again, this system is singular as there is no pivot for the second column. We skip to the third column and continue elimination. This gives

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

We obtain $x_3 = 0$ and $x_1 + x_2 = 1$. Thus the system has infinitely many solutions.

As a matter of fact, we have displayed all possibilities of the number of solutions to an $n \times n$ linear system. A linear system is either nonsingular or singular, depending on the number of pivots. A nonsingular linear system always has a unique solution. A singular system has either no solution or infinitely many solutions. We shall now see why.

Row Echelon Form

In general, through row operations (i), (ii), and (iii), an augmented matrix

$$\left[\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & b_n \end{array}\right] \tag{1.2.11}$$

associated with a linear system is reduced to a matrix in the form of

$$\left[\begin{array}{cccccccc|c} \blacksquare & * & * & * & \cdots & * & * & * & c_1 \\ 0 & 0 & \blacksquare & * & \cdots & * & * & * & c_2 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & \blacksquare & * & c_r \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & c_{r+1} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 & c_n \end{array}\right] \tag{1.2.12}$$

where the ■'s are nonzero and are called *pivots*. The matrix (1.2.12) is called the *row echelon form* of (1.2.11). (The word "echelon" means "steplike".) We shall now formulate our previous ideas.

Definition 1.2.4. A matrix is said to be in *row echelon form* if it has the following properties:

(i) If row k does not consist entirely of zeros, then the number of leading zeros in row $(k+1)$ is greater than that in row k.

(ii) All zero rows are below all nonzero rows.

Definition 1.2.5. Suppose a matrix is in row echelon form. The first nonzero entry of each nonzero row is called a *pivot*.

Remark. By definition 1.2.4, if a column of a matrix in row echelon form has a pivot, then the entries below a pivot are all zeros. Also, each column has at most one pivot. This is consistent with our previous idea.

Definition 1.2.6. The process of reducing a matrix to a row echelon form matrix using row operations (i), (ii), and (iii) is called *Gaussian elimination*. The resultant matrix is called the *row echelon form* of the original matrix.

Definition 1.2.7. An $n \times n$ square matrix is said to be *nonsingular* if its row echelon form has n pivots. It is said to be *singular* if its row echelon form has fewer than n pivots.

Possibilities of Number of Solutions to a Square Linear System

We now go back to discuss the possibilities of the number of solutions to a square linear system. The systems (1.2.11) and (1.2.12) are equivalent. That is, they have the same solution set. (1.2.12) has r pivots, which are in the first r rows, respectively. We now display all the possible cases.

1. If $r = n$, then (1.2.12) has full pivots and the system is nonsingular. The pivots are on the main diagonal and we can use back substitution to find a unique solution.

2. If $r < n$, then (1.2.12) has fewer than n pivots and the system is singular.

   (a) If all $c_{r+1}, ..., c_n$ are zeros, then the system is consistent and has infinitely many solutions. To see why, note that (1.2.12) is equivalent to

$$\begin{cases} a_{k_1}x_{k_1} + S_{k_1}(x) & = c_1 \\ a_{k_2}x_{k_2} + S_{k_2}(x) & = c_2 \\ \cdots\cdots & \\ a_{k_r}x_{k_r} + S_{k_r}(x) & = c_r \end{cases} \tag{1.2.13}$$

   where $a_{k_1} \neq 0, ..., a_{k_r} \neq 0$ are the pivots and $S_{k_1}(x), ..., S_{k_r}(x)$ are sums of variables with certain coefficients such that if x_j occurs in $S_{k_i}(x)$, then $j > k_i$. Since $r < n$, there are $(n-r)$ *free variables* apart from the r *lead variables* $x_{k_1}, ..., x_{k_r}$ $(n-r > 0)$. We can assign arbitrary values to these free variables. Then we solve for the lead variables $x_{k_r}, ..., x_{k_1}$, starting from the bottom equation and working back up:

$$x_{k_r} = -\frac{S_{k_r}(x) - c_r}{a_{k_r}}, \quad x_{k_{r-1}} = -\frac{S_{k_{r-1}}(x) - c_{r-1}}{a_{k_{r-1}}}, \quad \text{and so forth.}$$

   Since there is at least one free variable that we can assign an arbitrary value to, the system has infinitely many solutions.

   (b) If some c_i of $c_{r+1}, ..., c_n$ is nonzero, then the system is inconsistent.

Therefore, an $n \times n$ linear system must have no solution, exactly one solution, or infinitely many solutions. These are the only three possibilities.

Reduced Row Echelon Form

We can further simplify the row echelon form of a matrix by clearing out the entries above the pivots. In this process we are actually performing back substitution. Doing so enables us to directly solve the linear system and obtain a complete solution set. We illustrate this process with our previous examples.

Example 1.2.8. In example 1.2.1 we obtain the row echelon form

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & -1 \\ 0 & 0 & 3 & 0 \end{array}\right].$$

We start from the last row. We first make its pivot 1 using row operation (ii) (multiplying row 3 by $\frac{1}{3}$):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 2 & 4 & -1 \\ 0 & 0 & 1 & 0 \end{array}\right].$$

Then we use row 3 to zero out the entries above its pivot. We obtain

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{array}\right].$$

We continue this process:

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & -\frac{1}{2} \\ 0 & 0 & 1 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{3}{2} \\ 0 & 1 & 0 & -\frac{1}{2} \\ 0 & 0 & 1 & 0 \end{array}\right].$$

We immediately obtain the solution:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{3}{2} \\ -\frac{1}{2} \\ 0 \end{bmatrix}.$$

We can also do this for singular systems.

Example 1.2.9. For the matrix in example 1.2.2, we have

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & -1 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{array}\right].$$

The system is inconsistent because the last row is $\left[\begin{array}{ccc|c} 0 & 0 & 0 & -1 \end{array}\right]$.

Example 1.2.10. For the matrix in example 1.2.3, we get

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

We directly get $x_1 + x_2 = 1$ and $x = 3$. By letting the free variable $x_2 = \alpha$, we obtain the general solution

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1-\alpha \\ \alpha \\ 0 \end{bmatrix}, \qquad \text{where } \alpha \in \mathbb{R}.$$

The matrix we eventually obtain is called the *reduced row echelon form* of the original matrix. We give the formal definition below.

Definition 1.2.11. A matrix is said to be in *reduced row echelon form* if it satisfies the following conditions:

(i) The matrix is in row echelon form.

(ii) The pivots are all 1.

(iii) The entries above the pivots are all zeros.

Definition 1.2.12. The process of reducing a matrix to a matrix in reduced row echelon form using elementary row operations is called *Gauss-Jordan reduction*. The resultant matrix is called the *reduced row echelon form* of the original matrix.

Remark. Note that a matrix may have more than one row echelon form, but one and only one reduced row echelon form.

We end this section with an example of Gauss-Jordan reduction on a 5×5 system.

Example 1.2.13. Consider the linear system associated with the augmented matrix

$$\left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & 0 & 0 & 1 & -1 \\ -2 & -2 & 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 1 & 3 & -1 \\ 1 & 1 & 2 & 2 & 4 & 1 \end{array}\right]. \tag{1.2.14}$$

We perform Gauss-Jordan reduction, with pivots highlighted.

$$\left[\begin{array}{ccccc|c} \boxed{1} & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & 0 & 0 & 1 & -1 \\ -2 & -2 & 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 1 & 3 & -1 \\ 1 & 1 & 2 & 2 & 4 & 1 \end{array}\right] \to \left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & \boxed{1} & 1 & 2 & 0 \\ 0 & 0 & 2 & 2 & 5 & 3 \\ 0 & 0 & 1 & 1 & 3 & -1 \\ 0 & 0 & 1 & 1 & 3 & 0 \end{array}\right] \to \left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} & 3 \\ 0 & 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right]$$

$$\to \left[\begin{array}{ccccc|c} \boxed{1} & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & \boxed{1} & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} & 3 \\ 0 & 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 0 & 0 & -3 \end{array}\right].$$

So far we have finished Gaussian elimination. We continue.

$$\left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} & 3 \\ 0 & 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 0 & 0 & -3 \end{array}\right] \to \left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 0 & -2 \\ 0 & 0 & \boxed{1} & 1 & 0 & -6 \\ 0 & 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 0 & 0 & -3 \end{array}\right] \to \left[\begin{array}{ccccc|c} \boxed{1} & 1 & 0 & 0 & 0 & 4 \\ 0 & 0 & \boxed{1} & 1 & 0 & -6 \\ 0 & 0 & 0 & 0 & \boxed{1} & 3 \\ 0 & 0 & 0 & 0 & 0 & -4 \\ 0 & 0 & 0 & 0 & 0 & -3 \end{array}\right]. \tag{1.2.15}$$

Since (1.2.15) contains the rows $\left[\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & 0 & -3 \end{array}\right]$ and $\left[\begin{array}{ccccc|c} 0 & 0 & 0 & 0 & 0 & -4 \end{array}\right]$, then (1.2.14) has no solution.

However, if we change (1.2.14) to

$$\left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & 0 & 0 & 1 & -1 \\ -2 & -2 & 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 1 & 3 & 3 \\ 1 & 1 & 2 & 2 & 4 & 4 \end{array}\right],$$

then

$$\left[\begin{array}{ccccc|c} 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & 0 & 0 & 1 & -1 \\ -2 & -2 & 0 & 0 & 3 & 1 \\ 0 & 0 & 1 & 1 & 3 & 3 \\ 1 & 1 & 2 & 2 & 4 & 4 \end{array}\right] \to \cdots \to \left[\begin{array}{ccccc|c} \boxed{1} & 1 & 0 & 0 & 0 & 4 \\ 0 & 0 & \boxed{1} & 1 & 0 & -6 \\ 0 & 0 & 0 & 0 & \boxed{1} & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

We immediately obtain

$$\begin{cases} x_1 + x_2 = 4 \\ x_3 + x_4 = -6 \\ x_5 = 3 \end{cases} \quad \text{or equivalently,} \quad \begin{cases} x_1 = 4 - x_2 \\ x_3 = -6 - x_4 \\ x_5 = 3 \end{cases}.$$

By letting the free variables $x_2 = \alpha$ and $x_4 = \beta$, we obtain the general solution

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 4-\alpha \\ \alpha \\ -6-\beta \\ \beta \\ 3 \end{bmatrix}, \quad \text{where } \alpha, \beta \in \mathbb{R}.$$

1.2.2 Matrices and Matrix Arithmetic

We now give a more detailed introduction of matrices and operations on them. Suppose

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix} \tag{1.2.16}$$

is a matrix with m rows and n columns and its entries are real numbers. Then $A \in \mathbb{R}^{m\times n}$, where $\mathbb{R}^{m\times n}$ denotes the set of all $m \times n$ matrices whose entries are real numbers. We also write $A = (a_{ij})$ as a shorthand notation for (1.2.16).

We can represent A in terms of its *row vectors* $\vec{a}_1, ..., \vec{a}_m$:

$$A = \begin{bmatrix} \vec{a}_1 \\ \vdots \\ \vec{a}_m \end{bmatrix} \qquad \text{where} \quad \vec{a}_i = \begin{bmatrix} a_{i1} & \cdots & a_{in} \end{bmatrix} \in \mathbb{R}^{1\times n},\ i = 1, ..., m. \tag{1.2.17}$$

We can also represent A in terms of its *column vectors* $a_1, ..., a_n$:

$$A = \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix} \qquad \text{where} \quad a_j = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{bmatrix} \in \mathbb{R}^{m\times 1} = \mathbb{R}^m,\ j = 1, ..., n. \tag{1.2.18}$$

Matrix Examples

In this section we introduce some special matrices.

1. *Zero matrix.* A *zero matrix* is a matrix with all elements equal to 0. The zero matrix of size $m \times n$ is sometimes denoted by $O_{m\times n}$. But usually we will simply write O. In this case the size of the zero matrix must be determined from the context.

2. *Diagonal matrix.* A square matrix D is said to be a *diagonal matrix* if its off-diagonal entries are all zeros, that is, $D_{ij} = 0$ for all $i \neq j$. A commonly used notation is $D = \text{diag}(d_1, ..., d_n)$, which represents an $n \times n$ diagonal matrix D with diagonal entries $D_{ii} = d_i,\ i = 1, ..., n$.

3. *Identity matrix.* The *identity matrix* of size $n \times n$, denoted by I_n or simply I, is the $n \times n$ diagonal matrix whose diagonal entries are all 1. For example,
$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

4. *Triangular matrix.* A square matrix L is *lower triangular* if $L_{ij} = 0$ whenever $i < j$. A lower triangular matrix may take the shape
$$L = \begin{bmatrix} * & 0 & 0 \\ * & * & 0 \\ * & * & * \end{bmatrix}.$$
A square matrix U is *upper triangular* if $U_{ij} = 0$ whenever $i > j$. An upper triangular matrix may look like
$$U = \begin{bmatrix} \times & \times & \times & \times \\ 0 & \times & \times & \times \\ 0 & 0 & \times & \times \\ 0 & 0 & 0 & \times \end{bmatrix}.$$
If a matrix is either lower triangular or upper triangular, it is called *triangular*. A diagonal matrix is *always* lower triangular and upper triangular.

Matrix Arithmetic

We now introduce some operations on matrices.

Equality Two matrices $A = (a_{ij})$ and $B = (b_{ij})$ are said to be *equal* if they have the same size and if $a_{ij} = b_{ij}$ for every i and j. We denote it by $A = B$.

Scalar Multiplication Let $A = (a_{ij}) \in \mathbb{R}^{m\times n}$ be a matrix and α be a scalar. The product of A *multiplied* by α, denoted by αA, is a matrix in $\mathbb{R}^{m\times n}$ defined by
$$(\alpha A)_{ij} = \alpha a_{ij}.$$

Addition and Subtraction Suppose $A = (a_{ij})$ and $B = (b_{ij})$ are matrices of the same size, say $m \times n$. Their sum, denoted by $A + B$, is the matrix of the same size defined by
$$(A + B)_{ij} = a_{ij} + b_{ij}.$$
Similarly, we can define subtraction: $A - B$ is the $m \times n$ matrix defined by
$$(A - B)_{ij} = a_{ij} - b_{ij}.$$

Inner Product The *inner product* of a row vector $\vec{x} = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}$ of size n and a column vector $y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$ of size n, denoted by $\vec{x}y$, is a scalar defined as

$$\vec{x}y = \sum_{i=1}^{n} x_i y_i. \tag{1.2.19}$$

It is more common to define the inner product of two column vectors. Suppose

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad \text{and} \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}.$$

The *inner product* of x and y, usually denoted by $x^T y$ or $\langle x, y \rangle$, is defined as

$$x^T y = \sum_{i=1}^{n} x_i y_i. \tag{1.2.20}$$

(The "x^T" in $x^T y$ actually denotes the *transpose* of x and $x^T = \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix}$. So there is no conflict between the two definitions (1.2.19) and (1.2.20).)

Matrix-Vector Multiplication (Post-multiplication by a column vector)

Definition 1.2.14. Let

$$A = \begin{bmatrix} \vec{a}_1 \\ \vdots \\ \vec{a}_m \end{bmatrix} \in \mathbb{R}^{m \times n} \text{ be a matrix} \qquad \text{and} \qquad x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n \text{ be a vector.}$$

The product of *A post-multiplied by x* (which means A multiplied by x on the right), denoted by Ax, is an m-vector defined by

$$Ax = \begin{bmatrix} \vec{a}_1 x \\ \vdots \\ \vec{a}_m x \end{bmatrix}. \tag{1.2.21}$$

Example 1.2.15. Suppose

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 6 & 8 \\ 9 & 10 & 11 \end{bmatrix} \quad \text{and} \quad x = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}.$$

Then

$$Ax = \begin{bmatrix} (1)(1)+(2)(-1)+(3)(2) \\ (5)(1)+(6)(-1)+(8)(2) \\ (9)(1)+(10)(-1)+(11)(2) \end{bmatrix} = \begin{bmatrix} 5 \\ 15 \\ 21 \end{bmatrix}.$$

An alternative but equivalent definition of matrix-vector multiplication is as follows.

Definition 1.2.16. Let

$$A = \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix} \in \mathbb{R}^{m \times n} \text{ be a matrix} \qquad \text{and} \qquad x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n \text{ be a vector.}$$

The product of *A post-multiplied by x*, denoted by Ax, is an m-vector defined by

$$Ax = a_1 x_1 + \cdots + a_n x_n. \tag{1.2.22}$$

From this definition, we see that Ax is actually a linear combination of the columns of A.

Example 1.2.17. We use definition 1.2.16 to calculate Ax in example 1.2.15:

$$Ax = 1\begin{bmatrix}1\\5\\9\end{bmatrix} + (-1)\begin{bmatrix}2\\6\\10\end{bmatrix} + 2\begin{bmatrix}3\\8\\11\end{bmatrix} = \begin{bmatrix}5\\15\\21\end{bmatrix}.$$

Definitions 1.2.14 and 1.2.16 are equivalent because

$$x_1a_1 + \cdots + x_na_n = x_1\begin{bmatrix}a_{11}\\ \vdots \\ a_{m1}\end{bmatrix} + \cdots + x_n\begin{bmatrix}a_{1n}\\ \vdots \\ a_{mn}\end{bmatrix} = \begin{bmatrix}a_{11}x_1 + \cdots + a_{1n}x_n\\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n\end{bmatrix} = \begin{bmatrix}\vec{a}_1x\\ \vdots \\ \vec{a}_mx\end{bmatrix}.$$

After the introduction of matrix-vector multiplication, we see that the linear system

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases}$$

is equivalent to the matrix equation

$$\begin{bmatrix}a_{11} & a_{12} & \cdots & a_{1n}\\ a_{21} & a_{22} & \cdots & a_{2n}\\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn}\end{bmatrix}\begin{bmatrix}x_1\\x_2\\ \vdots \\ x_n\end{bmatrix} = \begin{bmatrix}b_1\\b_2\\ \vdots \\ b_m\end{bmatrix}, \text{ or equivalently, } Ax = b.$$

Complexity of Matrix-Vector Multiplication

Suppose $A \in \mathbb{R}^{m\times n}$ and $x \in \mathbb{R}^n$. Using definition 1.2.14, we see that calculating each entry of Ax requires n multiplications and $(n-1)$ additions, thus $(2n-1)$ flops in total. Since Ax has m entries in total, we conclude that the complexity of post-multiplying an $m \times n$ matrix by an n-vector is equal to $m(2n-1) = 2mn - m \sim 2mn$ flops.

If we use definition 1.2.16, we will still get the same result. Calculating each a_ix_i requires m multiplications and thus calculating all of them requires mn flops. Adding them up requires $m(n-1)$ flops. So the overall complexity is equal to $mn + m(n-1) = 2mn - m \sim 2mn$ flops.

As a special case, post-multiplying an $n \times n$ square matrix by an n-vector takes about $2n^2$ flops.

Matrix-Vector Multiplication (Pre-multiplication by a row vector)

Definition 1.2.18. Let

$$\vec{y} = \begin{bmatrix}y_1 & \cdots & y_m\end{bmatrix} \in \mathbb{R}^{1\times m} \text{ be a row vector} \qquad \text{and} \qquad A = \begin{bmatrix}\vec{a}_1\\ \vdots \\ \vec{a}_m\end{bmatrix} \in \mathbb{R}^{m\times n} \text{ be a matrix.}$$

The product of A *pre-multiplied by* $\vec{y}$ (which means A multiplied by $\vec{y}$ on the left), denoted by $\vec{y}A$, is a row vector of size n defined by

$$\vec{y}A = y_1\vec{a}_1 + \cdots + y_m\vec{a}_m.$$

From this definition, we see that $\vec{y}A$ is actually a linear combination of the rows of A.

The complexity of multiplying an $m \times n$ matrix by a row vector of size m is also $O(mn)$. We can think about this by taking transpose.

■

Note. I have included materials related to the row echelon form of a matrix, which was not mentioned in the original lecture. I have also included pre-multiplication of a matrix by a row vector, as this is needed for the discussion in the next lecture. I have also added complexity analysis for matrix-vector multiplication.

Chengpiao Huang

Chapter 2

Week Two

2.1 Monday (Prof. Tom Luo)

2.1.1 Null, and Left Null

After we have learned the concepts of linear combination and Gaussian elimination, we go back to solving (1.1.15) and see what implications it gives us. The matrix (1.1.15) is

$$[A|b] = \left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 1 & 7 \end{array}\right].$$

We apply Gaussian elimination on (1.1.15):

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 4 & -6 & 0 & -2 \\ -2 & 7 & 1 & 7 \end{array}\right] \begin{array}{c} R_1 \\ R_2 \\ R_3 \end{array} \rightarrow \left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 2 & 12 \end{array}\right] \begin{array}{c} R_1 \\ R_2' = R_2 - 2R_1 \\ R_3' = R_3 + R_1 \end{array}$$

$$\rightarrow \left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 0 & 0 \end{array}\right] \begin{array}{c} R_1 \\ R_2' = R_2 - 2R_1 \\ R_3'' = R_3' + R_2' \end{array} \tag{2.1.1}$$

Thus

$$R_3'' = R_3' + R_2' = (R_3 + R_1) + (R_2 - 2R_1) = -R_1 + R_2 + R_3 = 0. \tag{2.1.2}$$

From (2.1.2), we see that *a nontrivial linear combination of the rows of A gives* 0, or equivalently,

$$\vec{y}A = 0 \quad \text{for some } \vec{y} \neq 0. \tag{2.1.3}$$

The equation $Ax = b$ has infinitely many solutions as the last row of (2.1.1) is zero. Now consider the equation $Ax = 0$. The associated augmented matrix is just $[A|0]$ and is eventually reduced to (2.1.1) with the right part replaced by the zero vector. As the last row of the reduced augmented matrix is zero, $Ax = 0$ has a nontrivial solution. Thus *a nontrivial linear combination of the columns of A gives* 0, or equivalently,

$$Ax = 0 \quad \text{for some } x \neq 0. \tag{2.1.4}$$

It seems that (2.1.3) implies (2.1.4). In fact, they are equivalent. We have the following general theorem that holds for every square matrix.

Theorem 2.1.1. *For a square matrix A, $Ax = 0$ for some $x \neq 0$ if and only if $\vec{y}A = 0$ for some $\vec{y} \neq 0$. In other words, there exists a nontrivial linear combination of the rows of A gives* 0 *if and only if there is a nontrivial linear combination of the columns of A gives* 0*.*

Remark. Note that theorem 2.1.1 does not hold for non-square matrices in general.

2.1.2 Matrix Multiplication

General Formula for Matrix Multiplication

Suppose

$$A = \begin{bmatrix} \vec{a}_1 \\ \vdots \\ \vec{a}_m \end{bmatrix} \in \mathbb{R}^{l \times m} \quad \text{and} \quad B = \begin{bmatrix} b_1 & \cdots & b_n \end{bmatrix} \in \mathbb{R}^{m \times n}.$$

We can extend our definition of matrix-vector multiplication to matrix multiplication. We define the product of A and B, denoted by AB, as

$$AB = \begin{bmatrix} \vec{a}_1 b_1 & \cdots & \vec{a}_1 b_n \\ \vdots & \ddots & \vdots \\ \vec{a}_m b_1 & \cdots & \vec{a}_m b_n \end{bmatrix}.$$

Thus $AB \in \mathbb{R}^{l \times n}$ and the (i, j) entry of AB is given by

$$(AB)_{ij} = \vec{a}_i b_j = \sum_{k=1}^{m} a_{ik} b_{kj}.$$

Matrix Multiplication by Columns

By definition,

$$AB = A \begin{bmatrix} b_1 & \cdots & b_n \end{bmatrix} = \begin{bmatrix} Ab_1 & \cdots & Ab_n \end{bmatrix}.$$

Therefore, the jth column of AB depends on only A and the jth column of B, but not other columns of B. Since each Ab_i is a linear combination of the columns of A, we can interpret post-multiplying a matrix A by B as performing *column operations* on A.

Matrix Multiplication by Rows

By definition,

$$AB = \begin{bmatrix} \vec{a}_1 \\ \vdots \\ \vec{a}_m \end{bmatrix} B = \begin{bmatrix} \vec{a}_1 B \\ \vdots \\ \vec{a}_m B \end{bmatrix}.$$

Thus the ith row of AB depends on only B and the ith row of A, but not other rows of A. As each $\vec{a}_i B$ is a linear combination of B, we can interpret pre-multiplying a matrix B by A as performing *row operations* on B.

Complexity of Matrix Multiplication

Suppose $A \in \mathbb{R}^{l \times m}$ and $B \in \mathbb{R}^{m \times n}$. Let $C = AB \in \mathbb{R}^{l \times n}$. Computing each entry of C requires m multiplications and $(m-1)$ additions, thus $(2m-1) =$ flops in total. As there are nl entries in C, the overall complexity of calculating the product of an $l \times m$ matrix and an $m \times n$ matrix is equal to $nl(2m-1) = 2lmn - nl \sim 2lmn$ flops. As a special case, the complexity of matrix multiplication for two $n \times n$ square matrices is about $2n^3$ flops.

Block Multiplication

Suppose A and B are matrices such that the product AB is defined. To calculate AB, we can divide A and B into submatrices:

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}.$$

We can view the submatrices as scalars and apply the same matrix multiplication rule as for ordinary matrix multiplications to compute AB:

$$AB = \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix},$$

provided that the multiplications make sense. We can verify this rule using the definition of matrix multiplication.

2.1.3 Strassen Algorithm

The complexity of computing the product of two $n \times n$ matrices using the standard formula is $O(n^3)$. However, this is not the fastest way. Here we introduce a faster algorithm called *Strassen algorithm* for square matrix multiplication, which has the time complexity of only $O(n^{\log_2 7}) \approx O(n^{2.8074})$.

Base Case

We start with the base case, which is the multiplication of two 2×2 matrices. Suppose we want to compute the product of two 2×2 matrices

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} e & f \\ g & h \end{bmatrix}.$$

If we let

$$p_1 = a(f-h),\ p_2 = (a+b)h,\ p_3 = (c+d)e,\ p_4 = d(g-e),\ p_5 = (a+d)(e+h),$$

$$p_6 = (b-d)(g+h), \text{ and } p_7 = (a-c)(e+f),$$

it can be verified that

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} e & f \\ g & h \end{bmatrix} = \begin{bmatrix} p_4 + p_5 + p_6 - p_2 & p_1 + p_2 \\ p_3 + p_4 & p_1 + p_5 - p_3 - p_7 \end{bmatrix}.$$

Below we compare the numbers of additions and multiplications involved in Strassen algorithm with those in standard multiplication.

Table 2.1: Complexities of 2×2 Matrix Multiplication Algorithms

	Standard Multiplication	Strassen Algorithm
Multiplications	8	7
Additions	4	18

We see that Strassen algorithm is one multiplication fewer than standard multiplication, at the cost of much more additions. The reduction of multiplication, however, turns out to be the key to the reduction of time complexity.

Recursion

Suppose A, B, and C are $n \times n$ and $AB = C$. The idea of Strassen algorithm is to use block multiplication to recursively calculate AB. In each recursion, we break down each matrix into 4 smaller submatrices of the same size. For example, in the first recursion, we have

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix},$$

where the A_{ij}'s, B_{ij}'s and C_{ij}'s are all $\frac{n}{2} \times \frac{n}{2}$. Matrix multiplications are calculated using the formula in the base case, with submatrices in place of scalar entries. The recursion ends when we reach the base case.

Complexity Analysis

Let $T(n)$ be the number of arithmetic operations required to compute the product of two $n \times n$ matrices using Strassen algorithm. Using the results in table 2.1, we obtain

$$T(n) = \underbrace{7T\left(\frac{n}{2}\right)}_{\text{multiplications}} + \underbrace{18\left(\frac{n}{2}\right)^2}_{\text{additions}}.$$

Therefore

$$\begin{aligned} T(n) &= O(7^{\log_2 n}) + O(n^2) \\ &= O(2^{(\log_2 7)(\log_2 n)}) + O(n^2) \\ &= O(n^{\log_2 7}) + O(n^2) \\ &= O(n^{\log_2 7}) \\ &\approx O(n^{2.8704}). \end{aligned}$$

Modified Strassen Algorithm for Odd n

The reader may have noticed that Strassen algorithm works only when the orders n of the matrices are powers of 2. How do we modify the algorithm so that it works for *all* square matrices?

One solution is to pad the matrices with zeros, using the block matrix identity

$$\begin{bmatrix} A & O \\ O & O \end{bmatrix} \begin{bmatrix} B & O \\ O & O \end{bmatrix} = \begin{bmatrix} AB & O \\ O & O \end{bmatrix}.$$

There are two ways of adding zeros. The first one is to pad the original $n \times n$ matrices to $N \times N$ ones, where N is the closest power of 2 (that is, N such that $N = 2^k$ and $N/2 < n < N$). As $n < N < 2n$, the complexity is $O(N^{\log_2 7}) = O(n^{\log_2 7})$ flops. So this does not affect the complexity.

The other one is to pad the zeros recursively. Whenever a recursive call gets $m \times m$ matrices where m is odd, pad zeros such that the matrices become $(m+1) \times (m+1)$. The complexity is also unaffected because multiplying zeros does not take extra flops.

State of the Art

Currently the best algorithm for matrix multiplication can achieve the time complexity of $O(n^{2.373})$. The best lower bound is $O(n^2)$. If no subtraction is allowed in matrix multiplication, the best lower bound is $O(n^3)$.

2.1.4 Special Matrices

The Identity Matrix

Theorem 2.1.2. *Let X be an $n \times n$ matrix. If $XA = AX = A$ for every $A \in \mathbb{R}^n$, then $X = I$.*

Proof.

$$X = XI = I.$$

□

Elementary Matrices

Elementary matrices are matrices that perform elementary row operations. We now give the elementary matrices that correspond to the three types of elementary row operations. Recall that to manipulate the rows of a matrix, we perform pre-multiplications.

Type (i): Interchange two rows. Elementary matrices of type (i) take the form

$$E = \begin{array}{c} \\ \\ \\ \\ \\ \\ \end{array} \begin{array}{c} \begin{matrix} \quad & \quad & i & \quad & j & \quad \end{matrix} \\ \begin{bmatrix} 1 & & & & & & \\ & \ddots & & & & 0 & \\ & & 0 & \cdots & 1 & & \\ & & \vdots & \ddots & \vdots & & \\ & & 1 & \cdots & 0 & & \\ & 0 & & & & \ddots & \\ & & & & & & 1 \end{bmatrix} \end{array} \begin{array}{l} \\ \\ i \\ \\ j \\ \\ \\ \end{array} . \tag{2.1.5}$$

Pre-multiplying a matrix by (2.1.5) exchanges the ith row and jth row of the matrix.

Type (ii): Multiply a row by a nonzero constant. Elementary matrices of type (ii) take the form

$$\begin{bmatrix} 1 & & & & \\ & \ddots & & 0 & \\ & & c & & \\ & 0 & & \ddots & \\ & & & & 1 \end{bmatrix} i \tag{2.1.6}$$

where c is nonzero. Pre-multiplying a matrix by (2.1.6) has the effect of multiplying the ith row of the matrix by a nonzero scalar c.

Type (iii): Replace a row by its sum with a multiple of another row. Elementary matrices of type (iii) take the form

$$E_{ij} = \begin{array}{c} \\ \\ \\ \\ \end{array} \begin{array}{c} \begin{matrix} \quad & \quad & j & \quad \end{matrix} \\ \begin{bmatrix} 1 & & & \\ & \ddots & & 0 \\ & & 1 & \\ & -l_{ij} & & \ddots \\ & & & & 1 \end{bmatrix} \end{array} \begin{array}{l} \\ \\ \\ i \\ \\ \end{array} . \tag{2.1.7}$$

Pre-multiplying a matrix by (2.1.7) replaces (row i) of the matrix with [(row i) $- l_{ij}$(row j)].

The key observation is that *performing Gaussian elimination on a matrix is equivalent to pre-multiplying the matrix by a series of elementary matrices.* In Gaussian elimination, elementary matrices are all lower triangular.

2.1.5 Matrix Inverse

Definition 2.1.3. Suppose A is an $n \times n$ matrix. If there exists an $n \times n$ matrix B such that $AB = BA = I$, A is said to be *invertible*. The matrix B is called the *inverse* of A and is denoted by A^{-1}.

Theorem 2.1.4. *Let $A,\ B_1,\ B_2 \in \mathbb{R}^{n\times n}$. If $AB_1 = B_2A = I$, then $B_1 = B_2$.*

Proof.

$$B_2 = B_2I = B_2(AB_1) = (B_2A)B_1 = B_1.$$

□

The following theorem can be summarized with the simple formula $(AB)^{-1} = B^{-1}A^{-1}$.

Theorem 2.1.5. *Suppose A and B are $n \times n$ matrices.*

(i) If A and B are invertible, then AB is also invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

(ii) If AB is invertible, then A and B are also invertible.

We now prove the first one.

Proof of (i). Suppose A and B are invertible. Since $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I$ and $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I$, then AB is also invertible. □

We are currently not ready to prove (ii). In the next section, we will develop the tools we need to prove (ii).

2.1.6 Invertibility and Nonsingularity

In this section we explore the relationship between the invertibility and nonsingularity of a square matrix. We first recall the definition of a nonsingular matrix.

Definition 2.1.6. An $n \times n$ matrix is called *nonsingular* if its row echelon form has$n$ pivots. If its row echelon form has fewer than n pivots, it is called *singular*.

Theorem 2.1.7. *Let A be an $n \times n$ square matrix. The following statements are equivalent.*

(i) A is nonsingular.

(ii) For every $b \in \mathbb{R}^n$, the matrix equation $Ax = b$ always has a unique solution.

(iii) $Ax = 0$ implies $x = 0$.

Theorem 2.1.8. *Let A be an $n \times n$ square matrix. The following statements are equivalent.*

(i) A is singular.

(ii) If the matrix equation $Ax = b$ is consistent, then it has infinitely many solutions.

(iii) $Ax = 0$ has a nontrivial solution.

The reader may refer to the discussion in "Possibilities of Number of Solutions to a Square Linear System" subsection in section 1.2.1 for a sketch of proof.

We introduce another important theorem.

Theorem 2.1.9. *Two square matrices A and B are nonsingular if and only if their product AB is nonsingular.*

Proof. Suppose A and B are nonsingular. Consider the equation $(AB)x = 0$. This is equivalent to $A(Bx) = 0$. By the nonsingularity of A, $Bx = 0$. By the nonsingularity of B, $x = 0$. Hence $(AB)x = 0$ implies $x = 0$. Thus AB must be nonsingular.

Suppose now AB is nonsingular. If B were singular, then there would exist $x_0 \neq 0$ such that $Bx_0 = 0$. Then $(AB)x_0 = A(Bx_0) = 0$ with $x_0 \neq 0$. This contradicts the fact that AB is nonsingular. Thus B must be nonsingular.

If A were singular, then there would exist a row vector $\vec{y}_0$ such that $\vec{y}_0 A = 0$. But then $\vec{y}_0(AB) = (\vec{y}_0 A)B = 0$ with $\vec{y}_0 \neq 0$. This is a contradiction as AB is nonsingular. Therefore A must be nonsingular. Hence if AB is nonsingular, then A and B are also nonsingular. □

We now give a theorem which says the existence of either the left inverse or the right inverse is sufficient to imply invertibility.

Theorem 2.1.10. *Suppose A and B are two $n \times n$ matrices. If $AB = I$, then $BA = I$.*

Proof. As I is nonsinngular, so are A and B. Then their product BA is also nonsingular. Since $AB = I$, multiplying both sides by B on the left and A on the right gives $(BA)^2 = BA$, or equivalently,

$$BA(BA - I) = 0.$$

Since BA is nonsingular, every column of $(BA - I)$ is 0. Thus $BA - I = O$. Hence $BA = I$. □

Remark. Notice that so far we have not even mentioned or used the concept of the inverse of a matrix!

The following theorem reveals the connection between nonsingularity and invertibility.

Theorem 2.1.11. *Suppose A is an $n \times n$ square matrix. A is invertible if and only if A is nonsingular.*

Proof. Suppose A is invertible. Multiplying both sides of $Ax = 0$ by A^{-1} gives $x = 0$. Thus A is nonsingular.

Now suppose A is nonsingular. A is invertible if and only if there exists X such that $AX = I$. (Note that this relies on theorem 2.1.10.)

This is equivalent to solving

$$A \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} = \begin{bmatrix} e_1 & \cdots & e_n \end{bmatrix}. \tag{2.1.6}$$

Since A is nonsingular, we can always find exactly one solution $\tilde{x}_i$ to $Ax_i = e_i$ for each $i = 1, ..., n$. Then we have found A^{-1} by solving (2.1.6):

$$A^{-1} = \widetilde{X} = \begin{bmatrix} \tilde{x}_1 & \cdots & \tilde{x}_n \end{bmatrix}.$$

Thus A is invertible. □

We are now ready to prove the second part of theorem 2.1.5. In fact, we can reformulate theorem 2.1.5 as follows.

Theorem 2.1.12. *Two $n \times n$ matrices A and B are invertible if and only if their product AB is invertible.*

Proof. By theorem 2.1.11, nonsingularity and invertibility are equivalent. By theorem 2.1.9, Two square matrices of the same size are nonsingular if and only if their product is nonsingular. The proof is done. □

■

Note. There is another way to prove that $AB = I$ implies $BA = I$. We perform elementary row operations to reduce A to reduced row echelon form. No matter how we combine the rows of I, no row of I' consists entirely of zeros. Thus the reduced row echelon form of A is I. Thus B is the product of a series of elementary matrices and thus invertible. Therefore so is A.

Notice that this proof relies on part (i) of the theorem $(AB)^{-1} = B^{-1}A^{-1}$. Still, I consider the proof given in the text, which uses only nonsingularity, a more beautiful and elegant proof.

Chengpiao Huang

2.2 Wednesday (Prof. Tom Luo)

2.2.1 Applications of Matrix Multiplication

Picking out a row or a column. Pre-multiplying a matrix by e_i^T picks out the ith row of the matrix. Post-multiplying a matrix by e_j picks out the jth column of the matrix.

Bilinear form. Suppose $A \in \mathbb{R}^{m\times n}$, $x \in \mathbb{R}^m$, and $y \in \mathbb{R}^n$. Then $x^T Ay = \sum_{j=1}^{n}\sum_{i=1}^{m} a_{ij}x_i y_j$.

Calculating the sum. Suppose $A = (a_{ij}) \in \mathbb{R}^{m\times n}$. Let $\mathbf{1}_n$ denotes the n-vector whose entries are all equal to 1. Then $\mathbf{1}^T A\mathbf{1} = \sum_{j=1}^{n}\sum_{i=1}^{m} a_{ij}$ calculates the sum of all entries of A.

Calculating the mean. Let $A \in \mathbb{R}^{m\times n}$. Then $\left(\frac{1}{m}\mathbf{1}\right)^T A\left(\frac{1}{n}\mathbf{1}\right) = \frac{1}{mn}\sum_{j=1}^{n}\sum_{i=1}^{m} a_{ij}$ is the mean of the entries in A.

Inner product. If we take $A = I$, then $x^T Ay = x^T y$ calculates the inner product of x and y.

Rotation matrix. The matrix $R = \begin{bmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{bmatrix}$ rotates a vector $x = (x_1, x_2)^T$ by an angle of θ counterclockwise.

Flipping matrix. The $n \times n$ matrix F given by

$$F_{ij} = \begin{cases} 1 & \text{if } i + j = n+1 \\ 0 & \text{otherwise} \end{cases}$$

has the shape

$$\begin{bmatrix} 0 & & & 1 \\ & & \iddots & \\ & 1 & & \\ \iddots & & & \\ 1 & & & 0 \end{bmatrix}.$$

Pre-multiplying a matrix by F reverses the order of the rows of the matrix. Post-multiplying a matrix by F reverses the order of the columns of the matrix.

2.2.2 Computation of Matrix Inverse

In proving theorem 2.1.11 we proposed a method of computing the inverse of a matrix. We now discuss this method in detail.

To find the inverse of an $n \times n$ matrix A, we can solve the equation $AX = I$, which is equivalent to solving n matrix equations

$$Ax_1 = e_1, \quad Ax_2 = e_2, \quad \ldots\ldots, \quad Ax_n = e_n. \tag{2.2.1}$$

We can solve these n equations simultaneously by performing row reduction on the augmented matrix

$$\left[\begin{array}{c|ccc} A & e_1 & \cdots & e_n \end{array}\right] = \left[\begin{array}{c|c} A & I \end{array}\right]. \tag{2.2.2}$$

If we reduce A to I, then (2.2.2) becomes

$$\left[\begin{array}{c|ccc} I & \tilde{x}_1 & \cdots & \tilde{x}_n \end{array}\right]. \tag{2.2.3}$$

In (2.2.3), we can directly read off the solutions to the n equations, which are exactly $\tilde{x}_1, ..., \tilde{x}_n$. Thus to obtain the inverse of A, we only need to perform row operations on $\left[\begin{array}{c|c} A & I \end{array}\right]$ so that A is converted to I. Then what I becomes is exactly A^{-1}. That is, *the same series of row operations that converts A to I converts I to A^{-1}*.

We can also explain our conclusion in terms of matrix multiplications. Suppose

$$E_k \cdots E_1 A = I,$$

then post-multiplying both sides by A^{-1} yields

$$E_k \cdots E_1 I = A^{-1}.$$

2.2.3 Inverses of Triangular Matrices

We now study the inverse of a triangular matrix. We use a 4×4 lower triangular matrix as an example. Suppose we would like to compute the inverse of

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

To do so, we augment A with I and perform row reduction:

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & -1 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & -1 & 0 & 1 \end{array}\right]$$

$$\to \left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 1 \end{array}\right].$$

Hence

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

There are a few observations that are worth our attention.

1. The pivots of a triangular matrix are exactly its diagonal entries. Thus a triangular matrix is nonsingular if and only if its diagonal entries are all nonzero.

2. When we are performing Gaussian elimination on a lower triangular matrix A (which is in fact forward substitution), the zeros in the tail of each row are still zero because the entries above them are always zero. This also applies to the identity matrix augmented to A. Thus after row reduction on A, what the identity becomes, namely A^{-1}, is still lower triangular.

3. The diagonal entries of both A and A^{-1} are all equal to 1.

These observations lead to the following theorems. These theorems apply to both lower and upper triangular matrices. For simplicity, each theorem is stated for one of the two cases. To prove the other case, simply take the transpose.

Theorem 2.2.1. *A triangular matrix is invertible if and only if its diagonal entries are all nonzero.*

Corollary 2.2.2. *A diagonal matrix is invertible if and only if its diagonal entries are all nonzero.*

Theorem 2.2.3. *The inverse of an invertible lower triangular matrix is also lower triangular.*

Proof. We have already had an informal proof above. We now give a more rigorous proof. Suppose A is $n \times n$ and lower triangular and B is its inverse. Then

$$AB = A\begin{bmatrix} b_1 & \cdots & b_n \end{bmatrix} = \begin{bmatrix} Ab_1 & \cdots & Ab_n \end{bmatrix} = \begin{bmatrix} e_1 & \cdots & e_n \end{bmatrix} = I.$$

Thus $Ab_i = e_i$ for each i. As the diagonal entries of A are all nonzero, the first $(k-1)$ entries of b_k must be zeros. Thus B must be lower triangular. □

Theorem 2.2.4. *The product T of two upper triangular matrices U and R is also upper triangular. Each diagonal entry of T is the product of the corresponding diagonal entries of U and R, that is, $t_{ii} = u_{ii}r_{ii}$ for each i.*

Proof. By definition,

$$t_{ij} = \sum_{k=1}^{n} u_{ik}r_{kj}$$

Since $u_{ik} = 0$ when $i > k$ and since $r_{kj} = 0$ when $k > j$, then when $i > j$,

$$t_{ij} = \sum_{k=1}^{n} u_{ik}r_{kj} = \underbrace{\sum_{k=1}^{j} u_{ik}r_{kj}}_{=0} + \underbrace{\sum_{k=j+1}^{n} u_{ik}r_{kj}}_{=0} = 0.$$

Therefore T is upper triangular.

When $i = j$,

$$t_{ij} = t_{jj} = \sum_{k=1}^{n} u_{jk}r_{kj} = \underbrace{\sum_{k=1}^{j-1} u_{jk}r_{kj}}_{=0} + u_{jj}r_{jj} + \underbrace{\sum_{k=j+1}^{n} u_{jk}r_{kj}}_{=0} = u_{jj}r_{jj}.$$

□

Corollary 2.2.5. *Suppose A and B are $n \times n$ lower triangular matrices with diagonal entries all equal to 1. Then the diagonal entries of AB are all equal to 1.*

Corollary 2.2.6.

$$\operatorname{diag}(d_1, ..., d_n)\operatorname{diag}(d_1', ..., d_n') = \operatorname{diag}(d_1d_1', ..., d_nd_n').$$

Theorem 2.2.7. *Each diagonal entry of the inverse of a triangular matrix A is the reciprocal of the corresponding diagonal entry of A.*

Proof. Let a_i denote the ith diagonal entry of A and $\tilde{a}_i$ the ith diagonal entry of A^{-1}. Then a_i and $\tilde{a}_i$ are nonzero. As $AA^{-1} = I$, by theorem 2.2.4, $a_i\tilde{a}_i = 1$. Hence $\tilde{a}_i = a_i^{-1}$. □

Corollary 2.2.8.

$$(\operatorname{diag}(d_1, ..., d_n))^{-1} = \operatorname{diag}(d_1^{-1}, ..., d_n^{-1}),$$

provided that $\prod_{i=1}^{n} d_i \neq 0$.

So far we have developed many useful theorems related to triangular matrices. Later we will use these theorems.

2.2.4 LU Decomposition (I)

We motivate LU decomposition with an example. Suppose

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix}.$$

We perform Gaussian elimination on A:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & -6 & 0 \\ -2 & 7 & 2 \end{bmatrix} \xrightarrow{E_{21}A} \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ -2 & 7 & 2 \end{bmatrix} \xrightarrow{E_{31}E_{21}A} \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 8 & 3 \end{bmatrix} \xrightarrow{E_{32}E_{31}E_{21}A} \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix},$$

where

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad \text{and} \quad E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Thus a series of elementary row operations turn A into an upper triangular matrix U, i.e.

$$E_{32}E_{31}E_{21}A = U, \tag{2.2.4}$$

where

$$U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix}. \tag{2.2.5}$$

As the E_{ij}'s are invertible, we can rewrite it as

$$A = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}U. \tag{2.2.6}$$

Since every elementary matrix E_{ij} in Gaussian elimination is lower triangular and so is their product, this formula gives us a decomposition of A:

$$A = LU \tag{2.2.7}$$

where L is a lower triangular matrix and U is an upper triangular matrix. Notice that *the diagonal entries of U are exactly the pivots*.

Let us now calculate L:

$$L = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ \mathbf{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \mathbf{-1} & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \mathbf{-1} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ \mathbf{2} & 1 & 0 \\ \mathbf{-1} & \mathbf{-1} & 1 \end{bmatrix}. \tag{2.2.8}$$

The reader may have noticed that the entries below the diagonal are exactly the multipliers l_{ij}'s. Here the matrices come in right order so that their product can be written down directly by filling in the l_{ij}'s in the identity matrix. (Note that *the diagonal entries of L are always* 1.) However, if we do

$$EA = U, \tag{2.2.9}$$

we cannot fill in the entries directly:

$$E = E_{32}E_{31}E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}.$$

Note, however, that if the l_{ij}'s are in the same column in different matrices, we can still fill them in directly, regardless of the order of multiplication.

Therefore, when we are implementing LU factorization, we do need to perform Gaussian elimination to find U. However, to find L, we need only to keep record of the multipliers l_{ij}'s and fill them in the identity matrix to obtain L. There is no need to actually multiply the elementary matrices.

Complexity Analysis

To find U, we need to perform Gaussian elimination, which takes about $\frac{2}{3}n^3$ flops. In the process we have found the multipliers $-l_{ij}$'s and there is no extra computation required to find L. Therefore finding the LU factorization of an $n \times n$ matrix takes about $\frac{2}{3}n^3$ flops.

Applications of LU Decomposition

Once we have found the LU Factorization of an $n \times n$ matrix A, the matrix equation

$$Ax = b \tag{2.2.10}$$

becomes

$$LUx = b. \tag{2.2.11}$$

We can break down the problem of solving $Ax = b$ into two steps:

1. Solve $Ly = b$ using forward substitution.
2. Solve $Ux = y$ using backward substitution.

Each of the two steps takes about n^2 flops.

The advantage of LU factorization comes in when we need to solve *a series of* linear systems associated with the same coefficient matrix (these systems may not occur at the same time):

$$Ax_1 = b_1, \quad Ax_2 = b_2, \quad ..., \quad Ax_k = b_k \tag{2.2.12}$$

If we solve the equations one by one, it would take about $k(\frac{2}{3}n^3 + n^2) \sim \frac{2}{3}kn^3$ flops. However, if we use LU factorization, we first find the factorization, which takes about $\frac{2}{3}n^3$ flops, then we use the two-step process to solve the k equations, which takes about $2kn^2$ flops. Thus using LU factorization takes $\frac{2}{3}n^3 + 2kn^2 = O(\max\{k,n\}n^2)$ flops. This is much faster than applying Gaussian elimination individually, which takes about $\frac{2}{3}kn^3$ flops.

LU Decomposition for Singular Matrices

The row echelon form of a nonsingular matrix has full pivot. Thus in the LU decomposition of that matrix, the diagonal entries of U will all be nonzero. However, a singular matrix cannot have full pivots. In this case, some diagonal entry of U will be zero. As an example, consider the LU factorization of the following matrix.

$$B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 3 \\ 1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{1} & 1 & 1 \\ 0 & \mathbf{0} & 2 \\ 0 & 0 & \mathbf{1} \end{bmatrix} = LU.$$

2.2.5 LDU Decomposition

The LU decomposition of a nonsingular matrix yields U whose diagonal entries are nonzero. We can further factorize U into $U = D\widetilde{U}$, where the the diagonal entries of $\tilde{U}$ are 1 and D is a diagonal matrix whose diagonal entries are the pivots. As an example, we factorize the U matrix in (2.2.5):

$$U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 0 & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 2 & 0 & 0 \\ 0 & -8 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{D} \underbrace{\begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & \frac{1}{4} \\ 0 & 0 & 1 \end{bmatrix}}_{\tilde{U}} = D\widetilde{U}.$$

Then we have obtained the *LDU decomposition* of a nonsingular matrix A:

$$A = LD\widetilde{U} \tag{2.2.13}$$

where L is a lower triangular matrix, D is a diagonal matrix and $\widetilde{U}$ is an upper triangular matrix. The diagonal entries of L and $\widetilde{U}$ are all equal to 1 and the diagonal entries of D are the pivots of the row echelon form of A.

2.2.6 Permutation Matrices

So far we have discussed cases when row exchanges are not required. However, in many cases we need row exchanges to reduce A to its row echelon form. To do so, we need to introduce *permutation matrices*.

Elementary Permutation Matrices

The elementary permutation matrix P_{ij}, which is obtained by switching the ith row and jth row of the identity matrix, has the effect of exchanging the ith and jth rows of a matrix A when we pre-multiply A by P_{ij}. When we pre-multiply P_{ij} again, we switch back the two rows. So $P_{ij}^2 = I$. This gives the following theorem.

Theorem 2.2.9. *Every elementary permutation matrix P_{ij} is invertible, with $P_{ij}^{-1} = P_{ij}$.*

Note that P_{ij} can also be obtained by switching the ith and jth columns of the identity matrix. Thus $P_{ij}^T = P_{ij}$.

General Permutation Matrices

A permutation matrix is a matrix obtained by permuting the rows of the identity matrix. Every permutation matrix can be written as the product of a series of elementary permutation matrices. What is more, every permutation matrix can be expressed as the product of a series of matrices such that each of them exchanges a sequential pair of rows. So if P is a permutation matrix, then

$$P = P_{i_k,j_k} \cdots P_{i_1,j_1},$$

for some elementary permutation matrices $P_{i_k,j_k}, ..., P_{i_1,j_1}$. Then

$$P^{-1} = P_{i_1,j_1}^{-1} \cdots P_{i_k,j_k}^{-1} = P_{i_1,j_1}^T \cdots P_{i_k,j_k}^T = P^T.$$

2.2.7 PLU Decomposition

We now discuss the cases when we need row exchanges to reduce A to its row echelon form. Generally, we have

$$E_n P_n \cdots E_3 P_3 E_2 P_2 E_1 P_1 A = U. \tag{2.2.14}$$

However, it can be shown that we can transform the equation in such a way that it becomes

$$PA = LU. \tag{2.2.15}$$

If so, we can still factorize A into the product of a lower triangular matrix and an upper triangular matrix.

The procedure is as follows. We argue that there exists an elementary matrix $\widetilde{E}_i$ such that

$$P_{i+1} E_i = \widetilde{E}_i P_{i+1} \tag{2.2.16}$$

To see why, we write down E_i and P_{i+1} in block matrix notation:

$$E_i = \left[\begin{array}{c|c|c} I_i & & O \\ \hline O & |l_i & \\ \hline & & I_{n-i} \end{array}\right] \text{ where } l_i \text{ is an } (n-i)\text{-vector} \tag{2.2.17}$$

and

$$P_{i+1} = \left[\begin{array}{c|c} I_i & O \\ \hline O & \widetilde{P}_{i+1} \end{array}\right] \text{ where } \widetilde{P}_{i+1} \text{ is an } (n-i)\times(n-i) \text{ permutation matrix} \tag{2.2.18}$$

because P_{i+1} permutes row $(i+1)$ with some row below. Then

$$P_{i+1}E_i = \left[\begin{array}{c|c} I_i & O \\ \hline O & \tilde{P}_{i+1} \end{array}\right]\left[\begin{array}{c|c|c} I_i & & O \\ \hline O & |l_i & \\ \hline & & I_{n-i} \end{array}\right] = \left[\begin{array}{c|c|c} I_i & & O \\ \hline O & |\tilde{P}_{i+1}l_i & \\ \hline & & \tilde{P}_{i+1} \end{array}\right].$$

If such $\widetilde{E}_i$ exists, then

$$\widetilde{E}_i = P_{i+1}E_iP_{i+1}^{-1} = \left[\begin{array}{c|c|c} I_i & & O \\ \hline O & |\tilde{P}_{i+1}l_i & \\ \hline & & \tilde{P}_{i+1} \end{array}\right]\left[\begin{array}{c|c} I_i & O \\ \hline O & \tilde{P}_{i+1}^{-1} \end{array}\right] = \left[\begin{array}{c|c|c} I_i & & O \\ \hline O & |\tilde{P}_{i+1}l_i & \\ \hline & & I_{n-i} \end{array}\right]. \tag{2.2.19}$$

We have constructed such $\widetilde{E}_i$ that $P_{i+1}E_i = \widetilde{E}_iP_{i+1}$ and it is still an elementary matrix, with the l_{ij}'s swapped. We are now able to move all the permutation matrices to the right of the elementary matrices:

$$E_nP_n\cdots E_3P_3E_2P_2E_1P_1A = U \to E_nP_n\cdots E_3P_3E_2E_1'P_2P_1A = U \to E_nP_n\cdots E_3E_2'P_3E_1'P_2P_1A = U$$

$$\to E_nP_n\cdots E_3E_2'E_1''P_3P_2P_1A = U \to \cdots \to E_nE_{n-1}'\cdots E_1^{(n)}P_n\cdots P_1A = U.$$

Note that we do not have to actually multiply the permutation matrices and elementary matrices. We only have to permute the l_{ij}'s according to the permutation matrices. Moving the elementary matrices to the right, multiplying their inverses (we are actually filling in the entries), and multiplying the permutation matrices (we are actually switching rows), we eventually obtain the *PLU decomposition* of A:

$$PA = LU. \tag{2.2.20}$$

As we only have to switch for $1+2+\cdots+(n-1) \sim \frac{1}{2}n^2$ times, the overall complexity is still about $\frac{2}{3}n^3$ flops.

■

Note. I have added the subsection "LU Decomposition for Singular Matrices".

Chengpiao Huang

2.3 Friday (Prof. Tom Luo)

2.3.1 LU Decomposition (II)

Uniqueness of LDU Decomposition

Theorem 2.3.1. *Suppose a nonsingular matrix A can be decomposed as $A = LDU$, where L and U are lower and upper triangular matrices, respectively, with diagonal entries equal to 1 and D is a diagonal matrix. Then L, D, and U are unique.*

Proof. Suppose $A = L_1D_1U_1$ and $A = L_2D_2U_2$. Then

$$L_1D_1U_1 = L_2D_2U_2. \tag{2.3.1}$$

Multiplying both sides of the equation by L_1^{-1} on the left and U_2^{-1} on the right gives

$$D_1U_1U_2^{-1} = L_1^{-1}L_2D_2. \tag{2.3.2}$$

As $D_1U_1U_2^{-1}$ is an upper triangular matrix and $L_1^{-1}L_2D_2$ is a lower triangular matrix, then they are both diagonal matrices. As the diagonal entries of L_1, L_2^{-1}, U_1, and U_2^{-1} are all equal to 1, the diagonal entries of $D_1U_1U_2^{-1}$ are exactly those of D_1. Thus $D_1U_1U_2^{-1} = D_1$. The invertibility of D_1 implies $U_1U_2^{-1} = I$. Thus $U_1 = U_2$. Similarly, we obtain $L_1^{-1}L_2D_2 = D_2$ and thus $L_1 = L_2$. Then (2.3.2) becomes $D_1 = D_2$. Hence $L_1 = L_2$, $D_1 = D_2$ and $U_1 = U_2$. □

Limitations of Gaussian Elimination

LU decomposition provides us with a way to decompose a square matrix into the product of a lower triangular matrix and an upper triangular matrix. However, there are certain limitations of Gaussian elimination and LU decomposition.

Gaussian elimination ruins sparsity. In the field of big data, it is common to have matrices that store data which have the size of, say, one million by one million. Then there are one trillion (1T) data in a single matrix. The storage of a typical computer is 250GB. So a computer cannot even store a matrix of this kind! Fortunately, the matrices are typically sparse and only about one percent of the entries are nonzero. Though there are still a lot of nonzero entries (10^{10}), we can at least store the matrix. However, if we perform Gaussian elimination on sparse matrices, the zeros will be ruined and the matrices will become dense. But then we would have to store a lot more data. As a result, Gaussian elimination is not often used for large-scale problems.

Gaussian elimination is numerically unstable. Computers can only store a finite number of digits. Thus computer arithmetic is not exact. Suppose we perform Gaussian elimination and the pivot a_{11} is very close to zero. Then the multiplier $-a_{21}/a_{11}$ can get very large. When small and large numbers appear together, the small numbers will "disappear" as computers can only store a finite number of digits. Then arithmetic precision will be affected. So we need to avoid small pivots. A way to do this is to pick the largest number in a column or sub-column as the pivot. This method is called *pivoting*.

2.3.2 Power Method

As we have mentioned, Gaussian elimination ruins sparsity. Here we introduce a method called *power method* that maintains sparsity.

For every real $|x| < 1$, the power series of $(1-x)^{-1}$ is given by

$$(1-x)^{-1} = 1 + x + x^2 + \cdots + x^n + \cdots.$$

This formula also has a matrix version:

$$(I-A)^{-1} = I + A + A^2 + \cdots + A^n + \cdots . \tag{2.3.3}$$

We can use $S_n = I + A + A^2 + \cdots + A^n$ to approximate $(I-A)^{-1}$. To find S_{n+1}, we may use the formula $S_{n+1} = S_n + A^{n+1}$. However, as this method involves the computation of large powers of A, the complexity is high. We can use an alternative approach:

$$S_{n+1} = I + AS_n. \tag{2.3.4}$$

In (2.3.4), only *one* multiplication is involved. Further, if A is sparse, which is usually the case, AS_n is also sparse. So sparsity is preserved!

However, the validity of (2.3.3) has not been examined. We now discuss this problem.

1. If $A = \text{diag}(a_1, ..., a_n)$ is diagonal, it can be verified that (2.3.3) is true when $|a_i| < 1$ for each $i = 1, ..., n$.

2. If A is not diagonal but square and if we can decompose A as $A = P^{-1}DP$, where D is diagonal, then $A^n = P^{-1}D^nP$ for all nonnegative integers n. Also, $(I-A)^{-1} = P^{-1}(I-D)^{-1}P$. Thus (2.3.3) holds if and only if

$$(I-D)^{-1} = I + D + D^2 + \cdots + D^n + \cdots .$$

Therefore to make sure that (2.3.3) is valid, we may need to first scale the entries of A so that all diagonal entries of D are absolutely less than 1.

2.3.3 Matrix Transpose

Definition 2.3.2. Let A be an $m \times n$ matrix. The *transpose* of A, denoted by A^T, is an $n \times m$ matrix defined by

$$(A^T)_{ij} = A_{ji}. \tag{2.3.5}$$

Here we list some properties related to the transpose of a matrix.

Proposition 2.3.3. *Let A and B be matrices of the same size. Let C and D be matrices such that their product CD is defined. Let x be an n-vector. Then*

(i) $(A+B)^T = A^T + B^T$.

(ii) $(CD)^T = D^T C^T$.

(iii) (Inner product) $x^T x = \sum_{i=1}^{n} {x_i}^2$.

(iv) (Outer product)

$$xx^T = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \begin{bmatrix} x_1 & \cdots & x_n \end{bmatrix} = \begin{bmatrix} {x_1}^2 & \cdots & x_1 x_n \\ \vdots & \ddots & \vdots \\ x_n x_1 & \cdots & {x_n}^2 \end{bmatrix}$$

and

$$(xx^T)_{ij} = x_i x_j.$$

Definition 2.3.4. A matrix $A \in \mathbb{R}^{n \times n}$ is said to be *symmetric* if $A^T = A$.

Definition 2.3.5. A matrix $A \in \mathbb{R}^{n \times n}$ is said to be *skew symmetric* or *anti-symmetric* if $A^T = -A$.

Proposition 2.3.6. *The diagonal entries of an anti-symmetric matrix are all equal to zero.*

Proof. By definition, for every i, $a_{ii} = -a_{ii}$. Thus $a_{ii} = 0$ for every i. □

Theorem 2.3.7. *Every real square matrix can be written as the sum of a symmetric matrix and an anti-symmetric matrix.*

Proof. Let $A \in \mathbb{R}^{n\times n}$. Suppose

$$A = U + V \tag{2.3.6}$$

where U is symmetric and V is anti-symmetric. Then

$$A^T = U^T + V^T = U - V. \tag{2.3.7}$$

By (2.3.6) and (2.3.7),

$$U = \frac{A + A^T}{2} \quad \text{and} \quad V = \frac{A - A^T}{2}.$$

Since

$$\frac{A + A^T}{2} + \frac{A - A^T}{2} = A,$$

then A can indeed be written as the sum of a symmetric matrix and an anti-symmetric matrix. □

■

2.4 Saturday (Prof. Tom Luo)

2.4.1 General LU Decomposition

Recall that a square nonsingular matrix A can always be decomposed as $PA = LDU$. Let us move to a more general case.

We begin with a simple example, namely, Gaussian elimination on the matrix

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix}. \tag{2.4.1}$$

The process is as follows.

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix} \xrightarrow{E_{21}A} \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 3 \\ -1 & -3 & 3 & 4 \end{bmatrix} \xrightarrow{E_{31}E_{21}A} \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 6 & 6 \end{bmatrix} \xrightarrow{E_{32}E_{31}E_{21}A} \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{E_{22}E_{32}E_{31}E_{21}A} \begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{E_{12}E_{22}E_{32}E_{31}E_{21}A} \begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R$$

where

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \; E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \text{ and } E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \text{ are lower triangular matrices,}$$

$$E_{22} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ is a diagonal matrix, } E_{12} = \begin{bmatrix} 1 & -3 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \text{ is an upper triangular matrix,}$$

and R is the reduced row echelon form of A. We can therefore write A as

$$A = \underbrace{E_{32}^{-1}E_{31}^{-1}E_{21}^{-1}}_{L}\underbrace{E_{22}^{-1}}_{D}\underbrace{E_{12}^{-1}}_{U}R = LDUR. \tag{2.4.2}$$

Taking into account the permutation matrix, we can go further to a more general decomposition which applies to any matrix A:

$$PA = LDUR. \tag{2.4.3}$$

Note that if A is square and nonsingular, then $R = I$ and (2.4.2) becomes $PA = LDU$, which is consistent with what we have obtained earlier.

2.4.2 Matrix Transpose

In the previous lecture we have introduced the concept of the transpose of a matrix. We now explore more properties of matrix transpose.

Theorem 2.4.1. *Let A and B be such that their product AB is defined. Then*

$$(AB)^T = B^T A^T. \tag{2.4.4}$$

Theorem 2.4.2. *If A is invertible, then A^T is also invertible, and*

$$(A^T)^{-1} = (A^{-1})^T. \tag{2.4.5}$$

Proof. By theorem 2.1.10, we only need to check left inverse: $(A^{-1})^T A^T = (AA^{-1})^T = I^T = I$. □

Theorem 2.4.3. *Suppose A, B, C, D are matrices with appropriate sizes. Then*

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^T = \begin{bmatrix} A^T & C^T \\ B^T & D^T \end{bmatrix}.$$

Corollary 2.4.4. *Suppose an $n \times n$ matrix M can be written as $M = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ where A, B, C, and D are $\frac{n}{2} \times \frac{n}{2}$ submatrices. $M = M^T$ if and only if $A = A^T$, $D = D^T$, and $B^T = C$.*

Theorem 2.4.5. *If a nonsingular matrix A is symmetric, then its inverse A^{-1} is also symmetric.*

Proof. By theorem 2.4.2, $(A^{-1})^T = (A^T)^{-1} = A^{-1}$. □

Theorem 2.4.6. *If a symmetric nonsingular matrix A has LDU decomposition $A = LDU$, then $U = L^T$. Thus A can be decomposed as*

$$A = LDL^T. \tag{2.4.6}$$

Proof. $A = A^T$ gives $LDU = U^T D^T L^T = U^T D L^T$. As U^T is lower triangular and L^T is upper triangular, then by the uniqueness of LDU decomposition, $L^T = U$. □

2.4.3 Linear Systems and Beyond

Homogeneous Equation and Null Space

Recall that in section 2.4.1 we performed Gaussian elimination on

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix}$$

and obtained its reduced row echelon form

$$R = \begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

which has only two pivots. We are now interested in the structure of the solutions to the *homogeneous* equation $Ax = 0$. This gives rise to the concept of null space.

Definition 2.4.7. Let A be an $m \times n$ matrix. The *null space* of A, denoted by $N(A)$, is the set of all solutions to the equation $Ax = 0$. That is,

$$N(A) = \{x \in \mathbb{R}^n : Ax = 0\}. \tag{2.4.7}$$

It is obvious that the zero vector is always in the null space. Also, it can be verified that $N(A)$ is closed under linear combination.

As $PA = LDUR$ and P, L, D, and U are invertible, then the system $Ax = 0$ is equivalent to the system $Rx = 0$. Thus to study $Ax = 0$, we only have to study $Rx = 0$, which is easier. The equation $Rx = 0$ is equivalent to

$$\begin{cases} x_1 + 3x_2 \quad - x_4 = 0 \\ \qquad\quad x_3 + x_4 = 0 \end{cases}. \tag{2.4.8}$$

The variables that correspond to the pivot columns are called *pivot variables* and the other variables are called *free variables*. In (2.4.8), the pivot variables are x_1 and x_3, and the free variables are x_2 and

x_4. Notice that a pivot column always corresponds to a column of the identity matrix. To characterize the general solution, we represent the pivot variables in terms of the free variables:

$$\begin{cases} x_1 = -3x_2 + x_4 \\ x_3 = -x_4 \end{cases}. \tag{2.4.9}$$

Therefore the general solution is

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -3x_2 + x_4 \\ x_2 \\ -x_4 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}. \tag{2.4.10}$$

So every vector in $N(A)$ is a linear combination of $(-3,1,0,0)^T$ and $(1,0,-1,1)^T$. Intuitively, these two vectors *span* the whole null space and we write

$$N(A) = \text{Span}\left\{ \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

In some sense these two vectors are *independent* because they are not in the same direction. Intuition tells us that these two vectors form a *two-dimensional subspace* of $\mathbb{R}^4$.

We observe that

$$\begin{aligned} \dim N(A) &= \text{number of free variables} \\ &= n - \underbrace{\text{number of pivot variables}}_{=r} \\ &= n - r \\ &= n - \text{number of non-redundant equations}. \end{aligned}$$

Also, $m - r =$ number of redundant equations.

Inhomogeneous Equation and Column Space

Now let us turn to the inhomogeneous equation $Ax = b$. For ease of discussion, we introduce the concept of the *column space* of A.

Definition 2.4.8. Let $A = \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix}$ be an $m \times n$ matrix. The *column space* of A, denoted by $\text{Col}(A)$, is defined as the set of all linear combinations of the columns of A. That is to say,

$$\text{Col}(A) = \text{Span}\{a_1, ..., a_n\} = \{Ax \,|\, x \in \mathbb{R}^n\}. \tag{2.4.11}$$

Then the equation $Ax = b$ has a solution if and only if b lies in the column space of A. If $b \in \text{Col}(A)$ and $b' \in \text{Col}(A)$, then $\alpha b + \beta b' \in \text{Col}(A)$ for every scalar α and β. Thus the column space of a matrix is closed under linear combination. However, $Ax = b$ and $Ax' = b$ do not imply that $\alpha x + \beta x'$ is a solution to $Ax = b$. So the solution set of $Ax = b$, which is $\{x \in \mathbb{R}^n \,|\, Ax = b\}$, is not closed under linear combination.

Solution Set of Inhomogeneous Equation

With the introduction of null space and column space, we are prepared to characterize the solution set of the inhomogeneous equation $Ax = b$. Consider the augmented matrix $\left[\begin{array}{c|c} A & b \end{array}\right]$, where A is the matrix in (2.4.1) and $b \in \mathbb{R}^3$. After Gauss-Jordan reduction, the augmented matrix becomes

$$\left[\begin{array}{cccc|c} 1 & 3 & 0 & -1 & c_1 \\ 0 & 0 & 1 & 1 & c_2 \\ 0 & 0 & 0 & 0 & c_3 \end{array}\right]. \tag{2.4.12}$$

If $c_3 \neq 0$, then $Ax = b$ does not have a solution. If $c_3 = 0$, the system has infinitely many solution. We can, again, express the pivot variables in terms of free variables:

$$\begin{cases} x_1 = -3x_2 + x_4 + c_1 \\ x_3 = -x_4 + c_2 \end{cases}. \tag{2.4.13}$$

Therefore the general solution is

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -3x_2 + x_4 + c_1 \\ x_2 \\ -x_4 + c_2 \\ x_4 \end{bmatrix} = \underbrace{x_2 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}}_{N(A)} + \underbrace{\begin{bmatrix} c_1 \\ 0 \\ c_2 \\ 0 \end{bmatrix}}_{\text{special solution}}. \tag{2.4.14}$$

Comparing (2.4.10) and (2.4.14), we see that the general solution to $Ax = b$ can be expressed as an element in $N(A)$ plus a special solution. This leads to the following theorem.

Theorem 2.4.9. *Suppose $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$. Let $x_p \in \mathbb{R}^n$ be a particular solution to $Ax = b$. A vector $z \in \mathbb{R}^n$ is a solution to $Ax = b$ if and only if $z = x_p + x_n$ for some $x_n \in N(A)$.*

Proof. Suppose $z = x_p + x_n$, where x_p solves $Ax = b$ and $x_n \in N(A)$. Then $Az = A(x_p + x_n) = Ax_p + Ax_n = b + 0 = b$. So z is also a solution to $Ax = b$.

Suppose z is a solution to $Ax = b$. Then $A(z - x_p) = 0$. So $(z - x_p) \in N(A)$. Therefore $z = x_n + x_p$, where $x_n = z - x_p \in N(A)$. □

2.4.4 Vector Space

Our discussion of null space and column space motivates us to define something more general and fundamental, in which the operation of linear combination is well-defined. We give an informal definition of a *vector space*, usually denoted by V: a set over which linear combination is well-defined. That is, if $x, y \in V$, then $\alpha x + \beta y \in V$ for all scalars α and β. The elements in a vector space are called *vectors*.

Example 2.4.10. For each $n = 1, 2, 3, ..., \mathbb{R}^n$ is a vector space.

Example 2.4.11. The set

$$V = \left\{ \begin{bmatrix} a_1 \\ \vdots \\ a_n \\ \vdots \end{bmatrix} \;\middle|\; a_i \in \mathbb{R},\ i = 1, ..., n, ... \text{ and } \sum_{i=1}^{\infty} a_i^2 < \infty \right\}$$

of vectors with countably many entries and finite *energy* also defines a vector space.

To see why, we need to show that if $a, b \in V$, then $c = \alpha a + \beta b \in V$ for every $\alpha, \beta \in \mathbb{R}$.

$$\sum_{i=1}^{\infty} c_i^2 = \sum_{i=1}^{\infty} (\alpha a_i + \beta b_i)^2 = \alpha^2 \sum_{i=1}^{\infty} a_i^2 + \beta^2 \sum_{i=1}^{\infty} b_i^2 + 2\alpha\beta \sum_{i=1}^{\infty} a_i b_i.$$

Since $a, b \in V$, then $\sum_{i=1}^{\infty} a_i^2$ and $\sum_{i=1}^{\infty} b_i^2$ are both finite. So the first two terms are finite. Cauchy-Schwarz inequality gives

$$\left(\sum_{i=1}^{\infty} a_i b_i \right)^2 \leq \left(\sum_{i=1}^{\infty} a_i^2 \right) \left(\sum_{i=1}^{\infty} b_i^2 \right) < \infty.$$

Thus the third term is also finite. So $\sum_{i=1}^{\infty} c_i^2$ is finite. Therefore $c = \alpha a + \beta b \in V$. Hence V does define a vector space.

We can, by analogy, define inner product on V: the *inner product* of two vectors x and y in V is defined as

$$\langle x, y \rangle = \sum_{i=1}^{\infty} x_i y_i.$$

Cauchy-Schwarz inequality ensures that the inner product of any two vectors in V will not diverge. So this is a reasonable definition.

Example 2.4.12. The set

$$\left\{ f \,\middle|\, f : [0,1] \to \mathbb{R},\ \int_0^1 [f(x)]^2 \, dx < \infty \right\}$$

of functions with finite *energy* on $[0,1]$ is a vector space.

If $f, g \in V$, then for every $\alpha, \beta \in \mathbb{R}$,

$$\int_0^1 [\alpha f(x) + \beta g(x)]^2 \, dx = \alpha^2 \int_0^1 [f(x)]^2 \, dx + \beta^2 \int_0^1 [g(x)]^2 \, dx + \alpha\beta \int_0^1 f(x) g(x) \, dx.$$

By Cauchy-Schwarz inequality,

$$\left(\int_0^1 f(x) g(x) \, dx \right)^2 \leq \left(\int_0^1 [f(x)]^2 \, dx \right) \left(\int_0^1 [g(x)]^2 \, dx \right) < \infty.$$

Hence $\int_0^1 [\alpha f(x) + \beta g(x)]^2 \, dx < \infty$. So $\alpha f + \beta g \in V$. Thus V is indeed a vector space.

By analogy, we can define the inner product of two functions f and g in V:

$$\langle f, g \rangle = \int_0^1 f(x) g(x) \, dx.$$

Again, the inner product will never diverge by Cauchy-Schwarz inequality.

Remark. Here we see an example of how we can extend a concept by analogy. We can view a function as the limiting case of an Euclidean vector with a continuous index, namely x, which ranges over $[0,1]$. In the discrete case the inner product is defined using summation. In the continuous case, summation, very naturally, becomes integration.

■

Chapter 3

Week Three

3.1 Monday (Prof. Dongdong He)

3.1.1 Vector Space

Last lecture we gave an informal definition of vector space: a set over which linear combinations are well-defined. Here we define vector space more rigorously.

Definition 3.1.1. Let V be a set in which addition and scalar multiplication are defined. V is called a *vector space* if the following properties hold:

1. $u+v \in V$ for every $u,v \in V$.
2. $u+v = v+u$ for every $u,v \in V$.
3. $u+(v+w) = (u+v)+w$ for every $u,v,w \in V$.
4. There exists an element $0 \in V$ such that for every $v \in V$, $v+0 = v$.
5. For every $v \in V$, there exists a vector $-v \in V$ such that $v+(-v) = 0$.
6. $\alpha v \in V$ for every scalar α and every $v \in V$.
7. $\alpha(\beta v) = (\alpha\beta)v$ for all scalars α,β and every $v \in V$.
8. $(\alpha+\beta)v = \alpha v + \beta v$ for all scalars α,β and every $v \in V$.
9. $\lambda(u+v) = \lambda u + \lambda v$ for every scalar λ and all $u,v \in V$.
10. There exists a scalar 1 such that for every $v \in V$, $1v = v$.

The elements of V are called *vectors*.

Remark. Note that the elements of a vector space can be anything. They can of course be Euclidean vectors. But they can also be other kinds of objects such as matrices, functions.

Example 3.1.2. We now give a few examples of vector spaces.

(i) The n-dimensional Euclidean space

$$\mathbb{R}^n = \left\{ \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \;\middle|\; x_i \in \mathbb{R},\ i = 1,...,n \right\}.$$

(ii) The set of vectors with countably many entries and finite *energy*:

$$V = \left\{ \begin{bmatrix} a_1 \\ \vdots \\ a_n \\ \vdots \end{bmatrix} \;\middle|\; a_i \in \mathbb{R},\ i = 1, ..., n, ... \text{ and } \sum_{i=1}^{\infty} a_i^2 < \infty \right\}.$$

(iii) The set of all square integrable functions on $[0,1]$:

$$V = \left\{ f \;\middle|\; f : [0,1] \to \mathbb{R},\ \int_0^1 [f(x)]^2\, dx < \infty \right\}.$$

(iv) The set of all 2×3 real matrices:

$$\mathbb{R}^{2\times 3} = \left\{ \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \;\middle|\; a_{ij} \in \mathbb{R},\ i = 1,2,\ j = 1,2,3 \right\}.$$

(v) The set of all polynomials of degree no more than n:

$$P_n = \{p \mid\ p : \mathbb{R} \to \mathbb{R},\ p(t) = a_0 + a_1 t + a_2 t^2 + \cdots + c_n t^n\}.$$

Remark. If a set does not contain the zero vector, it is not a vector space.

3.1.2 Subspace

Definition 3.1.3. A subset S of a vector space V is called a *subspace* of V if it is also a vector space.

Proposition 3.1.4. *If S is a subspace of a vector space V, then*

(i) $0 \in S$ *and S is therefore non-empty.*

(ii) If $u, v \in S$, *then* $u + v \in S$.

(iii) If $u \in S$, *then* $\alpha u \in S$ *for every scalar* α.

Proof. This is immediate because a subspace is a vector space. □

3.1.3 Span

Definition 3.1.5. Let V be a vector space. The *span* of a set of vectors $\{v_1, ..., v_k\}$ in V, denoted by $\text{Span}\{v_1, ..., v_k\}$, is defined as the set of all linear combinations of $v_1, ..., v_k$.

Theorem 3.1.6. *The span of a set of vectors $\{v_1, ..., v_k\}$ in a vector space V is a vector space and a subspace of V.*

Proof. Suppose $u, w \in \text{Span}\{v_1, ..., v_k\}$. Then $u = \alpha_1 v_1 + \cdots + \alpha_k v_k$ and $w = \beta_1 v_1 + \cdots + \beta_k v_k$ for some scalars $\alpha_1, ..., \alpha_k, \beta_1, ..., \beta_k$. For all scalars λ and μ,

$$\lambda u + \mu w = \lambda(\alpha_1 v_1 + \cdots + \alpha_k v_k) + \mu(\beta_1 v_1 + \cdots + \beta_k v_k) = (\lambda\alpha_1 + \mu\beta_1)v_1 + \cdots + (\lambda\alpha_k + \mu\beta_k)v_k$$

is still a linear combination of $v_1, ..., v_k$. Thus $\lambda u + \mu w \in \text{Span}\{v_1, ..., v_k\}$. Therefore $\text{Span}\{v_1, ..., v_k\}$ is a vector space. Since $v_1, ..., v_k \in V$, by the definition of a vector space, $\text{Span}\{v_1, ..., v_k\} \subset V$. So $\text{Span}\{v_1, ..., v_k\}$ is a subspace of V. □

Definition 3.1.7. A set of vectors $\{v_1,...,v_k\}$ in a vector space V is said to *span* V if every vector in V is a linear combination of $v_1,...,v_k$, or equivalently, $V = \text{Span}\{v_1,...,v_k\}$. We also say $\{v_1,...,v_k\}$ is a *spanning set* for V.

Example 3.1.8. Consider an $m \times n$ matrix $A = \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix}$. The column space of A is defined as $\text{Col}(A) = \text{Span}\{a_1,...,a_n\}$. Thus $\{a_1,...,a_n\}$ is a spanning set for $\text{Col}(A)$.

Example 3.1.9. Consider $\mathbb{R}^3$. Let $e_1 = (1,0,0)^T$, $e_2 = (0,1,0)^T$, and $e_3 = (0,0,1)^T$. Then

(i) $\text{Span}\{e_1,e_2\} \neq \mathbb{R}^3$.

(ii) $\text{Span}\{e_1,e_2,e_3\} = \mathbb{R}^3$.

(iii) If we let $v = (1,1,1)^T$, then $\text{Span}\{e_1,e_2,e_3,v\} = \mathbb{R}^3$.

Form (ii) and (iii), we see that *the spanning set for a vector space is not unique*.

3.1.4 Linear Independence

Definition 3.1.10. Let V be a vector space. A set $\{v_1,...,v_k\}$ of vectors in V is said to be *linearly independent* if

$$c_1v_1 + \cdots + c_kv_k = 0 \quad \text{implies} \quad c_1 = \cdots = c_k = 0.$$

It is called *linearly dependent* otherwise, that is, if

$$\text{there exist } c_1,...,c_k \text{ not all zero} \quad \text{such that} \quad c_1v_1 + \cdots + c_kv_k = 0.$$

Example 3.1.11. The set

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

is linearly dependent because the equation

$$c_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} + c_4 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

is a homogeneous equation which involves 3 equations and 4 unknowns. Thus this equation must have a nonzero solution.

Proposition 3.1.12. *Suppose $a_1,...,a_k \in \mathbb{R}^n$. Let $A = \begin{bmatrix} a_1 & \cdots & a_k \end{bmatrix}$. The following statements are equivalent.*

(i) $\{a_1,...,a_n\}$ is linearly independent.

(ii) $Ax = 0$ implies $x = 0$.

Corollary 3.1.13. *A square matrix is nonsingular if and only if its columns are linearly independent.*

3.1.5 Basis and Dimension

Definition 3.1.14. Let V be a vector space. A set $\{v_1,...,v_n\}$ of vectors in V is said to be a *basis* of V if it satisfies the following conditions:

1. $\{v_1,...,v_n\}$ spans V.
2. $\{v_1,...,v_n\}$ is linearly independent.

Remark. The first condition in definition 3.1.14 requires that a basis cannot have too few vectors and the second condition requires that a basis cannot have too many vectors.

Theorem 3.1.15. *Let V be a vector space. Suppose $\{v_1,...,v_n\}$ forms a basis of V. If a set $\{u_1,...,u_m\}$ of vectors in V is linearly independent, then $m \leq n$.*

Proof. As $\{v_1,...,v_n\}$ forms a basis of V, then for every $j = 1,...,m$,

$$u_j = a_{1j}v_1 + \cdots + a_{nj}v_n = \sum_{i=1}^{n} a_{ij}v_i.$$

Consider the equation $c_1u_1 + \cdots + c_mu_m = 0$, which has only the trivial solution by the linear independence of $u_1,...,u_m$. The equation is equivalent to

$$c_1\sum_{i=1}^{n} a_{i1}v_i + \cdots + c_m\sum_{i=1}^{n} a_{im}v_i = (\sum_{j=1}^{m} c_ja_{1j})v_1 + \cdots + (\sum_{j=1}^{m} a_{nj}v_n).$$

As $v_1,...,v_n$ are linearly independent, we obtain a system of homogeneous linear equations:

$$\begin{bmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nm} \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix} = 0.$$

If $m > n$, then the system would have a nontrivial solution. This is a contradiction. Thus $m \leq n$. □

Theorem 3.1.16. *Let V be a vector space. All bases of V consist of the same number of vectors.*

Proof. Suppose $\{v_1,...,v_n\}$ and $\{u_1,...,u_m\}$ are two bases of V. By theorem 3.1.15, $n \leq m$ and $m \leq n$. Thus $m = n$. □

Definition 3.1.17. Let V be a vector space. The *dimension* of V, denoted by $\dim V$ or $\dim(V)$, is defined as the number of vectors in a basis of V.

Remark. Theorem 3.1.16 ensures the uniqueness of the dimension of a vector space.

Example 3.1.18.

(i) $\dim(\{0\}) = 0$.

(ii) As $\{e_1,...,e_n\}$ is a basis for $\mathbb{R}^n$, then $\dim\mathbb{R}^n = n$.

(iii) The dimension of the vector space

$$\mathbb{R}^{2\times 3} = \left\{ \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} \middle| \; a_{ij} \in \mathbb{R},\ i = 1,2,\ j = 1,2,3 \right\}$$

is equal to 6. We obtain a basis by letting each a_{ij} be one and others be zero in turn.

(iv) The dimension of P_n, which contains all polynomials of degree less than or equal to n, is equal to $n+1$. This is because $\{1,t,...,t^n\}$ is a basis of P_n.

■

Note. Some definitions are modified, such as the definition of the subspace of a vector space.

Chengpiao Huang

3.2 Wednesday (Prof. Tom Luo)

3.2.1 Four Fundamental Subspaces and Gaussian Elimination

Definition 3.2.1. Suppose $A \in \mathbb{R}^{m\times n}$.

(i) The *column space* of A, denoted by $\mathrm{Col}(A)$, the the set of all the linear combinations of the columns of A, i.e.

$$\mathrm{Col}(A) = \{Ax \mid x \in \mathbb{R}^n\}. \tag{3.2.1}$$

(ii) The *row space* of A, denoted by $\mathrm{Row}(A)$, is the set of all the linear combinations of the transposes of the rows of A. Thus

$$\mathrm{Row}(A) = \mathrm{Col}(A^T). \tag{3.2.2}$$

(iii) The *null space* of A, denoted by $N(A)$ or $\mathrm{Nul}(A)$, is the solution set of $Ax = 0$, i.e.

$$N(A) = \{x \in \mathbb{R}^n \mid Ax = 0\}. \tag{3.2.3}$$

(iv) The *left null space* of A is the set that contains all the $(\vec{y})^T$'s such that $\vec{y}A = 0$, or equivalently, the solution set of $A^T x = 0$. Thus the left null space is often denoted by $N(A^T)$.

We are interested in the effects of Gaussian elimination (or Gauss-Jordan reduction) on the four spaces. Suppose A is an $m \times n$ matrix. Performing Gaussian elimination on A is equivalent to pre-multiplying A by an $m \times m$ invertible matrix G. The reduced row echelon form of A is $B = GA$. As G is invertible, $A = G^{-1}B$.

Remark. In the following sections, we will overload the term Gaussian elimination. When we use this term, we may be actually referring to Gauss-Jordan reduction.

Gaussian Elimination Preserves Row Space

As $B = GA$, the rows of B are linear combinations of the rows of A. Thus the span of the rows of B lies in the span of the rows of A. As G is invertible, $A = G^{-1}B$. So the span of the rows of A lies in the rows of B. Hence $\mathrm{Row}(A) = \mathrm{Row}(B)$. This leads to the following theorem.

Theorem 3.2.2. *Gaussian elimination does not change the row space of a matrix. In other words, if A is an $m \times n$ matrix and $B = GA$ where G is an invertible $m \times m$ matrix, then* $\mathrm{Row}(A) = \mathrm{Row}(B)$.

Gaussian Elimination Preserves Null Space

As G is invertible, the system $Ax = 0$ is equivalent to the system $GAx = 0$. Thus applying Gaussian elimination does not change the null space. This is the very reason why we use Gaussian elimination.

Theorem 3.2.3. *Gaussian elimination does not change the null space of a matrix. In other words, if A is an $m \times n$ matrix and $B = GA$ where G is an invertible $m \times m$ matrix, then* $\mathrm{Nul}(A) = \mathrm{Nul}(B)$.

Gaussian Elimination Changes Column Space

Unlike what happens to the row space and the null space (actually nothing happens), Gaussian elimination does change the column space, as we illustrate in the following example.

Example 3.2.4. Recall the example

$$A = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix} \xrightarrow{\text{Gaussian elimination}} B = \begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

We see that if a vector is in $\text{Col}(B)$, its third entry must be 0. However, this is not the case for vectors in $\text{Col}(A)$, for the columns of A themselves do not even satisfy this condition! Thus we conclude that Gaussian elimination may change the column space.

3.2.2 Operations on Subspaces

In this section we discuss how set operations interact with vector spaces.

Sum

Definition 3.2.5. Suppose U_1 and U_2 are two subspaces of a vector space V. The *sum* of U_1 and U_2, denoted by $U_1 + U_2$, is defined as

$$U_1 + U_2 = \{u_1 + u_2 : u_1 \in U_1,\ u_2 \in U_2\}. \tag{3.2.4}$$

Remark. Notice that the constraint that U_1 and U_2 are subspaces of the *same* vector space ensures that vector addition and scalar multiplication are the same for U_1 and U_2.

Example 3.2.6. Suppose $V_1 = \left\{ t \begin{bmatrix} 1 \\ 0 \end{bmatrix} : t \in \mathbb{R} \right\}$ and $V_2 = \left\{ t \begin{bmatrix} 0 \\ 1 \end{bmatrix} : t \in \mathbb{R} \right\}$. Then $V_1 + V_2 = \mathbb{R}^2$.

Theorem 3.2.7. *Suppose U_1 and U_2 are two subspaces of a vector space V. Then $U_1 + U_2$ is a vector space. Further, it is a subspace of V.*

Proof. Suppose $u, v \in U_1 + U_2$. Then $u = u_1 + u_2$ and $v = v_1 + v_2$ for some $u_1, v_1 \in U_1$ and $u_2, v_2 \in U_2$. Then for all scalars α and β,

$$\alpha u + \beta v = \alpha(u_1 + u_2) + \beta(v_1 + v_2) = \underbrace{(\alpha u_1 + \beta v_1)}_{\in U_1} + \underbrace{(\alpha u_2 + \beta v_2)}_{\in U_2} \in U_1 + U_2.$$

Thus $U_1 + U_2$ is a subspace. As each element of U_1 or U_2 is an element of V, $U_1 + U_2 \subset V$ and is thus a subspace of V. □

Example 3.2.8. Suppose $A, B \in \mathbb{R}^{m \times n}$. Generally,

$$\text{Col}(A) + \text{Col}(B) \neq \text{Col}(A + B).$$

A counterexample is $\text{Col}(I_2) + \text{Col}(-I_2) = \text{Col}(I_2) = \mathbb{R}^2$ while $\text{Col}(I_2 + (-I_2)) = \text{Col}(O) = \{0\}$, where I_2 is the 2×2 identity matrix.

Intersection

Now we consider the intersection of two subspaces.

Theorem 3.2.9. *The intersection $U_1 \cap U_2$ of two subspaces U_1 and U_2 of a vector space V is a vector space and a subspace of V.*

Proof. Suppose $u, v \in U_1 \cap U_2$. Then u and v are in both U_1 and U_2. As U_1 is a vector space, then for all scalars λ and μ, $\lambda u + \mu v \in U_1$. As U_2 is also a vector space, $\lambda u + \mu v \in U_2$. Hence $\lambda u + \mu v \in U_1 \cap U_2$. Thus $U_1 \cap U_2$ is a vector space. As $U_1 \cap U_2 \subset V$, $U_1 \cap U_2$ is a subspace of V. □

Union

The union of two subspaces is not necessarily a vector space. As an example, consider the subspaces U_1 and U_2 in example 3.2.6. $U_1 \cup U_2$ is equivalent to the x-axis and the y-axis. However, $e_1, e_2 \in U_1 \cup U_2$ but $e_1 + e_2 \notin U_1 \cup U_2$. Thus $U_1 \cup U_2$ is not a vector space.

3.2.3 Linear Independence

Definition 3.2.10. Let V be a vector space. A set $\{v_1, ..., v_k\}$ of vectors in V is said to be *linearly independent* if

$$c_1 v_1 + \cdots + c_k v_k = 0 \quad \text{implies} \quad c_1 = \cdots = c_k = 0.$$

It is called *linearly dependent* otherwise, that is, if

$$\text{there exist } c_1, ..., c_k \text{ not all zero} \quad \text{such that} \quad c_1 v_1 + \cdots + c_k v_k = 0.$$

Theorem 3.2.11. *Suppose $v_1, ..., v_k \in \mathbb{R}^n$. Let $V = \begin{bmatrix} v_1 & \cdots & v_k \end{bmatrix}$. $\{v_1, ..., v_k\}$ is linearly independent if and only if $Vc = 0$ only has the trivial solution $c = 0$.*

Example 3.2.12. Consider the columns of B in example 3.2.4, where

$$b_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}, \quad b_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad b_4 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}.$$

Then

(i) $\{b_1\}$ is linearly independent.

(ii) $\{b_1, b_2\}$ is linearly dependent.

(iii) $\{b_1, b_3\}$ is linearly independent.

(iv) $\{b_1, b_3, b_4\}$ is linearly dependent.

We know that $\text{Col}(B) = \text{Span}\{b_1, b_2, b_3, b_4\}$. However, as b_2 is a multiple of b_1 and as b_4 is a linear combination of b_1 and b_3, we can discard b_2 and b_4, and the set $\{b_1, b_3\}$, which is linearly independent, still spans $\text{Col}(B)$. Here we have obtained a maximal linearly independent set that spans $\text{Col}(B)$.

Let's consider another example.

Example 3.2.13. Consider $P_3 = \{p | p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3,\ a_i \in \mathbb{R}\}$.

(i) Consider the dependence relationship of the vectors in $\{1+t, t+t^2, t^2+t^3\}$. If we let $\alpha_1(1+t) + \alpha_2(t+t^2) + \alpha_3(t^2+t^3) = 0$, we obtain $\alpha_1 + (\alpha_1+\alpha_2)t + (\alpha_2+\alpha_3)t^2 + \alpha_3 t^3 = 0$. Thus $\alpha_1 = \alpha_2 = \alpha_3 = 0$. So $\{1+t, t+t^2, t^2+t^3\}$ is linearly independent.

(ii) We now add another vector t^3 to the original set, which becomes $\{1+t, t+t^2, t^2+t^3, t^3\}$. Again, we consider the dependence relationship: $\alpha_1(1+t) + \alpha_2(t+t^2) + \alpha_3(t^2+t^3) + \alpha_4 t^3 = 0$. Then $\alpha_1 + (\alpha_1+\alpha_2)t + (\alpha_2+\alpha_3)t^2 + (\alpha_3+\alpha_4)t^3 = 0$. We again have $\alpha_1 = \alpha_2 = \alpha_3 = \alpha_4 = 0$. Therefore $\{1+t, t+t^2, t^2+t^3, t^3\}$ is linearly independent.

(iii) If we add one more vector 1 to the set in (ii), we obtain the set $\{1, 1+t, t+t^2, t^2+t^3, t^3\}$. This time the set is linearly dependent. To see why, consider $\alpha_0 \cdot 1 + \alpha_1(1+t) + \alpha_2(t+t^2) + \alpha_3(t^2+t^3) + \alpha_4 t^3 = 0$. We then have $(\alpha_0+\alpha_1) + (\alpha_1+\alpha_2)t + (\alpha_2+\alpha_3)t^2 + (\alpha_3+\alpha_4)t^3 = 0$. By letting $\alpha_0 = 1$, we obtain a nontrivial solution. Thus $\{1, 1+t, t+t^2, t^2+t^3, t^3\}$ is linearly dependent.

Remark. Two remarks can be made on example 3.2.13.

i. The concept of linear independence is defined on *every* vector space, not limited to Euclidean vector spaces.

ii. We cannot add too many vectors to an originally independent set and still claim linear independence! That is, there exists a maximal linearly independent set! In example 3.2.13, A linearly independent set can contain maximally 4 vectors. Adding other vectors to such a set will make the set linearly dependent. Equivalently, if $\{v_1, v_2, v_3, v_4\} \subset P_3$ is linearly independent, the every $v_5 \in P_3$ lies in Span$\{v_1, v_2, v_3, v_4\}$.

3.2.4 Basis and Dimension

The problem of finding a minimal spanning set leads to the definition of basis.

Definition 3.2.14. Let V be a vector space. A set $\{v_1, ..., v_n\}$ of vectors in V is said to be a *basis* of V if it satisfies the following conditions:

1. $\{v_1, ..., v_n\}$ spans V.

2. $\{v_1, ..., v_n\}$ is linearly independent.

Remark. In definition 3.2.14, the first condition requires that a basis must have sufficient coverage (cannot have too few vectors). The second condition requires that a basis cannot have too many vectors.

On Monday, we introduced an important theorem.

Theorem 3.2.15. *Let V be a vector space. All bases of V contain the same number of vectors.*

Example 3.2.16. We know that $\{(1,0,0)^T, (0,1,0)^T, (0,0,1)^T\}$ is a basis of $\mathbb{R}^3$. Another basis of $\mathbb{R}^3$ is $\{(1,0,1)^T, (1,1,0)^T, (1,0,0)^T\}$. If we consider the linear dependence relationship of the three vectors: $\alpha_1(1,0,1)^T + \alpha_2(1,1,0)^T + \alpha_3(1,0,0)^T = (0,0,0)^T$, by checking the third entry we get $\alpha_1 = 0$. Then the second entry gives $\alpha_2 = 0$. Hence $\alpha_1 = 0$. Thus the set is linearly independent. It can be verified that these three vectors span $\mathbb{R}^3$.

Theorem 3.2.15 gives rise to the concept of dimension.

Definition 3.2.17. Let V be a vector space. The *dimension* of V, denoted by $\dim V$ or $\dim(V)$, is defined as the number of vectors in a basis of V.

Theorem 3.2.18. *Let V be a vector space with $\dim V = n$.*

(i) The dimension of V is equal to the maximal number of linearly independent vectors in V. Equivalently, a set $\{v_1, ..., v_m\} \subset V$ such that $m > n$ is linearly dependent.

(ii) The dimension of V is equal to the minimal number of vectors in V that span V. Equivalently, a set $\{u_1, ..., u_k\} \subset V$ such that $k < n$ does not span V.

Proof of (i). This is by theorem 3.1.15. □

Proof of (ii). Suppose a set $\{u_1, ..., u_k\} \subset V$ spans V with $k < n$. Then $\dim V \leq k < n$. This is a contradiction. □

3.2.5 Dimensions and Bases of Four Fundamental Subspaces (I)

In this section we find out the dimensions and bases of the four spaces, namely $\mathrm{Col}(A)$, $\mathrm{Row}(A)$, $N(A)$, and $N(A^T)$ using the theorems we have developed in section 3.2.1. Before we do so, we need another powerful theorem, which says that *pre-multiplying a set of vectors by an invertible matrix does not change dependence relation*.

Theorem 3.2.19. *A set $\{v_1,...,v_k\} \subset \mathbb{R}^n$ is linearly independent if and only if $\{Av_1,...,Av_k\}$ is linearly independent, where $A \in \mathbb{R}^{m\times n}$ is an invertible matrix.*

Proof. Let $V = \begin{bmatrix} v_1 & \cdots & v_k \end{bmatrix}$. $\{v_1,...,v_k\} \subset \mathbb{R}^n$ is linearly independent if and only if $N(V) = \{0\}$, if and only if $N(AV) = \{0\}$, as Gaussian elimination does not change the null space. $N(AV) = \{0\}$ if and only if $\{Av_1,...,Av_k\}$ is linearly independent. □

Dimension and Basis of Row Space

In section 3.2.1, we have shown that Gaussian elimination does not change the row space of a matrix, that is, $\mathrm{Row}(A) = \mathrm{Row}(B)$, where $B = GA$ is the row echelon form of A. Thus to study the row space of A, we only need to study the row space of B, which is easier.

Recall that B is in the form

$$B = \begin{bmatrix} \blacksquare & * & * & * & \cdots & * & * & * \\ 0 & 0 & \blacksquare & * & \cdots & * & * & * \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & \blacksquare & * \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & 0 & \cdots & 0 & 0 & 0 \end{bmatrix}.$$

The row space of B is spanned by its nonzero rows. Consider the linear dependence relation

$$\lambda_1(\text{row } 1) + \cdots + \lambda_r(\text{row } r) = 0.$$

Now we check the first nonzero entry of row 1, which is the first pivot. By the definition of row echelon, the rows below have 0 at this position, then $\lambda_1 = 0$. After showing that $\lambda_1 = \cdots = \lambda_{k-1} = 0$, we check the kth pivot, which is the first nonzero entry in row k. By the definition of row echelon form, all entries below it in the same column are 0. Thus $\lambda_k = 0$. Hence all λ_i's are equal to zero. Thus the nonzero rows that span $\mathrm{Row}(B)$ are linearly independent. We have found a basis for $\mathrm{Row}(B) = \mathrm{Row}(A)$. Thus the dimension of $\mathrm{Row}(A)$ is equal to the number of nonzero rows in its row echelon form, which is equal to the number of pivots. We summarize our results below.

Theorem 3.2.20. *Let A be a matrix and B a row echelon form of A. Then*

(i) $\mathrm{Row}(A) = \mathrm{Row}(B)$.

(ii) The nonzero rows of B form a basis of $\mathrm{Row}(A)$.

(iii) $\dim\mathrm{Row}(A) = \dim\mathrm{Row}(B) =$ *number of non-zero rows of B = number of pivots in B.*

We now introduce the concept of the *rank* of a matrix, which will prove to be important later.

Definition 3.2.21. Let A be a matrix. The *rank* of A, denoted by $\mathrm{rank}(A)$ or $\mathrm{rank}A$, is defined as the dimension of the row space of A. In other words, $\mathrm{rank}(A) = \dim\mathrm{Row}(A)$.

Theorem 3.2.22. *The rank of a matrix A is equal to the number of pivots of its row echelon form.*

Theorem 3.2.23. *A square matrix A is nonsingular if and only if it has full rank.*

Dimension of Column Space

We have seen that if A is a matrix and $B = GA$ its row echelon form, then generally $\text{Col}(A) \neq \text{Col}(B)$. Surprisingly, $\dim \text{Col}(A) = \dim \text{Col}(B)$. We show the reason below.

Suppose $A = \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix}$ and $B = GA = \begin{bmatrix} b_1 & \cdots & b_n \end{bmatrix} = \begin{bmatrix} Ga_1 & \cdots & Ga_n \end{bmatrix}$, where G is invertible. From the columns of A, we can find some columns $a_{k_1}, ..., a_{k_r}$ such that they form a basis for $\text{Col}(A)$. The other columns are linear combinations of these r columns. As $B = GA$, where G is invertible, then by theorem 3.2.19, the linear dependence relation of the columns of B is the same as that of the columns of A. Thus $b_{k_1}, ..., b_{k_r}$ are still linearly independent and the other columns of B are still linear combinations of $b_{k_1}, ..., b_{k_r}$. Thus $\{b_{k_1}, ..., b_{k_r}\}$ is a basis for $\text{Col}(B)$. Hence $\dim \text{Col}(A) = \dim \text{Col}(B) = r$. Thus we have shown that *Gaussian elimination does not change the dimension of column space*.

Remark. Note that so far we have not known how to pick the columns $a_{k_1}, ..., a_{k_r}$ or $b_{k_1}, ..., b_{k_r}$. Neither do we know $\dim \text{Col}(A)$.

■

3.3 Friday (Prof. Tom Luo)

Recap

We summarize our main results in the last lecture.

Dimension and Basis. The *dimension* of a vector space V is equal to the maximal number of linearly independent vectors in V. A *basis* of V is a largest set of linearly independent vectors in V.

Gaussian Elimination and Fundamental Subspaces. The equation $Ax = 0$ leads to our discussion of $\text{Row}(A)$, $\text{Col}(A)$, and $N(A)$. Gaussian elimination on A is equivalent to pre-multiplying A by an invertible matrix G. Call the resultant matrix B. Then

$$\text{Row}(A) = \text{Row}(B), \quad N(A) = N(B), \quad \text{Col}(A) \neq \text{Col}(B) \text{ (in general)}$$

and

$$\dim\text{Row}(A) = \dim\text{Row}(B), \quad \dim N(A) = \dim N(B), \quad \dim\text{Col}(A) = \dim\text{Col}(B).$$

3.3.1 Dimensions and Bases of Four Fundamental Subspaces (II)

Dimensions of Row Space and Column Space are Equal

We begin with a claim that seems non-trivial.

Claim 3.3.1. *If B is the reduced row echelon form of A, then* $\dim\text{Row}(B) = \dim\text{Col}(B)$.

If this is true, then we have

$$\text{rank}(A) = \dim\text{Row}(A) = \dim\text{Row}(B) = \dim\text{Col}(B) = \dim\text{Col}(A),$$

which gives us

$$\dim\text{Row}(A) = \dim\text{Col}(A) \quad \text{for every } A \in \mathbb{R}^{m\times n}!$$

This is a surprising result!

As it turns out, the claim is true. To gain insights into what this claim means, let us first consider a simple example.

Example 3.3.2. Recall the the reduced row echelon form of the matrix A in section 2.4.1:

$$B = \begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The first and third columns are the *pivot columns*, namely columns in which there are pivots. The second and fourth columns correspond to the free variables. The key observation is that *each pivot column is a unit vector, or equivalently, a column of the identity matrix*.

Let us first consider the column space of B. Of course,

$$\text{Col}(B) = \text{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

However, as b_2 is a multiple of b_1 and $b_4 = -b_1 + b_3$, we have

$$\text{Col}(B) = \text{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Since b_1 and b_3, which are the pivot columns, are linearly independent, they form a basis for $\mathrm{Col}(B)$. Thus $\dim \mathrm{Col}(B) = 2 =$ the number of pivots.

Now let us consider the row space.

$$\mathrm{Row}(B) = \mathrm{Span}\left\{ \begin{bmatrix} 1 \\ 3 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

By checking the pivot positions (the shaded ones), we see that these two vectors are linearly independent. Therefore $\dim \mathrm{Row}(B) = 2 =$ the number of nonzero rows. As each nonzero rows has exactly one pivot, the dimension of $\mathrm{Row}(B)$ is equal to the number of pivots. Thus $\dim \mathrm{Row}(B) = \dim \mathrm{Col}(B) = 2$.

We now argue that $\dim \mathrm{Row}(B) = \dim \mathrm{Col}(B)$ is true for every matrix. On Wednesday we have already shown that $\dim \mathrm{Row}(B)$ is equal to the number of pivots of B. We now show that so is $\dim \mathrm{Col}(B)$. Suppose B has r nonzero rows and thus r pivots. By the definition of reduced row echelon form, the pivot columns are $e_1, .., e_r \in \mathbb{R}^m$, respectively. Thus the pivot columns are linearly independent. As row $(r+1)$ through m are all zeros, the $(r+1)$th through mth entries of each column b_i of B are all zeros. Thus each b_i is a linear combination of the pivot columns. Hence the r pivot columns span $\mathrm{Col}(B)$. We conclude that $\dim \mathrm{Col}(B) =$ the number of pivots $= \dim \mathrm{Row}(B)$. This is the first main theorem of this lecture.

Theorem 3.3.3. *For every $A \in \mathbb{R}^{m\times n}$, $\dim \mathrm{Col}(A) = \dim \mathrm{Row}(A) = \mathrm{rank}(A)$.*

Remark. We could have carried out the argument using only the row echelon form of A. As only b_{k_r}, among $b_{k_1}, ..., b_{k_r}$, has the rth entry nonzero (because this is the rth pivot), its linear combination coefficient must be zero. Then by working upward, from the $(r-1)$th pivot to the first pivot, we can inductively show that all linear combination coefficients are zero. This also gives us the conclusion that $\{b_{k_1}, ..., b_{k_r}\}$ is linearly independent.

We have just shown that the pivot columns span $\mathrm{Col}(B)$. This gives us a way to find a basis for $\mathrm{Col}(A)$. Recall from the last lecture that Gaussian elimination does not change dependence relation of columns. Thus if $b_{k_1}, ..., b_{k_r}$ are the pivot columns of B, then the corresponding columns $a_{k_1}, ..., a_{k_r}$ form a basis of $\mathrm{Col}(A)$.

The following theorem is immediate from theorem 3.3.3.

Theorem 3.3.4. *For every $A \in \mathbb{R}^{m\times n}$, $\mathrm{rank}(A) = \mathrm{rank}(A^T)$.*

Proof. As $\mathrm{Row}(A) = \mathrm{Col}(A^T)$, we have $\mathrm{rank}(A) = \dim \mathrm{Row}(A) = \dim \mathrm{Col}(A^T) = \mathrm{rank}(A^T)$. □

Corollary 3.3.5. *A square matrix A is nonsingular if and only if A^T is nonsingular.*

Dimension and Basis of Null Space

We now turn to the null space. We use the example above.

Example 3.3.6. As $N(A) = N(B)$, it suffices to study $N(B)$. The equation $Bx = 0$ gives

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -3x_2 + x_4 \\ x_2 \\ -x_4 \\ x_4 \end{bmatrix} = x_2 \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}.$$

Thus

$$N(B) = \text{Span}\left\{ \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix} \right\}.$$

We can check that these two vectors are linearly independent by comparing the free variable positions (the shaded ones). Thus they form a basis for $N(B)$. Thus $\dim N(A) = \dim N(B) = 2 =$ number of free variables.

We now show that $\dim N(A) = \dim N(B) =$ number of free variables for every matrix A. By renumbering the variables such that the pivot variables come first, we obtain, without loss of generality, the reduced system

$$\begin{bmatrix} I_r & F \\ O & O \end{bmatrix} \begin{bmatrix} x^p \\ x^f \end{bmatrix} = O. \tag{3.3.1}$$

In (3.3.1), x^p is the vector that contains the pivot variables and x^f is the vector that contains the free variables. The identity matrix contains the pivot columns and the matrix F, which is of size $r \times (n-r)$, corresponds to the free variables (Recall that each pivot column is a column of the identity matrix). (3.3.1) gives $x^p + Fx^f = O$. Thus the general solution is

$$x = \begin{bmatrix} x^p \\ x^f \end{bmatrix} = \begin{bmatrix} -Fx^f \\ x^f \end{bmatrix} = \begin{bmatrix} -F \\ I \end{bmatrix} x^f = x_1^f \begin{bmatrix} -f_1 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} + x_2^f \begin{bmatrix} -f_2 \\ 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix} + \cdots + x_{n-r}^f \begin{bmatrix} -f_{n-r} \\ 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix} \tag{3.3.2}$$

($f_1, f_2, ..., f_{n-r}$ are the columns of F). In the last equality of (3.3.2), we obtain a basis of $N(A)$. They are linearly independent because of the unit vector part (or equivalently, the identity submatrix in the original matrix). We conclude that $\dim N(A) =$ the number of free variables $= n - r$. This leads to the famous *rank-nullity theorem*, where the *nullity* of a matrix is defined as the dimension of its null space.

Theorem 3.3.7 (Rank-Nullity Theorem)**.** *If A is an $m \times n$ matrix, then its rank of plus its nullity is equal to n. Equivalently,*

$$\dim \text{Col}(A) + \dim N(A) = n. \tag{3.3.3}$$

The last equality in (3.3.2) shows us a way to find a basis for the null space of A. By setting $x_j^f = 1$ and other free variables equal to 0, we can pick out the jth vector in (3.3.2). Repeating this process for every $j = 1, ..., (n-r)$ yields a basis for $N(A)$.

By taking transpose, we obtain a similar relation between the row space and the left null space of a matrix.

Theorem 3.3.8. *If A is an $m \times n$ matrix, then*

$$\dim \text{Row}(A) + \dim N(A^T) = m. \tag{3.3.4}$$

Solution Set of Inhomogeneous Equation

We now discuss the solution set of the inhomogeneous equation $Ax = b$ (Note that this is not a vector space). For ease of discussion, we apply the same trick of renumbering the variables such that the pivot variables come first. This is done without loss of generality. The reduced system is then

$$\begin{bmatrix} I & F \\ O & O \end{bmatrix} \begin{bmatrix} x^p \\ x^f \end{bmatrix} = \begin{bmatrix} c^p \\ c^f \end{bmatrix} \tag{3.3.5}$$

(x and c may not have the same size). If $c^f \neq O$, then $Ax = b$ has no solution. We say that the system is *infeasible*.

Suppose $c^f = O$, then the system is consistent. (3.3.5) gives $x^p + Fx^f = c^p$. Thus the general solution to $Ax = b$ is

$$x = \begin{bmatrix} x^p \\ x^f \end{bmatrix} = \begin{bmatrix} -Fx^f + c^p \\ x^f \end{bmatrix} = \underbrace{\begin{bmatrix} -F \\ I \end{bmatrix} x^f}_{N(A)} + \underbrace{\begin{bmatrix} c^p \\ O \end{bmatrix}}_{x_p} \tag{3.3.6}$$

which is an element in the null space of A plus a special solution. This special solution is obtained by letting all free variables equal to 0. Thus the solution set of $Ax = b$ is equal to the sum of the null space of A and $\{x_p\}$, where x_p is a particular solution. This is consistent with theorem 2.4.10. We restate this theorem here.

Theorem 3.3.9. *Suppose $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$. Let $x_p \in \mathbb{R}^n$ be a special solution to $Ax = b$. A vector $z \in \mathbb{R}^n$ is a solution to $Ax = b$ if and only if $z = x_p + x_n$ for some $x_n \in N(A)$.*

3.3.2 Rank

Here we elaborate a little on the rank of a matrix. Recall that for a matrix A,

$$\text{rank}(A) = \text{number of pivot variables} = \text{rank}(A^T).$$

If A is $n \times n$, then $\text{rank}(A) = n$ if and only if A has n pivots, if and only if A is nonsingular, or equivalently, invertible. Another way of seeing this is by rank-nullity theorem, A has full rank if and only if its nullity is 0, if and only if $N(A) = \{0\}$, which is equivalent to A being nonsingular.

We further introduce two theorems.

Theorem 3.3.10. *For every matrix A,* $\text{rank}(A^TA) = \text{rank}(A)$.

Proof. Suppose A is $m \times n$. As A^TA is $n \times n$, by rank-nullity theorem, it suffices to show that $\dim N(A^TA) = \dim N(A)$. We can actually prove a stronger statement: $N(A^TA) = N(A)$. The reasoning is as below.

Suppose $Ax = 0$, then $A^TAx = A^T0 = 0$. Now suppose $A^TAx = 0$. Pre-multiplying both sides by x^T gives $x^TA^TAx = 0$. Thus $(Ax)^T(Ax) = \|Ax\|^2 = 0$. So $\|Ax\| = 0$. Hence $Ax = 0$. We have shown that $A^TAx = 0$ is equivalent to $Ax = 0$. Therefore $N(A^TA) = N(A)$. The proof is complete. □

Corollary 3.3.11. *For every matrix A,* $\text{rank}(A^TA) = \text{rank}(A) = \text{rank}(A^T) = \text{rank}(AA^T)$.

By proving that $Ax = 0$ is equivalent to $A^TAx = 0$, we obtain an important theorem:

Theorem 3.3.12. *Suppose A is an $m \times n$ matrix. The columns of A are linearly independent if and only if A^TA is nonsingular.*

The second theorem about rank is as follows.

Theorem 3.3.13. *Let A and B be matrices such that the product AB is defined. Then*

$$\text{rank}(AB) \leq \text{rank}(A) \quad \textit{and} \quad \text{rank}(AB) \leq \text{rank}(B).$$

There are multiple ways of proving this. We first present a proof which uses rank-nullity theorem.

Proof version I. We first show that $\text{rank}(AB) \leq \text{rank}(B)$. $Bx = 0$ implies $ABx = 0$. Thus $N(B) \subset N(AB)$ and thus $\dim N(B) \leq \dim N(AB)$. Since B and AB has the same number of columns, by rank-nullity theorem, $\text{rank}(B) \geq \text{rank}(AB)$.

Now we show that $\text{rank}(AB) \leq \text{rank}(A)$. We do this by taking transpose. We already know $\text{rank}(B^TA^T) \leq \text{rank}(A^T)$. This is equivalent to $\text{rank}(AB) \leq \text{rank}(A)$. □

In this proof we have made use of the following theorem.

Theorem 3.3.14. *Suppose V is a vector space of finite dimension. If S is a subspace of V, then* $\dim S \leq \dim V$.

Proof. We argue by contradiction. Suppose S had a basis that consisted of m vectors, where $m \geq \dim V$. As S is a subspace of V, these m vectors must be in V. But as $m \geq \dim V$, they would not be linearly independent and could not form a basis for S. This is a contradiction. Thus every basis of S must have size less than or equal to $\dim V$. □

We present another proof of theorem 3.3.13.

Proof version II. We first show that $\text{rank}(AB) \leq \text{rank}(A)$. The columns of AB are linear combinations of the columns of A. Thus $\text{Col}(AB) \subset \text{Col}(A)$. Therefore $\text{rank}(AB) = \dim \text{Col}(AB) \leq \dim \text{Col}(A) = \text{rank}(A)$.

Next we show $\text{rank}(AB) \leq \text{rank}(B)$. The rows of AB are linear combination of the rows of B. So $\text{Row}(AB) \subset \text{Row}(B)$. Hence $\text{rank}(AB) \leq \text{rank}(B)$. □

3.3.3 Linear Transformations

Definition 3.3.15. Let V and W be vector spaces. A mapping L from V to W is said to be *linear* if $L(\alpha_1 v_1 + \alpha_2 v_2) = \alpha_1 L(v_1) + \alpha_2 L(v_2)$ for all $v_1, v_2 \in V$ and all scalars α_1, α_2. The set of all linear mappings from V to W is denoted by $\mathcal{L}(V,W)$.

By definition, a linear transformation preserves linear combination structure.

Example 3.3.16. Consider $P_n = \{p | p : \mathbb{R} \mapsto \mathbb{R},\ p(t) = \sum_{i=0}^{n} a_i x^i,\ a_i \in \mathbb{R}\}$.

(i) Differentiation $\dfrac{d}{dt}$ from P_n to P_{n-1} defines a linear mapping.

(ii) Integration $\displaystyle\int_0^t \cdot\, du$ from P_n to P_{n+1} is also a linear mapping.

Example 3.3.17. The mapping $L : \mathbb{R}^3 \mapsto \mathbb{R}^2$ defined by

$$L((x_1, x_2, x_3)^T) = (x_1 x_2,\ x_2 x_3,\ x_1 x_3)^T$$

is not a linear mapping, as $L(2x) = 4L(x) \not\equiv 2L(x)$.

Theorem 3.3.18. *Suppose $L : V \mapsto W$ is linear. Then*

(i) $L(0_V) = 0_W$, *where* 0_V *and* 0_W *are the zero vectors in V and W, respectively.*

(ii) $L(\alpha u) = \alpha L(u)$ *for every scalar* α *and every* $u \in V$.

(iii) $L(u+v) = L(u) + L(v)$ *for all* $u, v \in V$.

Proof. The second and third statements are immediate from definition. For the first statement, as $L(0_V) = L(0_V + 0_V) = 2L(0_V)$, then $L(0_V) = 0_W$. □

Remark. The converse of (i) is not necessarily true. That is, there may be $u \neq 0_V$ such that $L(u) = 0_W$. As an example, the null space of a matrix is not necessarily $\{0\}$.

Example 3.3.19. The mapping $L : \mathbb{R}^n \mapsto \mathbb{R}^m$ defined by $L(x) = Ax$, where $A \in \mathbb{R}^{m \times n}$, is a linear mapping by the linearity of matrix multiplication.

As a matter of fact, every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ can be written in the form of $L(x) = Ax$.

Theorem 3.3.20. *A mapping $L : \mathbb{R}^n \mapsto \mathbb{R}^m$ is linear if and only if $L(x) = Ax$ for all $x \in \mathbb{R}^n$ for some $A \in \mathbb{R}^{m\times n}$.*

To prove this theorem, we need to know how to characterize a linear transformation. An intuitive idea is to start from a basis.

Claim 3.3.21. *A linear mapping from V to W is completely characterized by its actions on a basis of V. In other words, if $\{v_1,...,v_n\}$ is a basis of V, then $\{L(v_1),...,L(v_n)\}$ characterizes L.*

This is true because for every $x \in V$, $L(x) = L(x_1v_1 + \cdots + x_nv_n) = x_1L(v_1) + \cdots + x_nL(v_n)$. Thus to know the value of $L(x)$, we only need to know how x is represented in terms of a basis and how L acts on the basis.

Therefore, to characterize a linear mapping from $\mathbb{R}^n$ to $\mathbb{R}^m$, we choose the most natural basis $\{e_1,...,e_n\}$. Then for every $x \in \mathbb{R}^n$,

$$L(x) = L(x_1e_1 + \cdots + x_ne_n) = x_1L(e_1) + \cdots + x_nL(e_n) = \begin{bmatrix} L(e_1) & \cdots & L(e_n) \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = Ax$$

with $A = \begin{bmatrix} L(e_1) & \cdots & L(e_n) \end{bmatrix}$. We have thus proved theorem 3.3.20.

Kernel and Range

We now generalize the concept of null space and column space.

Definition 3.3.22. Let $L : V \mapsto W$ be a linear mapping.

(i) The *kernel* of L, denoted by $\ker(L)$, is the set of all $v \in V$ such that $L(v) = 0$, that is,

$$\ker(L) = \{v \in V : L(v) = 0\}. \tag{3.3.7}$$

(ii) The *range* of L, denoted by $\text{range}(L)$, is defined as

$$\text{range}(L) = \{w \in W : w = L(v),\ v \in V\}. \tag{3.3.8}$$

If $L : \mathbb{R}^n \mapsto \mathbb{R}^{\geqslant}$, then $L(x) = Ax$, where $A \in \mathbb{R}^{m\times n}$. Then $\ker(L) = N(A)$ and $\text{range}(L) = \text{Col}(A)$.

Linear Transformations on Coordinates

We illustrate how a linear transformation changes the coordinates with the examples of differentiation and integration.

Example 3.3.23. Consider differentiation (denoted by D)

$$\frac{d}{dt} : P_4 \mapsto P_3.$$

A basis for P_4 is $\{1,t,t^2,t^3,t^4\}$ and a basis for P_3 is $\{1,t,t,t^3\}$. Let $p_i(t) = t^i$. To characterize L, we only need to know how L acts on a basis of P_4. We have

$$\begin{aligned} D(p_0(t)) &= 0, \\ D(p_1(t)) &= p_0(t), \\ D(p_2(t)) &= 2p_1(t), \\ D(p_3(t)) &= 3p_2(t), \\ D(p_4(t)) &= 4p_3(t). \end{aligned}$$

Then for every $x \in P_4$,

$$x = x_1 p_0(t) + x_2 p_1(t) + x_3 p_2(t) + x_4 p_3(t) + x_5 p_4(t)$$

and

$$\begin{aligned} D(x) &= x_2 p_0(t) + 2x_3 p_1(t) + 3x_4 p_2(t) + 4x_5 p_3(t) \\ &= \begin{bmatrix} p_0 & p_1 & p_2 & p_3 \end{bmatrix} \begin{bmatrix} x_2 \\ 2x_3 \\ 3x_4 \\ 4x_5 \end{bmatrix} \\ &= \begin{bmatrix} p_0 & p_1 & p_2 & p_3 \end{bmatrix} \underbrace{\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}}_{A_{\text{diff}}} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \end{aligned}$$

We see that if the coordinates of $x \in P_4$ with respect to the basis $\{1, t, t^2, t^3, t^4\}$ are given by the vector $(x_1, x_2, x_3, x_4, x_5)^T$, then the coordinates of $D(x) \in P_3$ with respect to the basis $\{1, t, t^2, t^3\}$ are given by the vector $A_{\text{diff}}(x_1, x_2, x_3, x_4, x_5)^T$.

Example 3.3.24. Consider the integration (denoted by I)

$$\int_0^t \cdot \, du : P_3 \mapsto P_4.$$

For every $y \in P_3$,

$$y = y_1 p_0(t) + y_2 p_1(t) + y_3 p_2(t) + y_4 p_3(t)$$

and

$$\begin{aligned} I(y) &= y_1 p_1(t) + \frac{1}{2} y_2 p_2(t) + \frac{1}{3} y_3 p_3(t) + \frac{1}{4} y_4 p_4(t) \\ &= \begin{bmatrix} p_0 & p_1 & p_2 & p_3 & p_4 \end{bmatrix} \begin{bmatrix} 0 \\ y_1 \\ \frac{1}{2} y_2 \\ \frac{1}{3} y_3 \\ \frac{1}{4} y_4 \end{bmatrix} \\ &= \begin{bmatrix} p_0 & p_1 & p_2 & p_3 & p_4 \end{bmatrix} \underbrace{\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & \frac{1}{4} \end{bmatrix}}_{A_{\text{int}}} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix} \end{aligned}$$

Thus if the coordinates of $y \in P_3$ with respect to the basis $\{1, t, t^2, t^3\}$ are given by the vector $(y_1, y_2, y_3, y_4)^T$, then the coordinates of $I(y) \in P_4$ with respect to the basis $\{1, t, t^2, t^3, t^4\}$ are given by the vector $A_{\text{int}}(y_1, y_2, y_3, y_4)^T$.

Note that $A_{\text{diff}} A_{\text{int}} = I$ but $A_{\text{int}} A_{\text{diff}} \neq I$. This is reasonable because if we first differentiate then integrate, the constant term is lost.

■

Chapter 4

Week Four

4.1 Monday (Prof. Tom Luo)

4.1.1 Matrix Representation of Linear Transformation

Coordinate Vector

In this section we introduce the concept of *coordinate vectors*. We begin with an example.

Example 4.1.1. Let P_3 be the space of polynomials of degree no more than 3. We know that $\{1,t,t^2,t^3\}$ is a basis for P_3 and that every $p \in P_3$ can be written as $p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$. We can associate each $p \in P_3$ with a *coordinate vector* $(a_0,a_1,a_2,a_3)^T$ with respect to the basis $\{1,t,t^2,t^3\}$. Further, p is *uniquely* identified by its coordinate vector with respect to this basis.

We formulate our ideas in a more general setting.

Theorem 4.1.2. *Suppose V is a vector space with a basis $\{v_1,...,v_n\}$. Every vector in V can be written as a linear combination of $v_1,...,v_n$ uniquely up to order of terms.*

Definition 4.1.3. Suppose V is a vector space with an ordered basis $\mathcal{B} = \{v_1,...,v_n\}$. By theorem 4.1.2, every $x \in V$ can be written uniquely as $x = x_1 v_1 + \cdots + x_n v_n$. The vector $(x_1,...,x_n)^T$ of the coefficients is called the *coordinate vector of x with respect to the ordered basis $\mathcal{B}$* and is denoted by $[x]_{\mathcal{B}}$. Thus

$$[x]_{\mathcal{B}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}. \tag{4.1.1}$$

If $V = \mathbb{R}^n$, we can let $\mathbf{B} = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}$, then

$$x = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \mathbf{B}[x]_{\mathcal{B}}. \tag{4.1.2}$$

We see that $[\,\cdot\,]_{\mathcal{B}}$ defines a *coordinate mapping* $\varphi_{\mathcal{B}}$ that maps every vector in V to its coordinate vector in $\mathbb{R}^n$:

$$\varphi_{\mathcal{B}}(x) = [x]_{\mathcal{B}} \quad \text{for every } x \in V.$$

The coordinate mapping is a bijection.

Matrix Representation Theorem

Last week we saw that if $L:\mathbb{R}^n \mapsto \mathbb{R}^m$ is linear, then there exists an $m \times n$ matrix A such that $L(x) = Ax$, where $x \in \mathbb{R}^n$. We can extend this theorem to a more general case. However, for general vector spaces, matrix multiplication is not defined. Still, we know that in an n-dimensional vector space, if we fix a basis, then each vector is uniquely identified with its coordinate vector, which is in $\mathbb{R}^n$. We can perform matrix multiplication on coordinate vectors. So we operate on the coordinate vectors instead of the vectors themselves.

We now begin our discussion. Let V and W be vector spaces. Suppose L is a linear transformation from V to W. Let $\mathcal{B} = \{v_1, ..., v_n\}$ be a basis of V and $\mathcal{C} = \{w_1, ..., w_m\}$ be a basis of W. Suppose $x \in V$ has coordinate vector $[x]_\mathcal{B} = (x_1, ..., x_n)^T$. Then $L(x) \in W$. We raise the following question.

Question 4.1.4. How is $[L(x)]_\mathcal{C}$ related to $[x]_\mathcal{B}$?

To find out the answer, we need to represent $[L(x)]_\mathcal{C}$ in terms of $[x]_\mathcal{B}$. Like what we did before, we write x as the linear combination of the vectors in the basis $\mathcal{B}$. Let $\mathbf{B} = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}$ and $\mathbf{C} = \begin{bmatrix} w_1 & \cdots & w_n \end{bmatrix}$. If we define $L(\mathbf{B}) = \begin{bmatrix} L(v_1) & \cdots & L(v_n) \end{bmatrix}$, then we have

$$L(x) = L(\mathbf{B})[x]_\mathcal{B} = \mathbf{C}A[x]_\mathcal{B}$$

where $A = \begin{bmatrix} [L(v_1)]_\mathcal{C} & \cdots & [L(v_n)]_\mathcal{C} \end{bmatrix}$. By theorem 4.1.2, we conclude that $[L(x)]_\mathcal{C} = A[x]_\mathcal{B}$. We see that the matrix A has dual properties:

$$L(\mathbf{B}) = \mathbf{C}A \quad \text{and} \quad A[x]_\mathcal{B} = [L(x)]_\mathcal{C} \tag{4.1.3}$$

Notice the positions of A.

Theorem 4.1.5. *Let V and W be two vector spaces. Suppose $\mathcal{B} = \{v_1, ..., v_n\}$ is a basis for V and that $\mathcal{C} = \{w_1, ..., w_m\}$ is a basis for W. For every linear transformation $L: V \mapsto W$, there exists an $m \times n$ matrix A such that*

$$[L(x)]_\mathcal{C} = A[x]_\mathcal{B} \tag{4.1.4}$$

for every $x \in V$. This matrix A is given by $A = \begin{bmatrix} [L(v_1)]_\mathcal{C} & \cdots & [L(v_n)]_\mathcal{C} \end{bmatrix}$.

The matrix A is called *the matrix representing L with respect to the basis $\mathcal{B}$ and $\mathcal{C}$*. We can illustrate theorem 4.1.5 with a graph:

$$\begin{array}{ccc} V & \xrightarrow{\ L\ } & W \\ \downarrow{\scriptstyle \varphi_\mathcal{B}} & & \downarrow{\scriptstyle \varphi_\mathcal{C}} \\ \mathbb{R}^n & \xrightarrow{\ A\ } & \mathbb{R}^m \end{array} \tag{4.1.5}$$

Change of Matrix Representation over Change of basis

We now turn to another question. Let L be a linear transformation that maps an n-dimensional vector space V to itself. Suppose $\mathcal{B} = \{v_1, ..., v_n\}$ and $\mathcal{B}' = \{v_1', ..., v_n'\}$ are two bases for V. By theorem 4.1.5, there exist $n \times n$ matrices A and A' such that for every $x \in V$,

$$[L(x)]_\mathcal{B} = A[x]_\mathcal{B} \quad \text{and} \quad [L(x)]_{\mathcal{B}'} = A'[x]_{\mathcal{B}'}.$$

The question is as follows.

Question 4.1.6. How does the matrix representation of the same linear transformation change over change of basis? In other words, what is the relation between A and A'?

To answer this question, we shall first study how we change from one basis to another.

4.1.2 Change of Basis

Transition Matrix

We use the setting in the last section. Let

$$\mathbf{B} = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix} \quad \text{and} \quad \mathbf{B}' = \begin{bmatrix} v_1' & \cdots & v_n' \end{bmatrix}$$

As $\mathcal{B}$ and $\mathcal{B}'$ are both bases of V, then there must exist $n \times n$ matrices S_1 and S_2 such that

$$\mathbf{B} = \mathbf{B}' S_1 \quad \text{and} \quad \mathbf{B}' = \mathbf{B} S_2.$$

Therefore $\mathbf{B}' = \mathbf{B}' S_1 S_2$. By theorem 4.1.2, $S_1 S_2 = I$. Thus S_1 and S_2 are invertible and inverses of each other. Let $\underset{\mathcal{B}' \leftarrow \mathcal{B}}{S} = S_1 = S_2^{-1}$. Then

$$\mathbf{B} = \mathbf{B}' \underset{\mathcal{B}' \leftarrow \mathcal{B}}{S}. \tag{4.1.6}$$

This matrix $\underset{\mathcal{B}' \leftarrow \mathcal{B}}{S}$ is called the *transition matrix* from the basis $\mathcal{B}$ to the basis $\mathcal{B}'$ (because we are representing $\mathcal{B}$ in terms of $\mathcal{B}'$). It is immediate that $\underset{\mathcal{B}' \leftarrow \mathcal{B}}{S}^{-1} = \underset{\mathcal{B} \leftarrow \mathcal{B}'}{S}$.

If $x \in V$, then $x = \mathbf{B}[x]_{\mathcal{B}} = \mathbf{B}' \underset{\mathcal{B}' \leftarrow \mathcal{B}}{S} [x]_{\mathcal{B}}$. As $x = \mathbf{B}'[x]_{\mathcal{B}'}$, then by theorem 4.1.2,

$$\underset{\mathcal{B}' \leftarrow \mathcal{B}}{S} [x]_{\mathcal{B}} = [x]_{\mathcal{B}'}. \tag{4.1.7}$$

We see that $\underset{\mathcal{B}' \leftarrow \mathcal{B}}{S}$ is also a matrix with dual properties:

$$\mathbf{B} = \mathbf{B}' \underset{\mathcal{B}' \leftarrow \mathcal{B}}{S} \quad \text{and} \quad \underset{\mathcal{B}' \leftarrow \mathcal{B}}{S} [x]_{\mathcal{B}} = [x]_{\mathcal{B}'}. \tag{4.1.8}$$

Either of the equations above characterizes $\underset{\mathcal{B}' \leftarrow \mathcal{B}}{S}$.

Another Perspective

We can actually view the change of basis as a linear transformation. When we change from one basis to another, we are just representing the same vector in terms of different bases. So we can view this process as mapping a vector in V represented in terms of the basis $\mathcal{B}$ to itself represented in terms of $\mathcal{B}'$. The linear transformation involved is the identity mapping $\mathcal{I} : V \mapsto V$. The matrix representing $\mathcal{I}$ is the transition matrix $\underset{\mathcal{B}' \leftarrow \mathcal{B}}{S}$. In this case, the graph (4.1.5) becomes

$$\begin{array}{ccc} V & \xrightarrow{\mathcal{I}} & V \\ \downarrow{\scriptstyle \varphi_{\mathcal{B}}} & & \downarrow{\scriptstyle \varphi_{\mathcal{B}'}} \\ \mathbb{R}^n & \xrightarrow[\underset{\mathcal{B}' \leftarrow \mathcal{B}}{S}]{} & \mathbb{R}^n \end{array} \tag{4.1.9}$$

The first equation in (4.1.3) becomes $\mathcal{I}(\mathbf{B}) = \mathbf{B}' \underset{\mathcal{B}' \leftarrow \mathcal{B}}{S}$, which is just $\mathbf{B} = \mathbf{B}' \underset{\mathcal{B}' \leftarrow \mathcal{B}}{S}$, the first equation in (4.1.8).

4.1.3 Similarity Transformation

We can now answer our previous question: How does the matrix representation of the same linear transformation change over change of basis?

Suppose $\mathcal{B} = \{v_1, ..., v_n\}$ and $\mathcal{B}' = \{v'_1, ..., v'_n\}$ are two bases of a vector space V. Let L be a linear mapping from V to itself. Then there exist $n \times n$ matrices A and A' such that for every $x \in V$,

$$[L(x)]_{\mathcal{B}} = A[x]_{\mathcal{B}} \quad \text{and} \quad [L(x)]_{\mathcal{B}'} = A'[x]_{\mathcal{B}'} \tag{4.1.10}$$

Let $\underset{\mathcal{B}\leftarrow\mathcal{B}'}{S}$ be the transition matrix from $\mathcal{B}'$ to $\mathcal{B}$. Then

$$\underset{\mathcal{B}\leftarrow\mathcal{B}'}{S}[L(x)]_{\mathcal{B}'} = [L(x)]_{\mathcal{B}} \quad \text{and} \quad \underset{\mathcal{B}\leftarrow\mathcal{B}'}{S}[x]_{\mathcal{B}'} = [x]_{\mathcal{B}}. \tag{4.1.11}$$

Substituting (4.1.11) into (4.1.10), we obtain

$$\underset{\mathcal{B}\leftarrow\mathcal{B}'}{S}[L(x)]_{\mathcal{B}'} = A \underset{\mathcal{B}\leftarrow\mathcal{B}'}{S}[x]_{\mathcal{B}'}.$$

Thus

$$[L(x)]_{\mathcal{B}'} = \left(\underset{\mathcal{B}\leftarrow\mathcal{B}'}{S^{-1}} A \underset{\mathcal{B}\leftarrow\mathcal{B}'}{S}\right)[x]_{\mathcal{B}'}.$$

Since $[L(x)]_{\mathcal{B}'} = A'[x]_{\mathcal{B}'}$, then

$$A'[x]_{\mathcal{B}'} = \left(\underset{\mathcal{B}\leftarrow\mathcal{B}'}{S^{-1}} A \underset{\mathcal{B}\leftarrow\mathcal{B}'}{S}\right)[x]_{\mathcal{B}'}.$$

As this is true for every $x \in V$ and thus every $[x]_{\mathcal{B}'} \in \mathbb{R}^n$, we conclude that

$$A' = \underset{\mathcal{B}\leftarrow\mathcal{B}'}{S^{-1}} A \underset{\mathcal{B}\leftarrow\mathcal{B}'}{S}. \tag{4.1.12}$$

This gives us the following theorem.

Theorem 4.1.7. *Suppose $\mathcal{B} = \{v_1, ..., v_n\}$ and $\mathcal{B}' = \{v'_1, ..., v'_n\}$ are two ordered bases of a vector space V. Let L be a linear mapping from V to itself. Let S be the transition matrix from $\mathcal{B}'$ to $\mathcal{B}$. If A is the matrix representing L with respect to $\mathcal{B}$ and A' is the matrix representing L with respect to $\mathcal{B}'$, then $A' = S^{-1}AS$.*

We use a graph to better illustrate the idea.

$$\begin{array}{cccccc}
 & & \overset{\varphi_{\mathcal{B}}}{\frown} & & \overset{\varphi_{\mathcal{B}}}{\frown} & \\
\mathcal{B}: & V & \xrightarrow{L} & V & \mathbb{R}^n \xrightarrow{A} & \mathbb{R}^n \\
 & \mathcal{I}\uparrow & & \downarrow\mathcal{I} & S\uparrow & \downarrow S^{-1} \\
\mathcal{B}': & V & \xrightarrow{L} & V & \mathbb{R}^n \xrightarrow{A'} & \mathbb{R}^n \\
 & & \underset{\varphi_{\mathcal{B}'}}{\smile} & & \underset{\varphi_{\mathcal{B}'}}{\smile} &
\end{array} \tag{4.1.13}$$

where

S represents $\mathcal{I} : V_{(\mathcal{B}')} \mapsto V_{(\mathcal{B})}$, $\quad A$ represents $L : V_{(\mathcal{B})} \mapsto V_{(\mathcal{B})}$,

S^{-1} represents $\mathcal{I} : V_{(\mathcal{B})} \mapsto V_{(\mathcal{B}')}$, $\quad$ and $\quad A'$ represents $L : V_{(\mathcal{B}')} \mapsto V_{(\mathcal{B}')}$.

The mapping $L : V_{(\mathcal{B}')} \mapsto V_{(\mathcal{B}')}$ can be expressed as

$$(\mathcal{I} : V_{(\mathcal{B})} \mapsto V_{(\mathcal{B}')}) \circ (L : V_{(\mathcal{B})} \mapsto V_{(\mathcal{B})}) \circ (\mathcal{I} : V_{(\mathcal{B}')} \mapsto V_{(\mathcal{B})}).$$

Therefore $A' = S^{-1}AS$.

Conversely, if S is an invertible matrix and A is the matrix representation of a linear transformation $L : V \mapsto V$ with respect to the basis $\mathcal{B}$, then $A' = S^{-1}AS$ is the matrix representation of L with respect to the basis $\mathcal{B}'$ defined by $\mathbf{B}' = \mathbf{B}S$.

Definition 4.1.8. Let A be an $n \times n$ matrix. The matrix $S^{-1}AS$, where S is an $n \times n$ invertible matrix, is called a *similarity transformation* of A. An $n \times n$ matrix B is said to be *similar* to A if there exists an invertible matrix S such that $B = S^{-1}AS$.

Remark. It can be verified that "A is similar to B" defines an equivalence relation.

Based on the discussion above, we see that two matrices are similar if and only if they represent the same linear operator. (A *linear operator* is a linear transformation that maps a vector space to itself.)

A More General Case

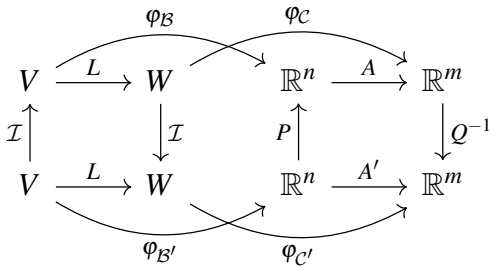
We can extend our discussion to a more general case.

Theorem 4.1.9. *Suppose that*

(i) $\mathcal{B}$ and $\mathcal{B}'$ are two ordered bases of a vector space V and P is the transition matrix from $\mathcal{B}'$ to $\mathcal{B}$.

(ii) $\mathcal{C}$ and $\mathcal{C}'$ are two ordered bases of a vector space W and Q is the transition matrix from $\mathcal{C}'$ to $\mathcal{C}$.

Suppose L is a linear transformation from V to W. If A is the matrix representing L with respect to $\mathcal{B}$ and $\mathcal{C}$ and A' is the matrix representing L with respect to $\mathcal{B}'$ and $\mathcal{C}'$, then $A' = Q^{-1}AP$.



(4.1.14)

4.1.4 Orthogonality

Orthogonality in Euclidean Spaces

In $\mathbb{R}^2$ and $\mathbb{R}^3$, to say that two vectors x and y are *orthogonal* is to say that their angle $\angle(x,y)$ is equal to $\frac{\pi}{2}$. Algebraically, suppose θ is the angle between x and the positive direction of the x-axis and φ the angle between y and the positive direction of the x-axis. Then $x = \|x\|(\cos\theta, \sin\theta)^T$ and $y = \|y\|(\cos\varphi, \sin\varphi)^T$. So $x^T y = \|x\|\|y\|\cos(\theta - \varphi)$. $x \perp y$ if and only if $\theta - \varphi = \pm\frac{\pi}{2}$. Thus $x \perp y$ if and only if $x^T y = 0$.

We extend the definition of orthogonality to $\mathbb{R}^n$. Two vectors x and y in $\mathbb{R}^n$ are said to be *orthogonal* if and only if $x^T y = 0$.

Orthogonality in General Vector Spaces

In a general vector space, $x^T y$ is no longer well-defined. To extend the concept of orthogonality to a general vector space, we need to introduce the notion of the so-called *inner product*, which is a generalization of $x^T y$. Two vectors are said to be orthogonal if their inner product is equal to zero.

The inner product of two vectors in a vector space is characterized by 3 axioms.

Definition 4.1.10. Let V be a vector space over $\mathbb{R}$. An *inner product* on V is a mapping from $V \times V$ to $\mathbb{R}$ which assigns each ordered pair $(x,y) \in V \times V$ a real number $\langle x,y\rangle$. Further, an inner product has to satisfy the following conditions.

(i) $\langle \alpha_1 x_1 + \alpha_2 x_2, y\rangle = \alpha_1 \langle x_1, y\rangle + \alpha_2 \langle x_2, y\rangle$ for all $x_1, x_2, y \in V$ and all $\alpha_1, \alpha_2 \in \mathbb{R}$.

(ii) $\langle x,y\rangle = \langle y,x\rangle$ for all $x, y \in V$.

(iii) $\langle x,x\rangle \geq 0$ for all $x \in V$; $\langle x,x\rangle = 0$ implies $x = 0$.

A vector space with an inner product is called an *inner product space*.

By axioms (i) and (ii), one can show that an inner product is bilinear.

Proposition 4.1.11. *An inner product on a vector space V over $\mathbb{R}$ is bilinear. That is,*

$$\langle x, \beta_1 y_1 + \beta_2 y_2\rangle = \beta_1 \langle x, y_1\rangle + \beta_2 \langle x, y_2\rangle$$

for all $x, y_1, y_2 \in V$ and all $\beta_1, \beta_2 \in \mathbb{R}$.

Proof. $\langle x, \beta_1 y_1 + \beta_2 y_2\rangle = \langle \beta_1 y_1 + \beta_2 y_2, x\rangle = \beta_1 \langle y_1, x\rangle + \beta_2 \langle y_2, x\rangle = \beta_1 \langle x, y_1\rangle + \beta_2 \langle x, y_2\rangle$. □

The reader may have noticed that there can be different inner products on the same vector space, as long as they satisfy the three axioms. For example, we can define the inner product on $\mathbb{R}^n$ as $\langle x,y\rangle = x^T D y$, where D is any diagonal matrix. It can be verified that this definition of inner product indeed satisfies the three axioms. As another example, a common definition of inner product on the space

$$\left\{ f \,\middle|\, f : [0,1] \to \mathbb{R},\ \int_0^1 [f(x)]^2\, dx < \infty \right\}$$

is

$$\langle f, g\rangle = \int_0^1 f(x) g(x)\, dx.$$

However, the formula

$$\langle f, g\rangle = \int_0^1 (x^2 + 1) f(x) g(x)\, dx.$$

also defines a legitimate inner product.

With the introduction of inner product, we can define orthogonality in a general vector space.

Definition 4.1.12. Let V be an inner product space. Two vectors $u, v \in V$ are said to be *orthogonal* if their inner product is equal to zero. In this case we write $u \perp v$.

Theorem 4.1.13. *Let V be an inner product space. If $u_1, ..., u_n \in V$ are mutually orthogonal and nonzero, then $\{u_1, ..., u_n\}$ is linearly independent.*

Proof. Suppose $c_1 u_1 + \cdots + c_n u_n = 0$. Taking inner product with u_i on both sides of the equation gives

$$\langle c_1 u_1 + \cdots + c_n u_n, u_i\rangle = \langle 0, u_i\rangle.$$

Thus $c_i \langle u_i, u_i\rangle = 0$. As u_i is nonzero, $\langle u_i, u_i\rangle > 0$. Hence $c_i = 0$. As this is true for every $i = 1, ..., n$, we conclude that $\{u_1, ..., u_n\}$ is linearly independent. □

With the introduction of inner product, we can extend the concept of the length of a vector.

Definition 4.1.14. Let V be an inner product space. The *norm* of a vector v in V, denoted by $\|v\|$, is defined as

$$\|v\| = \sqrt{\langle v, v\rangle}. \tag{4.1.15}$$

Theorem 4.1.15 (Cauchy-Schwarz Inequality)**.** *Let V be an inner product space. For all $x, y \in V$,*

$$\langle x, y\rangle^2 \leq \|x\|^2\|y\|^2. \tag{4.1.16}$$

Proof. Consider $f(t) = \langle x - ty, x - ty\rangle$, where $t \in \mathbb{R}$. As

$$0 \leq \langle x - ty, x - ty\rangle = \langle x, x\rangle + t^2\langle y, y\rangle - 2t\langle x, y\rangle = \|y\|^2t^2 - 2\langle x, y\rangle t + \|x\|^2,$$

for all $t \in \mathbb{R}$. Thus when $y \neq 0$,

$$f(t)_{\min} = f\left(\frac{\langle x, y\rangle}{\|y\|^2}\right) = \|y\|^2\left(\frac{\langle x, y\rangle}{\|y\|^2}\right)^2 - 2\langle x, y\rangle^2\left(\frac{\langle x, y\rangle}{\|y\|^2}\right) + \|x\|^2 = -\frac{\langle x, y\rangle^2}{\|y\|^2} + \|x\|^2 \geq 0.$$

Therefore $\langle x, y\rangle^2 \leq \|x\|^2\|y\|^2$. Equality holds if and only if $f(t) = 0$ for some $t \in \mathbb{R}$, i.e. $x = ty$ for some $t \in \mathbb{R}$.

When $y = 0$, both sides of (4.1.16) are equal to 0. We conclude that (4.1.16) holds for all $x, y \in V$. Equality holds if and only if $x = ty$ for some $t \in \mathbb{R}$ or $y = 0$. □

With Cauchy-Schwarz inequality, we see that for all nonzero vectors x and y in an inner product space, $\frac{\langle x, y\rangle}{\|x\| \cdot \|y\|}$ is between -1 and 1. We can therefore generalize the concept of the *angle* between two vectors.

Definition 4.1.16. Let V be an inner product space. The *angle* between two nonzero vectors x and y in V, which is denoted by $\angle(x, y)$ and whose range is $[0, \pi]$, is defined by

$$\cos\angle(x, y) = \frac{\langle x, y\rangle}{\|x\| \cdot \|y\|}. \tag{4.1.17}$$

or equivalently,

$$\angle(x, y) = \arccos\frac{\langle x, y\rangle}{\|x\| \cdot \|y\|}. \tag{4.1.18}$$

4.1.5 Orthogonal Subspaces

Definition 4.1.17. Let U and V be two subspaces of an inner product space W. Then inner product is well-defined on U and V. U and V are said to be *orthogonal* if and only if $u \perp v$ for all $u \in U$ and $v \in V$. If U and V are orthogonal, we write $U \perp V$.

Note that the concept of orthogonality of spaces is not always consistent with our intuitive idea of perpendicularity. For example, two perpendicular 2-dimensional plane in $\mathbb{R}^3$ are not orthogonal.

Proposition 4.1.18. *Suppose U and V are two subspaces of an inner product space. If $U \perp V$, then $U \cap V = \{0\}$.*

Proof. Suppose $x \in U \cap V$. Then $x \in U$ and $x \in V$. As $U \perp V$, then $\langle x, x\rangle = 0$. This implies $x = 0$. □

Theorem 4.1.19. *Suppose U and V are two subspaces of an inner product space. If $U \perp V$, then* $\dim(U + V) = \dim(U) + \dim(V)$.

Proof. Suppose $\{u_1,...,u_l\}$ is a basis of U and that $\{v_1,...,v_k\}$ is a basis of V. It suffices to show that $\{u_1,...,u_l,v_1,...,v_k\}$ is a basis for $U+V$.

Clearly, Span$\{u_1,...,u_l,v_1,...,v_k\} = U+V$. Now we show that $\{u_1,...,u_l,v_1,...,v_k\}$ is linearly independent. Suppose $\lambda_1u_1+\cdots+\lambda_lu_l+\mu_1v_1+\cdots+\mu_kv_k=0$. Then

$$U \ni \lambda_1u_1+\cdots+\lambda_lu_l = -\mu_1v_1-\cdots-\mu_kv_k \in V. \tag{4.1.19}$$

Therefore the vector in (4.1.19) is in $U\cap V=\{0\}$. Thus

$$\lambda_1u_1+\cdots+\lambda_lu_l=0 \quad \text{and} \quad -\mu_1v_1-\cdots-\mu_kv_k=0$$

Hence all λ_i's and μ_i's are equal to zero. So $\{u_1,...,u_l,v_1,...,v_k\}$ is linearly independent and thus forms a basis for $U+V$. We conclude that $\dim(U+V)=l+k=\dim(U)+\dim(V)$. □

Orthogonal Complement

Definition 4.1.20. Suppose U is a subspace of a finite-dimensional inner product space W. If there exists a subspace V of W such that $U\perp V$ and $\dim(W)=\dim(U)+\dim(V)$, then V is called an *orthogonal complement* of U in W and is denoted by $V=U^{\perp}$.

Theorem 4.1.21. *Suppose* $A\in\mathbb{R}^{m\times n}$. *Then* $\text{Row}(A)=(N(A))^{\perp}$ *in* $\mathbb{R}^n$ *and* $\text{Col}(A)=(N(A^T))^{\perp}$ *in* $\mathbb{R}^m$.

Proof. We only need to prove the first equality. To prove the second one, simply replace A by A^T. It is clear that $\text{Row}(A)$ and $N(A)$ are both subspaces of $\mathbb{R}^n$. By rank-nullity theorem, $\dim(\text{Row}(A))+\dim(N(A))=n$. It remains to show that $\text{Row}(A)\perp N(A)$. If $x\in\text{Row}(A)$, then $x=A^Tz$ for some $z\in\mathbb{R}^m$. If $y\in N(A)$, then $Ay=0$. Then $y^Tx=y^T(A^Tz)=(Ay)^Tz=0^Tz=0$. Thus $x\perp y$. Therefore $\text{Row}(A)\perp N(A)$. We conclude that $\text{Row}(A)=(N(A))^{\perp}$. □

■

Note. I have used different notations in the discussion of the matrix representation of a linear transformation (such as ${}_{\mathcal{B}\leftarrow\mathcal{B}'}S$ and $L(\mathbf{B})$) in order to make things clearer. (At least I hope so.) I have also added theorem 4.1.9 as a more general case.

Chengpiao Huang

4.2 Wednesday (Prof. Dongdong He)

In this lecture we will mainly focus on least squares problems.

4.2.1 Applications of the Fundamental Theorem of Matrix Subspaces

In the last lecture we proved the following theorem.

Theorem 4.2.1. *For $A \in \mathbb{R}^{m\times n}$, $\text{Row}(A) = (N(A))^{\perp}$ and $\text{Col}(A) = (N(A^T))^{\perp}$.*

We now give a few applications of this theorem.

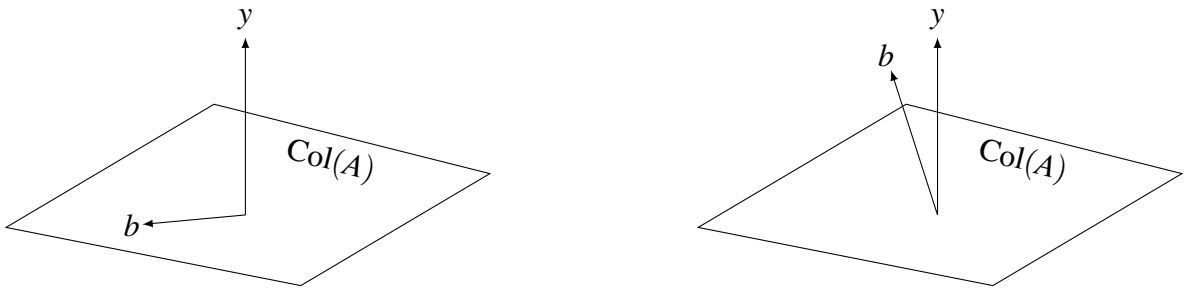
Theorem 4.2.2. *The matrix equation $Ax = b$ has a solution if and only if $y^TA = 0$ implies $y^Tb = 0$.*

This theorem is just saying that a linear system is consistent if and only if no matter how we combine the equations such that the left hand side gives zero, the right hand side always gives zero. An equivalent statement is as follows.

Theorem 4.2.3. *The matrix equation $Ax = b$ does not have a solution if and only if there exists y such that $y^TA = 0$ but $y^Tb \neq 0$.*

This is just theorem 1.1.1, the first theorem in this course. This is just to say that a linear system is inconsistent if and only if we can find a way to combine the equations such that the left hand side gives zero but the right hand side does not give zero.

We can interpret this theorem in terms of vector space.



$y^TA = 0$ if and only if $y \in N(A^T)$, or equivalently, $y \perp \text{Col}(A)$ by theorem 4.2.1. If $Ax = b$ has a solution, then b lies in $\text{Col}(A)$. Thus $y \perp b$, as is illustrated in the graph on the left. If $Ax = b$ does not have a solution, then b is not in $\text{Col}(A)$. Then b and y will not form a right angle, as is shown in the graph on the right. We prove this theorem formally.

Proof. $Ax = b$ has a solution if and only if $b \in \text{Col}(A) = (N(A^T))^{\perp}$, if and only if $y^Tb = 0$ for every $y \in N(A^T)$. □

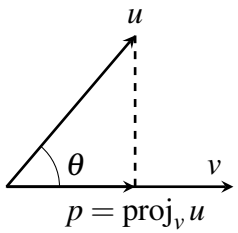
We now introduce an important theorem which will be used in solving least squares problems.

Theorem 4.2.4. *Let $A \in \mathbb{R}^{m\times n}$ and $b \in \mathbb{R}^m$. The equation $A^TAx = A^Tb$ always has a solution.*

Proof. We know $N(A^TA) = N(A)$. Thus $\text{Col}(A^TA) = N(A^TA)^{\perp} = N(A)^{\perp} = \text{Col}(A^T)$. For every $b \in \mathbb{R}^m$, $A^Tb \in \text{Col}(A^T) = \text{Col}(A^TA)$. Therefore there always exists $x \in \mathbb{R}^n$ such that $A^TAx = A^Tb$. □

4.2.2 Projection

Before we begin our discussion on least squares problems, we need one more tool, namely, projection.



Suppose we want to find the projection p of u onto v. Then

$$\|\operatorname{proj}_v u\| = \|u\|\cos\theta = \|u\| \cdot \frac{u^T v}{\|u\|\|v\|} = \frac{u^T v}{\|v\|}$$

and

$$\operatorname{proj}_v u = \|p\|\frac{v}{\|v\|} = \frac{u^T v}{\|v\|^2} v = v\frac{v^T u}{\|v\|^2} = \frac{vv^T}{v^T v} u.$$

We have obtained

$$\|\operatorname{proj}_v u\| = \frac{u^T v}{\|v\|} \quad \text{and} \quad \operatorname{proj}_v u = \frac{vv^T}{v^T v} u. \tag{4.2.1}$$

We see that applying the matrix $\dfrac{vv^T}{v^T v}$ to a vector u gives the projection of u into v.

4.2.3 Least Squares

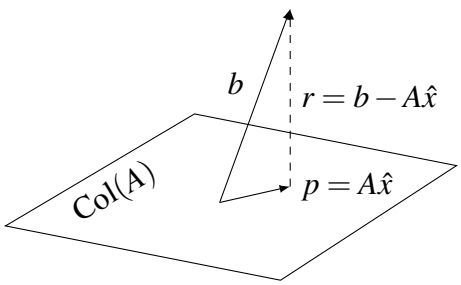
Suppose $A \in \mathbb{R}^{m\times n}$ is a tall matrix, i.e. $m > n$. Then the system $Ax = b$, where $b \in \mathbb{R}^m$, is overdetermined and does not have a solution in most cases. As a compromise, we look for an *approximate solution* $\hat{x}$ such that the residual $r = b - A\hat{x}$ is as small as possible. Our objective is thus

$$\min_{x \in \mathbb{R}^n} \quad \|Ax - b\|^2 \tag{4.2.2}$$

Such a problem is called a *least squares problem*. We now show that the optimal solution $\hat{x}$, which is called the *least squares solution*, must satisfy the equation $A^T Ax = A^T b$. We will present two approaches: first a geometric approach, then one via Calculus.

Geometric Approach

If $b \in \text{Col}(A)$, then $Ax = b$ has a solution. If $b \notin \text{Col}(A)$, then to find $\arg\min_x \|Ax - b\|^2$, we simply project b onto $\text{Col}(A)$. We illustrate with a graph.



We see that $\|r\|$ is the smallest when $r \perp \text{Col}(A)$. Therefore $\hat{r} = b - A\hat{x} \in N(A^T)$. Thus $A^T(b - A\hat{x}) = 0$. This gives

$$A^T A\hat{x} = A^T b. \tag{4.2.2}$$

The system of equations in (4.2.2) are called *normal equations*. By theorem 4.2.4, (4.2.2) must have a solution. If the columns of A are linearly independent, then $A^T A$ is invertible. In this case, we have a unique solution to the normal equations:

$$\hat{x} = (A^T A)^{-1} A^T b. \tag{4.2.3}$$

Projection Matrix

If the columns of A are linearly independent, then the projection of b onto $\text{Col}(A)$ is $p = A\hat{x} = A(A^T A)^{-1} A^T b$. Thus applying the *projection matrix*

$$P = A(A^T A)^{-1} A^T \tag{4.2.4}$$

to a vector b has the effect of projecting b onto $\text{Col}(A)$. The following result is immediate.

Proposition 4.2.5. *If $P = A(A^T A)^{-1} A^T$, then $P^2 = P$.*

This is intuitive because applying P once projects a vector onto $\text{Col}(A)$ and if we project a vector that is already in $\text{Col}(A)$ onto $\text{Col}(A)$, we get the vector itself.

Solution via Calculus

Let $f(x) = \|b - Ax\|^2$, $x \in \mathbb{R}^n$. We need to find $f_{\min}$. In Calculus, we know that $\hat{x}$ minimizes f only if $\nabla f(\hat{x}) = 0$. Since

$$f(x) = \|b - Ax\|^2 = (b - Ax)^T (b - Ax) = x^T A^T Ax - 2b^T Ax + \|b\|^2,$$

then

$$\nabla f(x) = 2A^T Ax - 2A^T b.$$

Setting $\nabla f(\hat{x}) = 0$ gives $A^T A\hat{x} = A^T b$.

Direct Verification

We now verify that if $\hat{x}$ satisfies the normal equations $A^T Ax = A^T b$, then $\hat{x}$ is indeed a least squares solution to $Ax = b$. For every $z \in \mathbb{R}^n$,

$$\begin{aligned}
\|Az - b\|^2 &= \|(Az - A\hat{x}) + (A\hat{x} - b)\|^2 \\
&= \|Az - A\hat{x}\|^2 + \|A\hat{x} - b\|^2 + 2(Az - A\hat{x})^T (A\hat{x} - b) \\
&= \|Az - A\hat{x}\|^2 + \|A\hat{x} - b\|^2 + 2(z - \hat{x})^T A^T (A\hat{x} - b) \\
&= \|Az - A\hat{x}\|^2 + \|A\hat{x} - b\|^2 \\
&\geq \|A\hat{x} - b\|^2.
\end{aligned}$$

Equality holds if and only if $A(z - \hat{x}) = 0$. If the columns of A are linearly independent, this condition is equivalent to $z = \hat{x}$. Thus the least squares solution to $Ax = b$ is unique when A has linearly independent columns.

4.2.4 Orthonormal Set and Orthogonal Matrix

Orthonormal Sets

Definition 4.2.6. A set $\{q_1,...,q_n\}$ of vectors in an inner product space V is said to be *orthonormal* if

$$\langle q_i, q_j\rangle = \delta_{ij} = \begin{cases} 0 & (i \neq j) \\ 1 & (i = j) \end{cases},$$

i.e. if the vectors are mutually orthogonal unit vectors. If $\{q_1,...,q_n\}$ is orthonormal and forms a basis of V, it is said to be a *orthonormal basis* of V.

Example 4.2.7.

(i) $\{e_1,...,e_n\}$ is an orthonormal basis for $\mathbb{R}^n$.

(ii) $\left\{\frac{1}{\sqrt{2}}(1,1)^T, \frac{1}{\sqrt{2}}(1,-1)^T\right\}$ is an orthonormal basis for $\mathbb{R}^2$.

(iii) $\left\{\frac{1}{\sqrt{2}}(1,1,0)^T, \frac{1}{\sqrt{2}}(1,-1,0)^T, (0,0,1)^T\right\}$ is an orthonormal basis for $\mathbb{R}^3$.

An advantage of using an orthonormal basis is that we can find the coordinate vector of a vector very easily. Suppose $\{q_1,...,q_n\}$ is an orthonormal basis for V. For every $x \in V$, suppose $x = x_1q_1 + \cdots + x_nq_n$, then

$$\langle x, q_i\rangle = \langle x_1q_1 + \cdots + x_nq_n, q_i\rangle = x_i.$$

Thus the coordinate vector of x is given by

$$\begin{bmatrix} \langle x, q_1\rangle \\ \vdots \\ \langle x, q_n\rangle \end{bmatrix} \tag{4.2.5}$$

Orthogonal Matrices

Definition 4.2.8. A matrix $Q \in \mathbb{R}^{n\times n}$ is said to be *orthogonal* if its columns are orthonormal.

Remark. There is no standard terminology for non-square matrices with orthonormal columns. We simply call them *matrices with orthonormal columns*.

We now consider the properties of matrices with orthonormal columns.

Theorem 4.2.9. *A matrix $Q \in \mathbb{R}^{m\times n}$ has orthonormal columns if and only if $Q^TQ = I$.*

Proof. Suppose $Q = \begin{bmatrix} q_1 & \cdots & q_n \end{bmatrix}$. Then

$$Q^TQ = \begin{bmatrix} q_1^T \\ \vdots \\ q_n^T \end{bmatrix} \begin{bmatrix} q_1 & \cdots & q_n \end{bmatrix} = \begin{bmatrix} q_1^Tq_1 & \cdots & q_1^Tq_n \\ \vdots & \ddots & \vdots \\ q_n^Tq_1 & \cdots & q_n^Tq_n \end{bmatrix} \tag{4.2.6}$$

By definition, the columns of Q are orthonormal if and only if $q_i^Tq_j = \delta_{ij}$, if and only if $Q^TQ = I$. □

Theorem 4.2.10. *A matrix $Q \in \mathbb{R}^{m\times n}$ has orthonormal columns if and only if $\langle Qx, Qy\rangle = \langle x, y\rangle$ for all $x, y \in \mathbb{R}^n$.*

Proof. Suppose the columns of Q are orthonormal. Then $Q^TQ = I$. Therefore for all $x, y \in \mathbb{R}^n$, $\langle Qx, Qy\rangle = \langle Q^TQx, y\rangle = \langle x, y\rangle$.

Now suppose $\langle Qx, Qy\rangle = \langle x, y\rangle$ for all $x, y \in \mathbb{R}^n$. Then $(Qx)^T(Qy) = x^TQ^TQy = x^Ty$. Letting $x = e_i$ and $y = e_j$ picks out the (i, j) entry of Q^TQ. Therefore

$$q_{ij} = e_i^Te_j = \begin{cases} 1 & (i = j) \\ 0 & (i \neq j) \end{cases}.$$

Thus $Q^TQ = I$. Therefore Q has orthonormal columns. □

Theorem 4.2.11. *A matrix $Q \in \mathbb{R}^{m\times n}$ has orthonormal columns if and only if it is norm-preserving, i.e. $\|Qx\| = \|x\|$ for all $x \in \mathbb{R}^n$.*

Proof. Suppose $Q \in \mathbb{R}^{m\times n}$ has orthonormal columns. Letting $x = y$ in theorem 4.2.10, we obtain $\|Qx\|^2 = \langle Qx, Qx\rangle = \langle x, x\rangle = \|x\|^2$. Thus $\|Qx\| = \|x\|$ for all $x \in \mathbb{R}^n$.

Suppose $\|Qx\| = \|x\|$ for all $x \in \mathbb{R}^n$. Then for all $x, y \in \mathbb{R}^n$,

$$(Qx)^T(Qy) = \frac{1}{4}(\|Q(x+y)\|^2 - \|Q(x-y)\|^2) = \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2) = x^Ty.$$

It follows from theorem 4.2.10 that Q has orthonormal columns. □

We summarize our results below.

Theorem 4.2.12. *Let $Q \in \mathbb{R}^{m\times n}$. The following statements are equivalent.*

(i) The columns of Q form an orthonormal set.

(ii) $Q^TQ = I$.

(iii) $\langle Qx, Qy\rangle = \langle x, y\rangle$ for all $x, y \in \mathbb{R}^n$.

(iv) $\|Qx\| = \|x\|$ for all $x \in \mathbb{R}^n$.

Corollary 4.2.13. *If Q is an $m \times n$ matrix with orthonormal columns, then applying Q to any $x, y \in \mathbb{R}^n$ preserves the angle of x and y, i.e.*

$$\angle(Qx, Qy) = \angle(x, y) \text{ for all } x, y \in \mathbb{R}^n.$$

Remark. We remind the reader that a matrix $Q \in \mathbb{R}^{m\times n}$ having orthonormal columns does not imply $QQ^T = I$ in general. In fact, if Q is non-square and has orthonormal columns ($m > n$), then Q^T, which is $n \times m$, cannot be orthogonal because an orthonormal set in $\mathbb{R}^n$ can contain at most n vectors. Then QQ^T can *never* be equal to I.

Now we consider orthogonal matrices. Suppose Q is orthogonal, then $Q^TQ = I$. As Q is square, $Q^TQ = I$ if and only if $QQ^T = I$. We have two conclusions here. The first one is that Q is invertible and that $Q^{-1} = Q^T$. The second one is $(Q^T)^TQ^T = I$. So Q^T is also orthogonal. We summarize our results below.

Theorem 4.2.14. *Let $Q \in \mathbb{R}^{n\times n}$. The following statements are equivalent.*

(i) Q is orthogonal.

(ii) $Q^TQ = I$.

(iii) $QQ^T = I$.

(iv) $Q^{-1} = Q^T$.

(v) Q^T is orthogonal.

(vi) $\langle Qx, Qy\rangle = \langle x, y\rangle$ for all $x, y \in \mathbb{R}^n$.

(vii) $\|Qx\| = \|x\|$ for all $x \in \mathbb{R}^n$.

4.2.5 Gram-Schmidt Algorithm

The objective of *Gram-Schmidt algorithm* is to find an orthonormal basis from a given set of vectors. We first consider the case where the given set of vectors is linearly independent.

Question 4.2.15. Given a linearly independent set of vectors $\{a_1,...,a_n\}$, can we find an orthonormal basis $\{q_1,...,q_n\}$ for Span$\{a_1,...,a_n\}$?

We briefly describe the idea of this algorithm. We start by normalizing a vector to get the first vector in the orthonormal basis. After obtaining $q_1,...,q_{k-1}$, to obtain q_k, we subtract from a_k its projection onto the span of the previous vectors $q_1,...,q_{k-1}$. The resultant vector $\widetilde{q}_k$ will be orthogonal to the previous $(k-1)$ vectors. Then we normalize $\widetilde{q}_k$ to get q_k. Repeating this process for $k=2,...,n$ yields an orthonormal basis $\{q_1,...,q_n\}$ for Span$\{a_1,...,a_n\}$. We give the algorithm below.

We are given a linearly independent set of vectors $\{a_1,...,a_n\}$. Let

$$\widetilde{q}_1 = a_1 \qquad \text{and} \qquad q_1 = \frac{\widetilde{q}_1}{\|\widetilde{q}_1\|} \tag{4.2.7}$$

For each $k=2,...,n$, after obtaining $q_1,...,q_{k-1}$, let

$$\widetilde{q}_k = a_k - \langle a_k, q_1\rangle q_1 - \cdots - \langle a_k, q_{k-1}\rangle q_{k-1} \quad \text{and} \quad q_k = \frac{\widetilde{q}_k}{\|\widetilde{q}_k\|}. \tag{4.2.8}$$

Then $\{q_1,...,q_n\}$ is an orthonormal basis for Span$\{a_1,...,a_n\}$.

Verification. We can verify that for each $k=2,...,n$, q_k is orthogonal to q_i for all $i=1,...,k-1$. As

$$\langle \widetilde{q}_k, q_i\rangle = \langle a_k - \langle a_k, q_1\rangle q_1 - \cdots - \langle a_k, q_{k-1}\rangle q_{k-1}, q_i\rangle = \langle a_k, q_i\rangle - \langle a_k, q_i\rangle = 0,$$

we conclude that $\widetilde{q}_k \perp q_i$, $i=1,...,k-1$. Thus $q_k \perp q_i$, $i=1,...,k-1$.

■

Note. In the "Least Squares" section, I have added the "Direct Verification" subsection to show that $\hat{x}$ is a least squares solution if and only if it satisfies the normal equations (i.e. the normal equations provide a sufficient and necessary condition for the least squares solution).

In the "Orthogonal Matrices" section, the original lecture focused on the properties of orthogonal matrices. I have included a more general case in the notes, namely, matrices with orthonormal columns. Also, only the "only if" parts of theorem 4.2.10 and 4.2.11 were discussed in the original lecture. Here I have included both the "if" and the "only if" parts.

Chengpiao Huang

Chapter 5

Week Five

5.1 Monday (Prof. Tom Luo)

5.1.1 Complex Inner Product

Recap

We first do a little review.

Inner Product. An inner product on a vector space over $\mathbb{R}$ is defined as a mapping $\langle\ ,\ \rangle$ from $V \times V$ to $\mathbb{R}$ characterized by three axioms. We say $u \perp v$ if and only if $\langle u, v\rangle = 0$.

Orthonormal Basis and Orthogonal Matrix. A set of vectors $\{q_1, ..., q_n\}$ is said to be orthonormal if $q_i \perp q_j$ when $i \neq j$ and $\|q_i\| = 1$ for all i. If $q_i \in \mathbb{R}^n$, then the matrix $Q = \begin{bmatrix} q_1 & \cdots & q_n \end{bmatrix}$ is called an orthogonal matrix. $\{q_1, ..., q_n\}$ is orthonormal if and only if $Q^T Q = I$. In this case, $Q^{-1} = Q^T$.

Inner Product Spaces over the Complex Field

In $\mathbb{C}^n$, the inner product of two vectors $x = (x_1, ..., x_n)^T$ and $y = (y_1, ..., y_n)^T$ is defined as

$$\langle x, y\rangle = \sum_{i=1}^{n} x_i \overline{y_i} = \bar{y}^T x \tag{5.1.1}$$

where $\overline{y_i}$ denotes the complex conjugate of y_i and $\bar{y} = (\overline{y_1}, ..., \overline{y_n})^T$ (notational convention: when we take the complex conjugate of a matrix, we take the complex conjugate of each entry). We need to take complex conjugate because the norm of a vector might be negative or even a complex number otherwise, which is not consistent with our intuitive idea. If we use the definition in (5.1.1), the norm of a vector in $\mathbb{C}^n$ will always be a non-negative real number:

$$\|x\| = \sqrt{\langle x, x\rangle} = \sqrt{\sum_{i=1}^{n} \overline{x_i} x_i} = \sqrt{\sum_{i=1}^{n} |x_i|^2}. \tag{5.1.2}$$

As a notational convenience, we write z^H for $\bar{z}^T$. Thus $\langle x, y\rangle = y^H x$ and $\|x\| = \sqrt{x^H x}$. When the entries of z are all real numbers, z^H is just z^T. Also note that $\langle y, x\rangle = \overline{\langle x, y\rangle}$ for all $x, y \in \mathbb{C}^n$.

Proposition 5.1.1. *Let $A, B, C \in \mathbb{C}^{m\times n}$. Let $E \in \mathbb{C}^{m\times n}$ and $F \in \mathbb{C}^{n\times l}$. Let $x \in \mathbb{C}^n$. Then*

(i) $(A+B)^H = A^H + B^H$.

(ii) $(C^H)^H = C$.

(iii) $(EF)^H = F^H E^H$.

(iv) $x^H x = \sum_{i=1}^{n} |x_i|^2$.

We can define inner product on a general vector space over $\mathbb{C}$.

Definition 5.1.2. Let V be a vector space over $\mathbb{C}$. An *inner product* on V is a mapping from $V \times V$ to $\mathbb{C}$ that assigns each ordered pair $(x,y) \in V \times V$ a complex number $\langle x,y \rangle$ such that

(i) $\langle \alpha_1 x_1 + \alpha_2 x_2, y \rangle = \alpha_1 \langle x_1, y \rangle + \alpha_2 \langle x_2, y \rangle$ for all $x_1, x_2, y \in V$ and all $\alpha_1, \alpha_2 \in \mathbb{C}$.

(ii) $\langle x,y \rangle = \overline{\langle y,x \rangle}$ for all $x,y \in V$.

(iii) $\langle x,x \rangle \geq 0$ for all $x \in V$; $\langle x,x \rangle = 0$ implies $x = 0$.

Proposition 5.1.3. *Let V be an inner product space over $\mathbb{C}$. For all $x, y_1, y_2 \in V$ and all $\alpha_1, \alpha_2 \in \mathbb{C}$,*

$$\langle x, \alpha_1 y_1 + \alpha_2 y_2 \rangle = \overline{\alpha_1} \langle x, y_1 \rangle + \overline{\alpha_2} \langle x, y_2 \rangle.$$

5.1.2 Orthonormal Sets and Unitary Matrices

In $\mathbb{C}^n$, we keep our original definition of orthogonality, only with the inner product changed.

Definition 5.1.4. A set $\{q_1, ..., q_n\}$ of vectors in $\mathbb{C}^n$ is said to be *orthonormal* if

$$\langle q_i, q_j \rangle = q_j^H q_i = \begin{cases} 0 & (i \neq j) \\ 1 & (i = j) \end{cases}.$$

In $\mathbb{C}^{n \times n}$, the analogue of an orthogonal matrix in $\mathbb{R}^{n \times n}$ is called a *unitary* matrix.

Definition 5.1.5. A matrix $U \in \mathbb{C}^{n \times n}$ is called *unitary* if its columns form an orthonormal basis in $\mathbb{C}^n$.

The following theorem is the complex version of theorem 4.2.14.

Theorem 5.1.6. *Suppose $U \in \mathbb{C}^{n \times n}$. The following statements are equivalent.*

(i) *U is unitary.*

(ii) $U^H U = I$.

(iii) $UU^H = I$.

(iv) $U^{-1} = U^H$.

(v) *U^H is unitary.*

(vi) $\langle Ux, Uy \rangle = \langle x,y \rangle$ *for all* $x,y \in \mathbb{C}^n$.

(vii) $\|Ux\| = \|x\|$ *for all* $x \in \mathbb{C}^n$.

We are interested in the following two properties of an orthogonal or unitary matrix. The first one is that it is norm-preserving, which is property (vii). The other one is that if v is in the span of an orthonormal basis $\{q_1, ..., q_n\}$, then we can write v as $v = \sum_{i=1}^{n} x_i q_i = Qx$, and thus the coordinate vector of v is

$$x = Q^H v = \begin{bmatrix} q_1^H v \\ \vdots \\ q_n^H v \end{bmatrix} = \begin{bmatrix} \langle v, q_1 \rangle \\ \vdots \\ \langle v, q_n \rangle \end{bmatrix} = \begin{bmatrix} \| \operatorname{proj}_{q_1} v \| \\ \vdots \\ \| \operatorname{proj}_{q_2} v \| \end{bmatrix}.$$

We see that the matrix Q^H defines a coordinate mapping that preserves norm:

$$\begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \xrightarrow{Q^H} Q^H v = \begin{bmatrix} \langle v, q_1 \rangle \\ \vdots \\ \langle v, q_n \rangle \end{bmatrix}$$

and

$$\|v\|^2 = \sum_{i=1}^{n} |\langle v, q_i \rangle|^2 .$$

If v is not in the span of $\{q_1, ..., q_n\}$, then we would still get some component left of v after we project v onto $\text{Span}\{q_1, ..., q_n\}$. Thus more generally, for every vector v, $\|v\|^2 \geq \sum_{i=1}^{n} |\langle v, q_i \rangle|^2$. However, if $v \in \text{Span}\{q_1, ..., q_n\}$, then the equality must hold. This gives the following theorem.

Theorem 5.1.7. *Let $\{q_1, ..., q_m\}$ be an orthonormal set in $\mathbb{C}^n$. For every $v \in \mathbb{C}^n$,*

$$\|v\|^2 \geq \sum_{i=1}^{m} |\langle v, q_i \rangle|^2. \tag{5.1.3}$$

Equality holds if and only if $v \in \text{Span}\{q_1, ..., q_m\}$.

The theorem also holds in $\mathbb{R}^n$.

Parseval's Theorem / Rayleigh's Identity

In Fourier analysis, we have a theorem analogous to theorem 5.1.7, namely, *Perseval's theorem* (also known as *Rayleigh's identity*).

Theorem 5.1.8 (Parseval's Theorem)**.** *Let $L^2[-\pi, \pi]$ be the space of square-integrable functions on $[-\pi, \pi]$ with inner product*

$$\langle f, g \rangle = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \overline{g(x)}\, dx. \tag{5.1.4}$$

Suppose $f \in L^2[-\pi, \pi]$ has Fourier series

$$\frac{a_0}{\sqrt{2}} + \sum_{k=1}^{\infty} [a_k \cos(kx) + b_k \sin(kx)], \tag{5.1.5}$$

then

$$\|f\|^2 = |a_0|^2 + \sum_{k=1}^{\infty} (|a_k|^2 + |b_k|^2). \tag{5.1.6}$$

Analogous to theorem 5.1.7, for a general function f with Fourier series (5.1.5), we would expect $\|f\|^2$ to be greater than or equal to the sum of the squares of its Fourier coefficients. Equality holds if and only if f lies in the span of $\{\frac{1}{\sqrt{2}}, \cos(x), \sin(x), \cos(2x), \sin(2x), ...\}$. Thus Parseval's theorem tells us that every function in $L^2[-\pi, \pi]$ is in the span of $\{\frac{1}{\sqrt{2}}, \cos(x), \sin(x), \cos(2x), \sin(2x), ...\}$. In other words, the Fourier series of every function in $L^2[-\pi, \pi]$ must converge.

■

Note. I have added some properties of the Hermitian operator and also some properties of unitary matrices.

Chengpiao Huang

5.2 Wednesday (Prof. Tom Luo)

5.2.1 QR Factorization via Gram-Schmidt Algorithm

Here we drop the condition that the given set of vectors is linearly independent and consider a more general version of Gram-Schmidt algorithm.

Given a set $\{a_1,...,a_n\}$ of vectors, we need to find an orthonormal basis $\{q_{k_1},...,q_{k_r}\}$ for Span$\{a_1,...,a_n\}$. We do the following. For each $j=1,...,n$, suppose we have obtained an orthonormal set $\{q_{k_1},...,q_{k_{r-1}}\}$ from $\{a_1,...,a_{j-1}\}$, let

$$\widetilde{q}_{k_r} = a_j - \langle a_j, q_{k_1}\rangle q_{k_1} - \cdots - \langle a_j, q_{k_{r-1}}\rangle q_{k_{r-1}}. \qquad \text{(Orthogonalization)}$$

If $\widetilde{q}_{k_r} = 0$, skip $\widetilde{q}_{k_r}$. Otherwise let

$$q_{k_r} = \frac{\widetilde{q}_{k_r}}{\|\widetilde{q}_{k_r}\|}. \qquad \text{(Normalization)}$$

This yields an orthonormal basis $\{q_{k_1},...,q_{k_m}\}$ for Span$\{a_1,...,a_n\}$.

If $\{a_1,...,a_n\}$ is linearly independent, then each $\widetilde{q}_k$ is nonzero and we would obtain an orthonormal basis $\{q_1,...,q_n\}$ which consists of n vectors. In $\mathbb{R}^n$, we can use this basis to form an $n\times n$ orthogonal matrix

$$Q = \begin{bmatrix} q_1 & \cdots & q_n \end{bmatrix}. \qquad (5.2.1)$$

If we let

$$A = \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix}, \qquad (5.2.2)$$

then we obtain

$$\underbrace{\begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix}}_{A:\ n\times n} = \underbrace{\begin{bmatrix} q_1 & \cdots & q_n \end{bmatrix}}_{Q:\ n\times n} \underbrace{\begin{bmatrix} \|a_1\| & \langle a_2,q_1\rangle & \langle a_3,q_1\rangle & \cdots & \langle a_n,q_1\rangle \\ 0 & \|\widetilde{q}_2\| & \langle a_3,q_2\rangle & \cdots & \langle a_n,q_2\rangle \\ 0 & 0 & \|\widetilde{q}_3\| & \cdots & \langle a_n,q_3\rangle \\ 0 & 0 & 0 & \cdots & \langle a_n,q_4\rangle \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \|\widetilde{q}_n\| \end{bmatrix}}_{R:\ n\times n} \qquad (5.2.3)$$

where Q is orthogonal and R is upper triangular. This is called the *QR factorization* of A. More generally, A can be $m\times n$, then Q will be $m\times n$ and R will be $n\times n$, provided that the columns of A are linearly independent.

Complexity Analysis

We now calculate the complexity of Gram-Schmidt algorithm. Suppose A is $m\times n$ and has linearly independent columns. We analyze the complexity of finding q_k $(k=2,3,4,...)$. Obtaining each $\langle a_k,q_i\rangle q_i$ requires $(2m-1)$ flops for inner product and m flops for scalar multiplication, thus $(3m-1)$ flops in total. We need to find $(k-1)$ such vectors, so finding all $\langle a_k,q_i\rangle q_i$'s require $(3m-1)(k-1)$ flops. Subtracting all $\langle a_k,q_i\rangle q_i$'s from a_j requires $(k-1)m$ flops. Thus getting $\widetilde{q}_k$ (orthogonalization) requires $(4m-1)(k-1)$ flops. Normalization requires $3m$ flops. Thus getting each q_k requires around $4mk$ flops. Thus obtaining Q requires around

$$3m + \sum_{k=2}^{n} 4mk \sim 2mn^2 \text{ flops}$$

are required. As there is no extra calculation needed for obtaining Q, the overall complexity of finding the QR factorization of a matrix via Gram-Schmidt algorithm is about $2mn^2$ flops. If the matrix is square, i.e. if A is $n\times n$, then the complexity is about $2n^3$ flops.

Solving Linear Systems via QR Factorization

QR factorization provides us with a new way to solve a square linear system $Ax = b$. We first find the QR factorization of A. Then $Ax = b$ becomes $QRx = b$. We can therefore break down the process of solving $Ax = b$ into two stages:

1. Solve $Qy = b$ using the formula $y = Q^T b$ (because Q is orthogonal).

2. Solve $Rx = y$ using back-substitution.

The first step takes about $2n^2$ flops and the second step takes about n^2 flops. Thus the two steps take $3n^2$ flops in total.

Advantage of QR Compared to LU

QR factorization is generally more preferred than LU factorization. An important reason is that the matrix Q, which has orthonormal columns, is norm-preserving. Thus normally we would not expect large numbers to appear in R, in $y = Q^T b$, and thus in the whole process. As a matter of fact, QR factorization is regarded as one of the most important factorizations of a matrix.

5.2.2 Determinants

We are about to move to our next big topic, namely, the theory of eigenvalues and eigenvectors, where we will see two more matrix decompositions called eigen-decomposition and singular value decomposition, respectively. However, to prepare ourselves for this topic, we shall first study the theory of *determinants*, a classical but no longer central topic in linear algebra.

Mathematicians in the Old Days

Before computer was invented, mathematicians in the old days considered a problem solved only after they had found an explicit closed-form solution. As we shall see later, determinants do provide explicit formulas for $A^{-1}b$ and A^{-1}.

Characterization of the Determinant

We shall start by giving major properties of the determinant instead of giving a formal definition directly. The determinant of a matrix A is a scalar denoted by $\det(A)$ or $|A|$.

Property 1. The determinant of a scalar is the scalar itself. That is, if a is a scalar, then $\det(a) = a$.

Property 2. The determinant of a 2×2 matrix is given by $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$.

Property 3 (Defining Property). The determinant of the identity matrix is equal to 1, i.e. $\det(I) = 1$.

Property 4 (Defining Property). A row exchange changes the sign of the determinant of a matrix, but not its absolute value.

Property 5 (Defining Property). The determinant of a matrix depends linearly on the first row of a matrix. For example,

$$\begin{vmatrix} a+a' & b+b' \\ c & d \end{vmatrix} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} + \begin{vmatrix} a' & b' \\ c & d \end{vmatrix} \quad \text{and} \quad \begin{vmatrix} ta & tb \\ c & d \end{vmatrix} = t \begin{vmatrix} a & b \\ c & d \end{vmatrix}.$$

By property 4, the determinant of a matrix depends linearly on *any* row of a matrix. Note, however, that this does not imply $|A+B| = |A| + |B|$ or $|tA| = t|A|$. If A is $n \times n$, then $|tA| = t^n|A|$.

In fact, these three defining properties *completely* and *uniquely* determines the determinant of any matrix. (Later we will derive an explicit formula for the determinant of a matrix using only these three properties.)

Property 6. If 2 rows of a matrix A are identical, then $\det(A) = 0$.

Proof. Let A' be the matrix obtained by exchanging the two identical rows of A. Then $A' = A$ and thus $\det(A') = \det(A)$. By property 3, $\det(A') = -\det(A)$. Hence $\det(A) = 0$. □

Property 7. Adding a constant multiple of one row to another does not change the determinant of a matrix.

Proof. By property 4, we only have to prove that adding a constant multiple of row k ($k \geq 2$) to the first row does not change the determinant of a matrix. By linearity,

$$\begin{vmatrix} \vec{a}_1 + c\vec{a}_k \\ \vdots \\ \vec{a}_k \\ \vdots \end{vmatrix} = \begin{vmatrix} \vec{a}_1 \\ \vdots \\ \vec{a}_k \\ \vdots \end{vmatrix} + c \begin{vmatrix} \vec{a}_k \\ \vdots \\ \vec{a}_k \\ \vdots \end{vmatrix} = \begin{vmatrix} \vec{a}_1 \\ \vdots \\ \vec{a}_k \\ \vdots \end{vmatrix} + 0 = \begin{vmatrix} \vec{a}_1 \\ \vdots \\ \vec{a}_k \\ \vdots \end{vmatrix}$$

where the second equality is by property 6. □

So far we have completely determined how the three types of elementary row operations can affect the determinant of a matrix: exchanging two rows changes the sign of the determinant, multiplying a row by a constant has the effect of scaling the determinant by the same constant, and adding a constant multiple of one row to another does not change the determinant. Thus if we reduce a matrix to its row echelon form U using only row operations of type (i) and type (iii), then $|\det(A)| = |\det(U)|$.

Property 8. If A has a row consisting entirely of zeros, then $\det(A) = 0$.

Proof. Adding a multiple of a row to the zero row does not change the determinant of A. As the resultant matrix has two identical rows, $\det(A) = 0$. □

Property 9. If A is a triangular matrix, then $\det(A) = \prod_{i=1}^{n} a_{ii}$, that is, the determinant of A is equal to the product of its diagonal entries.

Proof. We prove the statement only for lower triangular matrices. We apply only row operations of type (iii) to reduce A to its row echelon form:

$$A = \begin{bmatrix} a_{11} & & 0 \\ & \ddots & \\ * & & a_{nn} \end{bmatrix} \to \begin{bmatrix} a_{11} & & 0 \\ & \ddots & \\ 0 & & a_{nn} \end{bmatrix} = U.$$

By property 7, $\det(A) = \det(U)$. If some a_{ii} is zero, then U will have a row consisting entirely of zeros, then $\det(U) = 0$ and the equality automatically holds. If U has full pivots, then by linearity, $\det(U) = \left(\prod_{i=1}^{n} a_{ii}\right) \det(I) = \prod_{i=1}^{n} a_{ii}$. Thus $\det(A) = \prod_{i=1}^{n} a_{ii}$. □

Property 10. A matrix A is singular if and only if $\det(A) = 0$.

Proof. We can reduce A to its row echelon form U using only elementary row operations (i) and (iii). A is singular if and only if U has a zero row, if and only if $\det(U) = 0$. Since $|\det(A)| = |\det(U)|$, then A is singular if and only if $\det(A) = 0$. □

Equivalently, A is non-singular if and only if $\det(A) \neq 0$. As a corollary, $\det(A) = \pm \prod_{i=1}^{n}$ pivots.

Property 11 (Product Rule). For all square matrices A and B of the same size, $\det(AB) = \det(A)\det(B)$.

Proof. If either of A and B is singular, then both sides will be 0 and the equality automatically holds. Thus without loss of generality, we can assume that B is non-singular. So $\det(B) \neq 0$. To show that $\det(AB) = \det(A)\det(B)$, it suffices to show that $\mathrm{d}(A) = \frac{\det(AB)}{\det(B)}$ and $\det(A)$ are the same. To do so, we only need to verify that $\mathrm{d}(A)$ satisfies the three defining properties.

For property 3, $\mathrm{d}(I) = \frac{\det(B)}{\det(B)} = 1$. For property 4, suppose A' is the matrix obtained by exchanging the ith and jth rows of A. Then $A'B$ is equal to AB with the ith and jth rows exchanged. Therefore

$$\mathrm{d}(A') = \frac{\det(A'B)}{\det(B)} = \frac{-\det(AB)}{\det(B)} = -\mathrm{d}(A).$$

For property 5,

$$\mathrm{d}\left(\begin{bmatrix} \alpha\vec{a}_1 + \beta\vec{b}_1 \\ \vdots \end{bmatrix}\right) = \frac{\det\left(\begin{bmatrix} \alpha\vec{a}_1 + \beta\vec{b}_1 \\ \vdots \end{bmatrix} B\right)}{\det(B)} = \frac{\det\left(\begin{bmatrix} \alpha\vec{a}_1 B + \beta\vec{b}_1 B \\ \vdots \end{bmatrix}\right)}{\det(B)}$$

$$= \frac{\alpha\det\left(\begin{bmatrix} \vec{a}_1 \\ \vdots \end{bmatrix} B\right) + \beta\det\left(\begin{bmatrix} \vec{b}_1 \\ \vdots \end{bmatrix} B\right)}{\det(B)} = \alpha\,\mathrm{d}\left(\begin{bmatrix} \vec{a}_1 \\ \vdots \end{bmatrix}\right) + \beta\,\mathrm{d}\left(\begin{bmatrix} \vec{b}_1 \\ \vdots \end{bmatrix}\right).$$

Therefore $\mathrm{d}(A)$ satisfies properties 3, 4, and 5. Hence $\mathrm{d}(A) = \frac{\det(AB)}{\det(B)} = \det(A)$. Therefore $\det(AB) = \det(A)\det(B)$. □

Property 12. For every square matrix A, $\det(A^T) = \det(A)$.

Proof. If A is singular, then A^T is also singular. Then the equality holds.

Suppose A is nonsingular. We consider the LDU decomposition of A: $PA = LDU$. Thus $A^T P^T = U^T D^T L^T = U^T D L^T$. By product rule, $\det(A)\det(P) = \det(D) \neq 0$ and $\det(A^T)\det(P^T) = \det(D) \neq 0$. It suffices to show that for every permutation matrix P, $\det(P) = \det(P^T)$.

As P is norm-preserving, then $P^T P = I$. Taking determinants on both sides yields

$$\det(P)\det(P^T) = 1.$$

As P is obtained by permuting the rows of I, $|\det(P)| = |\det(I)| = 1$. So either $\det(P) = \det(P^T) = 1$ or $\det(P) = \det(P^T) = -1$. This completes the proof. □

By property 12, the "column" versions of properties 4, 5, 6, 7, and 8 are also true.

Formula for 2×2 Determinants

We now derive the formula for 2×2 determinants.

$$\begin{aligned}
\begin{vmatrix} a & b \\ c & d \end{vmatrix} &= \begin{vmatrix} a & 0 \\ c & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & d \end{vmatrix} \\
&= \begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix} + \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} + \begin{vmatrix} 0 & b \\ c & 0 \end{vmatrix} + \begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix} \\
&= \begin{vmatrix} a & 0 \\ c & 0 \end{vmatrix} + \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix} - \begin{vmatrix} c & 0 \\ 0 & b \end{vmatrix} + \begin{vmatrix} 0 & b \\ 0 & d \end{vmatrix} \\
&= 0 + ad - bc + 0 \\
&= ad - bc.
\end{aligned}$$

■

Note. I have added the complexity analysis of the QR factorization of a matrix via Gram-Schmidt algorithm.

Chengpiao Huang

5.3 Friday (Prof. Tom Luo)

Last lecture we introduced the three defining properties that completely and uniquely define the determinant:

(i) $\det(I) = 1$.

(ii) A row exchange changes the sign of the determinant.

(iii) The determinant of a matrix is linear in the first row of the matrix.

These three properties yield useful results, such as $\det(AB) = \det(A)\det(B)$ and $\det(A^T) = \det(A)$.

5.3.1 Usages of the Determinant

We now give some usages of determinants.

Judging Invertibility. A is invertible if and only if $\det(A) \neq 0$.

Geometric Meaning. $|\det(A)|$ is equal to the volume of the parallelepiped determined by the rows of A. To verify that this is true, we only have to check the three defining properties.

Product of Pivots. The determinant of a nonsingular matrix is equal to the product of the pivots of its row echelon form, i.e. if A is nonsingular, then $\det(A) = \pm\prod_{i=1}^{n}$ pivots.

Finding Inverses. We will soon see that each entry of the inverse of a matrix can be expressed as the ratio of the determinants of some matrices. This gives a closed-form expression for every entry of the inverse.

We now give another important property of the determinant.

Theorem 5.3.1. *If A is invertible, then* $\det(A^{-1}) = \dfrac{1}{\det(A)}$.

Sketch of Proof. $AA^{-1} = I$. Take determinants on both sides and apply product rule. □

5.3.2 Closed-Form Expression for the Determinant

We now derive the closed-form expression for the determinant of a matrix. Last lecture we have found that

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc. \tag{5.3.1}$$

To find the determinant of a 3×3 matrix, we apply the same method. We expand the determinant row by row as the sum of determinants such that each one of them has at most one non-zero entry in each row. For example, the first step is

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = \begin{vmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} 0 & a_{12} & 0 \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} + \begin{vmatrix} 0 & 0 & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}.$$

We can keep expanding the determinants, but eventually we will end up with $3^3 = 27$ determinants. To simplify the calculation, note that only the determinants with *exactly (or at most) one non-zero*

entry in each row and each column are of interest, as the others either have a zero row or a zero column and are thus equal to 0. Each of the determinants of interest corresponds to a permutation matrix. Therefore we only have to consider $3! = 6$ determinants:

$$\begin{aligned}
\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} &= \begin{vmatrix} a_{11} & & \\ & a_{22} & \\ & & a_{33} \end{vmatrix} + \begin{vmatrix} & a_{12} & \\ & & a_{23} \\ a_{31} & & \end{vmatrix} + \begin{vmatrix} & & a_{13} \\ a_{21} & & \\ & a_{32} & \end{vmatrix} \\
&+ \begin{vmatrix} a_{11} & & \\ & & a_{23} \\ & a_{32} & \end{vmatrix} + \begin{vmatrix} & a_{12} & \\ a_{21} & & \\ & & a_{33} \end{vmatrix} + \begin{vmatrix} & & a_{13} \\ & a_{22} & \\ a_{31} & & \end{vmatrix} \\
&= a_{11}a_{22}a_{33}\overbrace{\begin{vmatrix} 1 & & \\ & 1 & \\ & & 1 \end{vmatrix}}^{1} + a_{12}a_{23}a_{31}\overbrace{\begin{vmatrix} & 1 & \\ & & 1 \\ 1 & & \end{vmatrix}}^{1} + a_{13}a_{21}a_{32}\overbrace{\begin{vmatrix} & & 1 \\ 1 & & \\ & 1 & \end{vmatrix}}^{1} \\
&+ a_{11}a_{23}a_{32}\underbrace{\begin{vmatrix} 1 & & \\ & & 1 \\ & 1 & \end{vmatrix}}_{-1} + a_{12}a_{21}a_{33}\underbrace{\begin{vmatrix} & 1 & \\ 1 & & \\ & & 1 \end{vmatrix}}_{-1} + a_{13}a_{22}a_{31}\underbrace{\begin{vmatrix} & & 1 \\ & 1 & \\ 1 & & \end{vmatrix}}_{-1} \\
&= \sum_{(\alpha,\beta,\gamma)\in S_3} (-1)^{\sigma(\alpha,\beta,\gamma)} a_{1\alpha}a_{2\beta}a_{3\gamma}
\end{aligned}$$

where S_3 is the set of all the permutations of $\{1,2,3\}$ and $\sigma(\alpha,\beta,\gamma)$ is the number of pairwise permutations needed to restore (α,β,γ) to $(1,2,3)$, or equivalently, the number of column exchanges required to restore a permutation matrix to the identity.

We can generalize this formula to the general case. Suppose A is $n \times n$, then

$$\det(A) = \sum_{(j_1,\ldots,j_n)\in S_n} (-1)^{\sigma(j_1,\ldots,j_n)} a_{1,j_1} \cdots a_{n,j_n} \tag{5.3.2}$$

where S_n is the set of all the permutations of $\{1,\ldots,n\}$ and $\sigma(j_1,\ldots,j_n)$ is the number of pairwise permutations required to restore $(j_1,\ldots,j_n)$ to $(1,\ldots,n)$, which is equal to the number of column exchanges required to restore a permutation matrix to the identity matrix.

The formula (5.3.2) does give a closed-form expression for $\det(A)$. However, $n!$ terms are involved in the summation and it is practically impossible to calculate the determinant using this formula (the complexity is $O(n!)$, which is of higher order than any exponential). This is one important reason why the theory of determinants is no longer central in linear algebra. In practice, in order to calculate $\det(A)$, we first reduce A to its row echelon form, then directly apply the formula $\det(A) = \pm\prod_{i=1}^{n}$ pivots. The complexity of this method is only $O(n^3)$.

5.3.3 Permutations

In this section we examine permutations more detailedly.

Definition 5.3.2. Let p be a permutation of $\{1,\ldots,n\}$. The number $\tau(p)$ is defined as the number of pairs (i,j) in p such that $i > j$. If $\tau(p)$ is even, then p is called an *even permutation*. If $\tau(p)$ is odd, then p is called an *odd* permutation.

Definition 5.3.3. Let p be a permutation of $\{1,\ldots,n\}$. The number $\sigma(p)$ is defined as the number of pairwise permutations required to restore p to $(1,\ldots,n)$.

Now we will see how different operations affect the parity of a permutation (namely, the fact of a permutation being odd or even).

Swapping a Pair

As we will see below, swapping any pair will always change the parity of a permutation.

Theorem 5.3.4. *Swapping a pair of neighboring numbers in a permutation changes the parity of the permutation.*

Proof. If we swap the pair (i, j) in $p = (..., i, j, ...)$, then only the order of the pair (i, j) is reversed and the other pairs are left unchanged. Thus if we let p' be the resultant permutation, then

$$|\tau(p) - \tau(p')| = 1.$$

□

Theorem 5.3.5. *Swapping any one pair of numbers in a permutation changes the parity of the permutation.*

Proof. Suppose we swap the pair (i, j) in $p = (..., i, k_1, ..., k_t, j, ...)$. This is equivalent to doing a series of neighbor swaps:

$$(i, k_1), ..., (i, k_t), (i, j), (k_t, j), ..., (k_1, j).$$

There are $(2t + 1)$ neighbor swaps in total. By theorem 5.3.4, $\tau(p')$ is equal to the sum of $\tau(p)$ and an odd number.

Another way of viewing this is that $(2k+1)$ pairs are reversed: $(i, k_1), ..., (i, k_i), (i, j), (k_t, j), ..., (k_1, j)$. Thus $\tau(p')$ is equal to $\tau(p)$ plus an odd number. □

Deleting a Number

Things are more complicated if we delete a number in a permutation.

We first consider deleting the first number in a permutation. Suppose $p = (j, \beta, \gamma, ...)$ is a permutation of $\{1, ..., n\}$ and that $p' = (\beta, \gamma, ...)$ is obtained by deleting the first number in p. As all but $(j - 1)$ numbers behind j in p are larger than j, we conclude that $\tau(p) = \tau(p') + (j - 1)$.

Theorem 5.3.6. *Suppose $p = (j, \beta, \gamma, ...)$ is a permutation of $\{1, ..., n\}$ and that $p' = (\beta, \gamma, ...)$ is obtained by deleting the first number in p. Then*

$$\tau(p) = \tau(p') + (j - 1).$$

We now consider deleting the kth number in a permutation. Suppose $p = (j_1, ..., j_{k-1}, \gamma, \delta, ...)$ is a permutation of $\{1, ..., n\}$ and that $p' = (j_1, ..., j_{k-1}, \delta, ...)$ is obtained by deleting the kth number γ in p. Let $p'' = (\gamma, j_1, ..., j_{k-1}, \delta, ...)$, which is obtained by performing $(k - 1)$ neighbor swaps on p. Then by theorem 5.3.4,

$$(-1)^{\tau(p)} = (-1)^{\tau(p'')+(k-1)}.$$

As $\tau(p'') = \tau(p') + (\gamma - 1)$, then

$$(-1)^{\tau(p'')} = (-1)^{\tau(p')+(\gamma-1)}.$$

Hence

$$(-1)^{\tau(p)} = (-1)^{\tau(p')+\gamma+k}. \tag{5.3.3}$$

We conclude that p and p' have the same parity if and only if γ and k have the same parity

Relation between σ and τ

The reader may have noticed that for a permutation p, $\sigma(p)$ can take infinitely many values while $\tau(p)$ is precisely defined and is unique. We now examine the relation between $\sigma(p)$ and $\tau(p)$.

We know that $\sigma(1,...,n)$ is an even number. Now consider p, a permutation of $\{1,...,n\}$. By theorem 5.3.3, each pairwise permutation changes the parity of $\tau(p)$. If $\tau(p)$ is odd, then we need an odd number of pairwise permutations to restore p to $(1,...,n)$. If $\tau(p)$ is even, then we need an even number of pairwise permutations to restore p to $(1,...,n)$. Thus, we have proved that for every permutation p of $(1,...,n)$, the values of $\sigma(p)$ must have the same parity and that $\sigma(p)$ and $\tau(p)$ have the same parity. Hence

$$(-1)^{\tau(p)} = (-1)^{\sigma(p)}. \tag{5.3.4}$$

5.3.4 Cofactor Expansion of the Determinant

We now find another formula for the determinant of a matrix, namely, the *cofactor expansion*. We will derive this formula based on the formula (5.3.2). Let S_n be the set of all the permutations of $\{1,...,n\}$. By (5.3.4),

$$\det(A) = \sum_{p\in S_n}\left[(-1)^{\sigma(p)}\prod_{k=1}^{n} a_{k,j_k}\right] = \sum_{p\in S_n}\left[(-1)^{\tau(p)}\prod_{k=1}^{n} a_{k,j_k}\right]$$

where $p=(j_1,...,j_n)$. By deleting the ith entry j_i of p, we obtain a permutation $p'=(j_1,...,j_{i-1},j_{i+1},...,j_n)$ of $A_{j_i}=\{1,...,n\}\backslash\{j_i\}$. By (5.3.3),

$$(-1)^{\tau(p)} = (-1)^{\tau(p')+i+j_i} = (-1)^{\tau(p')}(-1)^{i+j_i}. \tag{5.3.5}$$

Thus

$$\begin{aligned}
\det(A) &= \sum_{p\in S_n}\left[(-1)^{\tau(p')}(-1)^{i+j_i}\prod_{k=1}^{n} a_{k,j_k}\right] \\
&= \sum_{p\in S_n}\left\{(-1)^{i+j_i}a_{i,j_i}\left[(-1)^{\tau(p')}\prod_{k=1,k\neq i}^{n} a_{k,j_k}\right]\right\} \\
&= \sum_{j_i=1}^{n}\sum_{p'\in S(A_{j_i})}\left\{(-1)^{i+j_i}a_{i,j_i}\left[(-1)^{\tau(p')}\prod_{k=1,k\neq i}^{n} a_{k,j_k}\right]\right\} \\
&= \sum_{j_i=1}^{n}\left\{(-1)^{i+j_i}a_{i,j_i}\left[\sum_{p'\in S(A_{j_i})}(-1)^{\tau(p')}\prod_{k=1,k\neq i}^{n} a_{k,j_k}\right]\right\} \\
&= \sum_{j=1}^{n}(-1)^{i+j}a_{ij}\det(M_{ij}).
\end{aligned}$$

where $S(A_{j_i})$ is the set of all the permutations over the set A_{j_i} and M_{ij} is the matrix obtained by deleting the ith row and jth column of A. The last step is by noting that the summation inside is just $\det(M_{ij})$. We have now obtained a recursive formula for the determinant of a matrix.

Definition 5.3.7. Let A be a square matrix.

(i) The (i,j) *minor* of A, denoted by M_{ij}, is defined as the matrix obtained by deleting the ith row and jth column of A.

(ii) The (i,j) *cofactor* of A, denoted by C_{ij}, is a scalar defined by $C_{ij}=(-1)^{i+j}\det(M_{ij})$.

Theorem 5.3.8. *Suppose A is an $n \times n$ matrix. Let C_{ij} denote its (i,j) cofactor. For every $m = 1, \ldots, n$,*

$$\det(A) = \sum_{j=1}^{n} a_{mj} C_{mj}. \tag{5.3.6}$$

5.3.5 Solving for Matrix Inverse via Determinants

We begin with an interesting observation. We know that $\det(A) = \sum_{j=1}^{n} a_{1j} C_{1j}$. What is the value of $\sum_{j=1}^{n} a_{2j} C_{1j}$? The reader may have noticed that it is the cofactor expansion (along the first row) of the determinant of the matrix

$$B = \begin{bmatrix} \vec{a}_2 \\ \vec{a}_2 \\ \vdots \\ \vec{a}_n \end{bmatrix},$$

which is obtained by replacing the first row of A by its second row. However, as B has two identical rows, $\det(B) = 0$. Thus $\sum_{j=1}^{n} a_{2j} C_{1j} = 0$.

As a generalization,

$$\underbrace{\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} C_{11} & \cdots & C_{n1} \\ \vdots & \ddots & \vdots \\ C_{1n} & \cdots & C_{nn} \end{bmatrix}}_{C^T} = \det(A) I. \tag{5.3.7}$$

Therefore if $\det(A) \neq 0$, then A is invertible and

$$A^{-1} = \frac{C^T}{\det(A)}. \tag{5.3.8}$$

The (i,j) entry of A^{-1} is thus

$$(A^{-1})_{ij} = \frac{C_{ji}}{\det(A)} = (-1)^{i+j} \frac{\det(M_{ji})}{\det(A)}. \tag{5.3.9}$$

5.3.6 Crammer's Rule

We now give an explicit formula for the solution to $Ax = b$, where A is a square matrix. Suppose

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix},$$

then

$$\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 & 0 & & \\ x_2 & 1 & & \\ \vdots & & \ddots & \\ x_n & & 0 & 1 \end{bmatrix} = \begin{bmatrix} b & a_2 & \cdots & a_n \end{bmatrix}. \tag{5.3.10}$$

Taking determinants on both sides of (5.3.9) yields

$$\det(A) x_1 = \det(B_1),$$

where B_1 is the matrix obtained by replacing the first column of A by b. Therefore, if we let B_i be the matrix obtained by replacing the ith column of A by b, then for each $i = 1, ..., n$,

$$x_i = \frac{\det(B_i)}{\det(A)}. \tag{5.3.11}$$

We have obtained a closed-form solution to $Ax = b$. This method is known as *Crammer's rule*.

■

Chapter 6

Week Six

6.1 Monday (Prof. Tom Luo)

6.1.1 Vandermonde Matrix

Recall that the determinant of a matrix A is a multi-linear function of the entries of A and can be expanded across any row or column:

$$\det(A) = \sum_{j=1}^{n} a_{mj} C_{mj},$$

where C_{mj} is the (m, j) cofactor of A.

As an example of how determinants can be applied, we consider an important matrix called the *Vandermonde matrix*. The 2×2 and 3×3 Vandermonde matrices are

$$V_2 = \begin{bmatrix} 1 & 1 \\ \lambda_1 & \lambda_2 \end{bmatrix}, \quad \text{and} \quad V_3 = \begin{bmatrix} 1 & 1 & 1 \\ \lambda_1 & \lambda_2 & \lambda_3 \\ \lambda_1^2 & \lambda_2^2 & \lambda_3^2 \end{bmatrix}, \quad \text{respectively.}$$

Thus $\det(V_2) = \lambda_2 - \lambda_1$. The general formula for the $n \times n$ Vandermonde matrix is given by

$$V_n = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \lambda_1 & \lambda_2 & \cdots & \lambda_n \\ \lambda_1^2 & \lambda_2^2 & \cdots & \lambda_n^2 \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_1^{n-1} & \lambda_2^{n-1} & \cdots & \lambda_n^{n-1} \end{bmatrix}. \tag{6.1.1}$$

The Vandermonde matrix has important applications in polynomial interpolation. Suppose we are given n data points $(\lambda_1, f(\lambda_1)), ..., (\lambda_n, f(\lambda_n))$ and we would like to fit the data with the polynomial $f(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_{n-1} x^{n-1}$. This gives us the following linear system:

$$\begin{cases} f(\lambda_1) = a_0 + a_1 \lambda_1 + \cdots + a_{n-1} \lambda_1^{n-1} \\ f(\lambda_2) = a_0 + a_1 \lambda_2 + \cdots + a_{n-1} \lambda_2^{n-1} \\ \quad \vdots \\ f(\lambda_n) = a_0 + a_1 \lambda_n + \cdots + a_{n-1} \lambda_n^{n-1} \end{cases}. \tag{6.1.2}$$

We can rewrite (6.1.2) in matrix form:

$$\underbrace{\begin{bmatrix} 1 & \lambda_1 & \lambda_1^2 & \cdots & \lambda_1^{n-1} \\ 1 & \lambda_2 & \lambda_2^2 & \cdots & \lambda_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \lambda_n & \lambda_n^2 & \cdots & \lambda_n^{n-1} \end{bmatrix}}_{V_n^T} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ \vdots \\ a_{n-1} \end{bmatrix} = \begin{bmatrix} f(\lambda_1) \\ f(\lambda_2) \\ \vdots \\ f(\lambda_n) \end{bmatrix}. \tag{6.1.3}$$

A sufficient condition for (6.1.3) to be solvable is that V^T is invertible, which is equivalent to $\det(V_n^T) = \det(V_n) \neq 0$. A necessary condition for $\det(V_n) \neq 0$ is that all the λ_i's are distinct. What is more interesting is that this is also a sufficient condition. In fact, we have the following theorem.

Theorem 6.1.1. *Let V_n be the $n \times n$ Vandermonde matrix ($n \geq 2$). Then*

$$\det(V_n) = \prod_{1 \leq i < j \leq n} (\lambda_j - \lambda_i). \tag{6.1.4}$$

Proof. We prove by induction. When $n = 2$, (6.1.4) clearly holds. Suppose when $n = k-1$ ($k = 3, 4, 5, ...$), (6.1.4) holds. When $n = k$, consider the function

$$g(x) = \begin{vmatrix} 1 & 1 & \cdots & 1 & 1 \\ \lambda_1 & \lambda_2 & \cdots & \lambda_{k-1} & x \\ \lambda_1^2 & \lambda_2^2 & \cdots & \lambda_{k-1}^2 & x^2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \lambda_1^{k-2} & \lambda_2^{k-2} & \cdots & \lambda_{k-1}^{k-2} & x^{k-2} \\ \lambda_1^{k-1} & \lambda_2^{k-1} & \cdots & \lambda_{k-1}^{k-1} & x^{k-1} \end{vmatrix}.$$

Expanding $g(x)$ along the last column, we obtain $g(x) = \det(V_{k-1})x^{k-1} + c_{k-2}x^{k-2} + \cdots + c_0$. So $g(x)$ is of degree $(k-1)$ and has $(k-1)$ roots. As $\lambda_1, ..., \lambda_{k-1}$ are $(k-1)$ roots of $g(x)$, then $g(x) = \det(V_{k-1})(x - \lambda_1) \cdots (x - \lambda_{k-1})$ (Note that this will hold whether or not the λ_i's are all distinct). Therefore

$$\begin{aligned} \det(V_k) &= g(\lambda_k) \\ &= \det(V_{k-1})(\lambda_k - \lambda_1) \cdots (\lambda_k - \lambda_{k-1}) \\ &= \left[\prod_{1 \leq i < j \leq k-1} (\lambda_j - \lambda_i) \right] (\lambda_k - \lambda_1) \cdots (\lambda_k - \lambda_{k-1}) \\ &= \prod_{1 \leq i < j \leq k} (\lambda_j - \lambda_i). \end{aligned}$$

This completes the proof. □

Corollary 6.1.2. *The $n \times n$ Vandermonde matrix ($n \geq 2$) is invertible if and only if the λ_i's are distinct.*

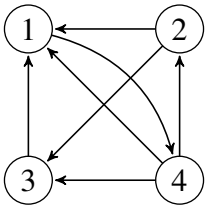
6.1.2 PageRank Algorithm

We shall now enter the topic of eigenvectors and eigenvalues. We give the definitions below.

Definition 6.1.3. Suppose A is a square matrix. If $Ax = \lambda x$ for some scalar λ and some vector $x \neq 0$, then λ is called an *eigenvalue* of A and x is called an *eigenvector of A corresponding to the eigenvalue λ*. The set of all the eigenvalues of A is denoted by $\text{eig}(A)$.

As an example to motivate the study of eigenvectors and eigenvalues, we introduce the *PageRank algorithm*, the first web search algorithm developed by Larry Page, one of the co-founders of Google. This algorithm provides a way to measure the relative importance of web pages.

We can model a web by a *web graph*, which consists of vertices and edges. The vertices represent web pages and the edges represent hyperlinks. As an example, consider the following graph:



We can measure the relative importance of a page by the weighted average of the importance of the pages pointing to it. Let R_i be the relative importance of page i. Then it is reasonable to let

$$R_1 = \frac{R_2}{2} + \frac{R_3}{1} + \frac{R_4}{3}$$

where the weight of each R_i is set to be the reciprocal of the number of edges pointing from R_i, based on the assumption that each edge pointing from R_i transfers an equal part of the importance of R_i. This gives us the following dynamical system:

$$\underbrace{\begin{bmatrix} R_1^{(t+1)} \\ R_2^{(t+1)} \\ R_3^{(t+1)} \\ R_4^{(t+1)} \end{bmatrix}}_{R^{(t+1)}} = \underbrace{\begin{bmatrix} 0 & \frac{1}{2} & 1 & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{3} \\ 0 & \frac{1}{2} & 0 & \frac{1}{3} \\ 1 & 0 & 0 & 0 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} R_1^{(t)} \\ R_2^{(t)} \\ R_3^{(t)} \\ R_4^{(t)} \end{bmatrix}}_{R^{(t)}} \tag{6.1.5}$$

where t is the iteration index. Thus $R^{(t+1)} = AR^{(t)}$. Note that each column of A corresponds to a page and that the entries in each column of A sum up to 1. Also, the entries of A are non-negative. A matrix with these two properties is called a *stochastic matrix*. We can initialize R, which represents a probability distribution, as

$$R^{(0)} = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right)^T.$$

Note that $\mathbf{1}^T R^{(0)} = 1$, where $\mathbf{1} = (1,1,1,1)^T$. We sum up the entries of $R^{(t+1)}$:

$$\mathbf{1}^T R^{(t+1)} = \mathbf{1}^T (AR^{(t)}) = (\mathbf{1}^T A) R^{(t)} = \mathbf{1}^T R^{(t)} = \cdots = 1$$

because the entries in each column of A sum up to 1. Thus $R^{(t+1)}$ still represents a probability distribution.

Now suppose $R^{(t)}$ converges to R^* as $t \to \infty$ (it in fact does, regardless of the initial distribution, as long as A is a stochastic matrix), then taking limits on both sides of $R^{(t+1)} = AR^{(t)}$ gives

$$R^* = AR^*.$$

Therefore R^* is an eigenvector of A corresponding to the eigenvalue 1. Thus to rank the pages, we only need to find R^*.

There are two ways to find R^*. The first one is to perform iterations using the formula $R^{(t+1)} = AR^{(t)}$ until $R^{(t+1)}$ is stable. This comes down to computing matrix power. The second one is to directly solve the system $(I - A)R^* = 0$. However, this system is a large-scale system, which is difficult to solve in general. As for the first one, since we do not need to get a precise result, normally we just have to do 4 or 5 iterations, which involve only matrix multiplications, which are easier to do. Therefore in practice we use the first method to find R^*.

6.1.3 Eigenvectors and Eigenvalues (I)

We now go back to our discussion of eigenvectors and eigenvalues. The key formula is $Ax = \lambda x$, $x \neq 0$. There are several questions we can ask: *When does a matrix have eigenvectors and eigenvalues? If it does, how many eigenvectors and eigenvalues does it have? How do we find them? What are the implications and applications?*

We begin with an example.

Example 6.1.4. Suppose

$$A = \begin{bmatrix} 4 & -2 \\ 1 & 1 \end{bmatrix} \quad \text{and} \quad x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

As

$$Ax = \begin{bmatrix} 6 \\ 3 \end{bmatrix} = 3x,$$

then 3 is an eigenvalue of A with $x = (2,1)^T$ as a corresponding eigenvector.

Proposition 6.1.5. *If x is an eigenvector of A corresponding to an eigenvalue λ, then for every $\alpha \neq 0$, αx is also an eigenvector of A corresponding to the eigenvalue λ.*

Proof. For all $\alpha \neq 0$, $A(\alpha x) = \alpha(Ax) = \alpha(\lambda x) = \lambda(\alpha x)$. □

By proposition 6.1.5, we see that an eigenvector gives a *direction* such that applying A does not change this direction.

Characteristic Polynomial

By definition, λ is an eigenvalue of A if and only if $Ax = \lambda x$ for some $x \neq 0$, if and only if $(\lambda I - A)x = 0$ for some $x \neq 0$, if and only if $\det(\lambda I - A) = 0$. The determinant $\det(\lambda I - A)$ is called the *characteristic polynomial* of A and is denoted by $p_A(\lambda)$. Thus

$$p_A(\lambda) = \det(\lambda I - A) = \begin{bmatrix} \lambda - a_{11} & -a_{12} & \cdots & -a_{1n} \\ -a_{21} & \lambda - a_{22} & \cdots & -a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ -a_{n1} & -a_{n2} & \cdots & \lambda - a_{nn} \end{bmatrix} = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_0. \quad (6.1.6)$$

By letting $\lambda = 0$, we obtain the constant term of $p_A(\lambda)$, which is $\det(-A) = (-1)^n \det(A)$.

We now find another expression for $p_A(\lambda)$. As $p_A(\lambda)$ is of degree n, it has exactly n roots $\lambda_1, ..., \lambda_n$ (counting multiplicity). Thus we can write

$$p_A(\lambda) = (\lambda - \lambda_1) \cdots (\lambda - \lambda_n). \quad (6.1.7)$$

These n roots $\lambda_1, ..., \lambda_n$ are exactly the eigenvalues of A (some of them may be equal). So an $n \times n$ matrix A has n eigenvalues (some of which may be equal). By (6.1.7), the constant term of $p_A(\lambda)$ is $(-1)^n \prod_{i=1}^n \lambda_i$. Comparing the constant terms of (6.1.6) and (6.1.7), we conclude that

$$\prod_{i=1}^{n} \lambda_i = \det(A). \quad (6.1.8)$$

Theorem 6.1.6. *The product of the eigenvalues of a matrix is equal to the determinant of the matrix.*

Corollary 6.1.7. *A square matrix is invertible if and only if all of its eigenvalues are non-zero.*

We now find the coefficient of the term λ^{n-1}. By (6.1.7), this coefficient is $(-\sum_{i=1}^{n}\lambda_i)$. To find the coefficient of λ^{n-1} using (6.1.6), we expand $\det(\lambda I - A)$. We first expand along the first row. We get $\det(\lambda I - A) = (\lambda - a_{11})C_{11} + (-a_{12})C_{12} + \cdots + (-a_{1n})C_{1n}$. Each $(-a_{1j})C_{1j}$ $(j \geq 2)$ involves at most $(n-2)$ λ's because eliminating the first row and the jth column $(j \neq 1)$ already eliminates 2 λ's. So none of them contributes to the term λ^{n-1}. Thus we only need to consider the term $(\lambda - a_{11})C_{11}$. We continue to expand along row 2, then row 3, ..., and eventually row n. By repeating the same argument, we see that the term λ^{n-1} can only come from the term $(\lambda - a_{11})\cdots(\lambda - a_{nn})$ in the expansion. Therefore the coefficient of the term λ^{n-1} is $(-\sum_{i=1}^{n} a_{ii})$. Comparing the coefficients of λ^{n-1} obtained from (6.1.6) and (6.1.7), we conclude that

$$\sum_{i=1}^{n}\lambda_i = \sum_{i=1}^{n} a_{ii}. \tag{6.1.9}$$

(6.1.9) involves the sum of the diagonal entries of A. This quantity is called the *trace* of A, and is denoted by $\mathrm{Tr}(A)$. We summarize our main results below.

Theorem 6.1.8. *If $\lambda_1, \ldots, \lambda_n$ are the n eigenvalues of an $n \times n$ matrix A, then*

$$\prod_{i=1}^{n}\lambda_i = \det(A) \quad \textit{and} \quad \sum_{i=1}^{n}\lambda_i = \mathrm{Tr}(A).$$

In practice, to compute the eigenvalues of a matrix, we need to solve a polynomial equation. However, for polynomial equations with degree greater than or equal to 5, we do not have a closed-form solution involving only ordinary arithmetic operations $(+, -, \times, \div)$ and radicals $(\sqrt{\ })$ (proved by Abel and Galois independently). But with the help of computers, we can use numerical algorithms (such as Newton's method) to find approximate solutions.

Finding Eigenvectors

After we have found the eigenvalues of A, we can find the eigenvectors of A. A vector $x \neq 0$ is an eigenvector of A corresponding to the eigenvalue λ if and only if $(\lambda I - A)x = 0$, if and only if $x \in N(\lambda I - A)$. Thus we only need to the examine $N(\lambda I - A)$, which is called the *eigenspace* of A. Note that there may be linearly independent eigenvectors corresponding to the same eigenvalue.

Eigenvalues of Similar Matrices

The following theorem states that similar matrices have the same eigenvalues.

Theorem 6.1.9. *If two square matrices A and B are similar, then $p_A(\lambda) = p_B(\lambda)$. Thus A and B have exactly the same eigenvalues.*

Proof. As A and B are similar, then there exists an invertible matrix S such that $A = S^{-1}BS$. Then

$$\begin{aligned} p_A(\lambda) &= \det(\lambda I - A) \\ &= \det(S^{-1}(\lambda I - B)S) \\ &= \det(S^{-1})\det(\lambda I - B)\det(S) \\ &= \det(\lambda I - B) \\ &= p_B(\lambda). \end{aligned}$$

The eigenvalues of a matrix are exactly the roots of its characteristic polynomial. As $p_A(\lambda) = p_B(\lambda)$, A and B must have exactly the same eigenvalues. □

Remark. This theorem also tells us that we cannot use merely a characteristic polynomial to identify a matrix or the corresponding eigenvectors.

We end this lecture with two more important results.

Theorem 6.1.10. *Suppose A is an invertible matrix. λ is an eigenvalue of A if and only if λ^{-1} is an eigenvalue of A^{-1}.*

We have shown before that the eigenvalues of an invertible matrix must be non-zero. So there is no problem with λ^{-1}.

Proof. $Ax = \lambda x$ if and only if $(\lambda^{-1}A^{-1})Ax = (\lambda^{-1}A^{-1})\lambda x$, or equivalently, $\lambda^{-1}x = A^{-1}x$. □

Theorem 6.1.11. *If $\lambda_1, ..., \lambda_k$ are k distinct eigenvalues of an $n \times n$ matrix A ($k \leq n$), and $x_1, ..., x_k$ are the corresponding eigenvectors, respectively, then $\{x_1, ..., x_k\}$ is linearly independent.*

Proof. Consider the equation

$$\alpha_1 x_1 + \cdots + \alpha_k x_k = 0. \tag{6.1.10}$$

Applying $A, A^2, ..., A^{k-1}$ to (6.1.10) yield the following linear system:

$$\begin{cases} \alpha_1 x_1 + \cdots + \alpha_k x_k = 0 \\ \alpha_1 \lambda_1 x_1 + \cdots + \alpha_k \lambda_k x_k = 0 \\ \quad \vdots \\ \alpha_1 \lambda_1^{k-1} x_1 + \cdots + \alpha_k \lambda_k^{k-1} x_k = 0 \end{cases}. \tag{6.1.11}$$

This is equivalent to

$$\begin{bmatrix} \alpha_1 x_1 & \alpha_2 x_2 & \cdots & \alpha_k x_k \end{bmatrix} \underbrace{\begin{bmatrix} 1 & \lambda_1 & \lambda_1^2 & \cdots & \lambda_1^{k-1} \\ 1 & \lambda_2 & \lambda_2^2 & \cdots & \lambda_2^{k-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \lambda_k & \lambda_1^2 & \cdots & \lambda_k^{k-1} \end{bmatrix}}_{V_k^T} = 0. \tag{6.1.12}$$

As the λ_i's are distinct, V_k^T is non-singular. Hence $\begin{bmatrix} \alpha_1 x_1 & \alpha_2 x_2 & \cdots & \alpha_k x_k \end{bmatrix} = 0$. As each x_i is non-zero, we conclude that $\alpha_1 = \cdots = \alpha_k = 0$. Therefore $\{x_1, ..., x_k\}$ is linearly independent. □

■

6.2 Wednesday (Prof. Tom Luo)

6.2.1 Eigenvectors and Eigenvalues (II)

Last lecture we introduced the concepts of eigenvalues and eigenvectors. We now give an example.

Example 6.2.1. The rotation matrix in $\mathbb{R}^2$ is given by

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}. \tag{6.2.1}$$

Pre-multiplying a vector by (6.2.1) has the effect of rotating the vector counterclockwise by an angle of θ. Consider the case when $\theta = \dfrac{\pi}{2}$. In this case, the rotation matrix is

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

We now find its eigenvalues and eigenvectors.

$$\det(\lambda I - A) = \begin{vmatrix} \lambda & 1 \\ -1 & \lambda \end{vmatrix} = \lambda^2 + 1 = 0$$

gives $\lambda_1 = i$ and $\lambda_2 = -i$. We find that

$$N(\lambda_1 I - A) = \left\{ \alpha \begin{bmatrix} 1 \\ -i \end{bmatrix} \;\middle|\; \alpha \in \mathbb{C} \right\} \quad \text{and} \quad N(\lambda_2 I - A) = \left\{ \beta \begin{bmatrix} 1 \\ i \end{bmatrix} \;\middle|\; \beta \in \mathbb{C} \right\}.$$

It is not surprising that no real-valued vector is an eigenvector of A, as applying A to any real-valued vector will rotate the vector in $\mathbb{R}^2$ by $\frac{\pi}{2}$ and thus change the direction of the vector.

Number of Linearly independent eigenvectors

We now turn to an interesting question.

Question 6.2.2. How many linearly independent eigenvectors can a matrix have (at most)?

In example 6.2.1, we have seen that A has two distinct eigenvalues and can have two linearly independent eigenvectors. However, an $n \times n$ matrix may have fewer than n eigenvectors, as we illustrate in the following example.

Example 6.2.3. Suppose

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Then

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - 1 & -1 \\ 0 & \lambda - 1 \end{vmatrix} = (\lambda - 1)^2 = 0$$

gives $\lambda_1 = \lambda_2 = 1$. We see that the two eigenvalues of A are the same. $(I - A)x = 0$ gives $x = \alpha(1,0)^T$, where $\alpha \in \mathbb{R}$. So the eigenspace of A has dimension 1. Thus A has only one linearly independent eigenvector.

It is natural to ask when an $n \times n$ matrix has n linearly independent eigenvectors. Last lecture we have given a theorem which says that if a matrix has n distinct eigenvalues, then it must have n linearly independent eigenvectors.

Theorem 6.2.4. *If an $n \times n$ matrix has n distinct eigenvalues, then the corresponding eigenvectors $x_1, ..., x_n$ are linearly independent.*

However, the converse is not necessarily true. That is, a matrix may have n linearly independent eigenvectors but fewer than n distinct eigenvalues. As an example, consider the identity matrix. For every $x \in \mathbb{R}^n$, $Ix = x$. Thus $\text{eig}(I) = \{1\}$. But I has n linearly independent vectors. The following proposition gives us a way to construct matrices with fewer distinct eigenvalues than linearly independent eigenvectors.

Proposition 6.2.5. *The diagonal entries of a triangular matrix are exactly its eigenvalues.*

Proof. $p_A(\lambda) = \det(\lambda I - A) = \prod_{i=1}^{n}(\lambda - a_{ii})$. □

Thus if two diagonal entries of an $n \times n$ diagonal matrix are the same, then this matrix will have fewer than n distinct eigenvalues. However, it will still have n linearly independent vectors, say, $\{e_1, ..., e_n\}$.

6.2.2 Eigen-decomposition

Suppose $A \in \mathbb{R}^{n\times n}$ has n linearly independent eigenvectors $s_1, ..., s_n$ corresponding to the n eigenvalues $\lambda_1, ..., \lambda_n$ (some of which may be equal), respectively. If we let

$$S = \begin{bmatrix} s_1 & \cdots & s_n \end{bmatrix}, \tag{6.2.2}$$

then we have

$$A \underbrace{\begin{bmatrix} s_1 & \cdots & s_n \end{bmatrix}}_{S:\, n\times n} = \begin{bmatrix} As_1 & \cdots & As_n \end{bmatrix} = \begin{bmatrix} \lambda_1 s_1 & \cdots & \lambda_n s_n \end{bmatrix} = \underbrace{\begin{bmatrix} s_1 & \cdots & s_n \end{bmatrix}}_{S:\, n\times n} \underbrace{\begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}}_{\Lambda:\, n\times n \text{ diagonal}}.$$

Thus $AS = S\Lambda$. As the columns of S are linearly independent, then S is invertible. This gives

$$A = S\Lambda S^{-1} \tag{6.2.3}$$

where Λ is a diagonal matrix whose diagonal entries are the eigenvalues of A and S is an invertible matrix whose columns are the eigenvectors of A. The decomposition $A = S\Lambda S^{-1}$ is called the *eigen-decomposition* of A.

By (6.2.3), we see that A is similar to a diagonal matrix Λ. Thus A and Λ have the same set of eigenvalues. Another implication is that this gives us a way to change the basis so that *the linear transformation originally represented by A can now be represented by a diagonal matrix* Λ, with S being the transition matrix from the new basis to the old basis.

Conversely, if A can be factorized as in (6.2.3), then $As_i = \lambda_i s_i$ for each i. So the diagonal entries of Λ are the eigenvalues of A, and the columns of S are the corresponding eigenvectors of A. As S is invertible, then the eigenvectors of A must be linearly independent. This leads to the following result.

Definition 6.2.6. A square matrix A is said to be *diagonalizable* if A can be decomposed as $A = S\Lambda S^{-1}$, where Λ is a diagonal matrix and S is an invertible matrix.

Theorem 6.2.7. *An $n \times n$ matrix is diagonalizable if and only if it has n linearly independent eigenvectors.*

Example 6.2.8. The eigen-decomposition of the matrix A in example 6.2.1 is given by

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix} \begin{bmatrix} i & 0 \\ 0 & -i \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix}^{-1}$$

Remark. Three remarks can be made.

(i) In example 6.2.8, A is a real-valued matrix, but S and Λ are complex-valued. Thus the eigen-decomposition of a real-valued matrix can be complex-valued.

(ii) Not every matrix is diagonalizable. Only those with n linearly independent eigenvectors have eigen-decompositions.

(iii) A diagonalizable matrix has more than one eigen-decomposition.

6.2.3 Applications of Eigen-decomposition

Matrix Function

Suppose A is diagonalizable with $A = S\Lambda S^{-1}$. Then for each $k = 1,2,3,...,$

$$A^k = S\Lambda^k S^{-1} = S \begin{bmatrix} \lambda_1^k & & \\ & \ddots & \\ & & \lambda_n^k \end{bmatrix} S^{-1}. \tag{6.2.4}$$

We see that with eigendecomposition, matrix power can be easily computed.

We now consider the problem of finding $(I-A)^{-1}$, assuming that A is diagonalizable with $|\lambda_i| < 1$ for each $i = 1,...,n$. We have

$$\begin{aligned}
(I-A)^{-1} &= \left(S(I-\Lambda)S^{-1}\right)^{-1} \\
&= S(I-\Lambda)^{-1}S^{-1} \\
&= S \begin{bmatrix} \frac{1}{1-\lambda_1} & & \\ & \ddots & \\ & & \frac{1}{1-\lambda_n} \end{bmatrix} S^{-1} \\
&= S \begin{bmatrix} \sum_{k=0}^{\infty} \lambda_1^k & & \\ & \ddots & \\ & & \sum_{k=0}^{\infty} \lambda_n^k \end{bmatrix} S^{-1} \\
&= \sum_{k=0}^{\infty} S\Lambda^k S^{-1} \\
&= \sum_{k=0}^{\infty} A^k.
\end{aligned}$$

Thus we have the following formula:

$$(I-A)^{-1} = I + A + A^2 + \cdots \tag{6.2.5}$$

provided that A is diagonalizable with all eigenvalues absolutely less than 1.

We can similarly define the e^A, $\sin(A)$, etc., where A is a matrix. However, we do not apply the function to each entry of A. Instead, we apply the function to the eigenvalues of A.

Difference Equations: Fibonacci Numbers

The Fibonacci sequence $\{F_n\}$ is defined by the recurrence equation

$$F_{n+1} = F_n + F_{n-1} \ (n \geq 2) \quad \text{with } F_0 = 0 \text{ and } F_1 = 1. \tag{6.2.6}$$

To find out how fast F_n grows, we can write (6.2.6) in matrix form:

$$\begin{bmatrix} F_{n+1} \\ F_n \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}}_{A} \begin{bmatrix} F_n \\ F_{n-1} \end{bmatrix} = \cdots = A^n \begin{bmatrix} F_1 \\ F_0 \end{bmatrix}.$$

To compute A^n, we make use of formula $A^n = S\Lambda^n S^{-1}$. $\det(\lambda I - A) = 0$ gives

$$\begin{vmatrix} \lambda - 1 & -1 \\ -1 & \lambda \end{vmatrix} = \lambda^2 - \lambda - 1 = 0.$$

Thus $\lambda_1 = \frac{1+\sqrt{5}}{2}$ and $\lambda_2 = \frac{1-\sqrt{5}}{2}$. Hence

$$\Lambda = \begin{bmatrix} \frac{1+\sqrt{5}}{2} & 0 \\ 0 & \frac{1-\sqrt{5}}{2} \end{bmatrix}.$$

Therefore

$$A^n = S \begin{bmatrix} \left(\frac{1+\sqrt{5}}{2}\right)^n & 0 \\ 0 & \left(\frac{1-\sqrt{5}}{2}\right)^n \end{bmatrix} S^{-1}.$$

6.2.4 Spectral Theorem

Generally, the eigen-decomposition has two imperfections:

(i) Not every matrix has an eigen-decomposition.

(ii) The matrices S and Λ may not be real-valued.

However, the eigen-decomposition of a symmetric matrix has many good properties, as we show in the following theorem.

Theorem 6.2.9 (Spectral Theorem)**.** *Every real symmetric matrix is diagonalizable. The diagonalization is real-valued and the eigenvectors can be chosen to be orthonormal. That is to say, if $A \in \mathbb{R}^{n\times n}$ is symmetric, then A can be factorized as*

$$A = Q\Lambda Q^T \tag{6.2.7}$$

where $Q \in \mathbb{R}^{n\times n}$ is orthogonal and $\Lambda \in \mathbb{R}^{n\times n}$ is diagonal. Equivalently, A can be written as

$$A = \sum_{i=1}^{n} \lambda_i q_i q_i^T, \tag{6.2.8}$$

where $\lambda_1, \ldots, \lambda_n$ are the eigenvalues of A with corresponding orthonormal eigenvectors $q_1, \ldots, q_n$, respectively.

This theorem has further implications. Each $q_i q_i^T$ is a rank-one projection matrix. Recall that every $x \in \mathbb{R}^n$ can be written as $x = \sum_{i=1}^{n} (q_i^T x) q_i$. By (6.2.8),

$$Ax = \sum_{i=1}^{n} \lambda_i (q_i^T x) q_i. \tag{6.2.9}$$

This tells us that applying A to x is equivalent to applying the eigenvalues λ_i to the corresponding projections $(q_i^T x)q_i$, respectively.

We divide the proof into three parts.

Proposition 6.2.10 (Spectral Theorem (Part I))**.** *Suppose λ is an eigenvalue of a real symmetric matrix A. Then λ is real-valued and A has a real-valued eigenvector corresponding to λ.*

Proof. Suppose $A \in \mathbb{R}^{n\times n}$ and $A^T = A$. Let λ be an eigenvalue of A. Then $Ax = \lambda x$ for some nonzero $x \in \mathbb{C}^n$. As $x^H Ax = x^H(\lambda x) = \lambda \|x\|^2$, then

$$\lambda = \frac{x^H Ax}{\|x\|^2}.$$

Since $\overline{x^H Ax} = (x^H Ax)^H = x^H A^H x = x^H Ax$, then $x^H Ax \in \mathbb{R}$. Therefore $\lambda \in \mathbb{R}$.

Next we show that A has a real-valued eigenvector corresponding to λ. Suppose $x = x_{\text{real}} + i x_{\text{imag}}$, where $x_{\text{real}}, x_{\text{imag}} \in \mathbb{R}^n$. $A(x_{\text{real}} + i x_{\text{imag}}) = \lambda(x_{\text{real}} + i x_{\text{imag}})$ implies

$$Ax_{\text{real}} = \lambda x_{\text{real}} \quad \text{and} \quad Ax_{\text{imag}} = \lambda x_{\text{imag}}.$$

As x is non-zero, then at least one of x_{real} and x_{imag} is non-zero. Then the non-zero one is a real eigenvector of A corresponding to the eigenvalue λ. Another way of proving this is to note that the coefficient matrix in $(\lambda I - A)x = 0$ is real-valued. Thus this equation has a real-valued non-zero solution. □

Proposition 6.2.11 (Spectral Theorem (Part II))**.** *Two real eigenvectors of a real symmetric matrix corresponding to two distinct eigenvalues are orthogonal.*

Proof. Let A be a real symmetric matrix. Suppose λ_1 and λ_2 are two distinct eigenvalues of A and that x_1 and x_2 are two real-valued eigenvectors corresponding to λ_1 and λ_2, respectively. As

$$x_1^T Ax_2 = x_1^T(Ax_2) = x_1^T(\lambda_2 x_2) = \lambda_2(x_1^T x_2)$$

and

$$x_1^T Ax_2 = (A^T x_1)^T x_2 = (Ax_1)^T x_2 = (\lambda_1 x_1)^T x_2 = \lambda_1(x_1^T x_2),$$

then $\lambda_1(x_1^T x_2) = \lambda_2(x_1^T x_2)$. As $\lambda_1 \neq \lambda_2$, then $x_1^T x_2 = 0$. Thus x_1 and x_2 are orthogonal. □

Proposition 6.2.11 implies that if the eigenvalues of a real symmetric matrix are all distinct, then the eigenvectors of the matrix can be chosen to be orthonormal. The following theorem shows that even if some of the eigenvalues of a real symmetric matrix are equal, the eigenvectors can still be chosen to be orthonormal.

Theorem 6.2.12 (Spectral Theorem (Part III))**.** *If $A \in \mathbb{R}^{n\times n}$ is a symmetric matrix, then A can be factorized as $A = Q\Lambda Q^T$, where $Q \in \mathbb{R}^{n\times n}$ is orthogonal and $\Lambda \in \mathbb{R}^{n\times n}$ is diagonal.*

Proof. We prove by induction. When $n = 1$, $A \in \mathbb{R}$ must be diagonal and $A = 1AI^T$ (1 is orthonormal as a vector and orthogonal as a matrix). Suppose when $n = k-1$ ($k = 2,3,...$) the statement holds. We now show that this statement is true when $n = k$.

Suppose $A \in \mathbb{R}^{n\times n}$ is symmetric. Then A has a real-valued eigenvalue λ_1 and a real-valued eigenvector s_1 corresponding to λ_1 such that $\|s_1\| = 1$. We can extend $\{s_1\}$ to an orthonormal basis $\{s_1,...,s_n\}$ of $\mathbb{R}^n$. Let S be the $n \times n$ real matrix whose columns are $s_1,...,s_n$. Then S is orthogonal. The first row of S^TAS is

$$s_1^TAS = (A^Ts_1)^TS = (As_1)^TS = \lambda_1 s_1^T \begin{bmatrix} s_1 & \cdots & s_n \end{bmatrix} = \lambda_1 e_1^T.$$

Since S^TAS is symmetric, then the first column of S^TAS is $\lambda_1 e_1$. Thus

$$S^TAS = \left[\begin{array}{c|ccc} \lambda_1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \widetilde{A} & \\ 0 & & & \end{array}\right] \tag{6.2.10}$$

where $\widetilde{A}$ is an $(n-1)\times(n-1)$ real symmetric matrix (by the symmetry of A). By assumption, there exists an $(n-1)\times(n-1)$ real orthogonal matrix $\widetilde{Q}$ and an $(n-1)\times(n-1)$ real diagonal matrix $\widetilde{\Lambda}$ such that $\widetilde{A} = \widetilde{Q}\widetilde{\Lambda}\widetilde{Q}^T$. Thus

$$S^TAS = \left[\begin{array}{c|ccc} \lambda_1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \widetilde{Q}\widetilde{\Lambda}\widetilde{Q}^T & \\ 0 & & & \end{array}\right] = \underbrace{\left[\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \widetilde{Q} & \\ 0 & & & \end{array}\right]}_{T} \underbrace{\left[\begin{array}{c|ccc} \lambda_1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \widetilde{\Lambda} & \\ 0 & & & \end{array}\right]}_{\Lambda} \underbrace{\left[\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & \widetilde{Q}^T & \\ 0 & & & \end{array}\right]}_{T^T}.$$

Then $T \in \mathbb{R}^{n\times n}$ is orthogonal and $\Lambda \in \mathbb{R}^n$ is diagonal. Hence

$$A = (ST)\Lambda(ST)^T = Q\Lambda Q^T$$

with $Q = ST$ orthogonal and Λ diagonal. This completes the proof. □

Remark. Note that to prove the spectral theorem, we only need proposition 6.2.10 and theorem 6.2.12.

■

6.3 Friday (Prof. Tom Luo)

Last lecture we proved that every real symmetric matrix A has a real-valued eigen-decomposition and that the eigenvectors of A can be chosen to be orthonormal. Thus A can be written as $A = \sum_{i=1}^{n} \lambda_i q_i q_i^T$, where $q_i^T q_j = \delta_{ij}$. We now see how it is connected with analytic geometry.

6.3.1 Quadratic Forms

We begin by considering quadratic functions on $\mathbb{R}^2$. The quadratic function $ax^2 + bxy + cy^2 = 1$ defines several types of curves. Below are some examples.

(i) The equation $x^2 + y^2 = 1$ represents a circle.

(ii) The equation $\frac{x^2}{2} + \frac{y^2}{5} = 1$ represents an ellipse.

(iii) The equation $\frac{x^2}{2} - \frac{y^2}{5} = 1$ represents a hyperbola.

(iv) The equation $x^2 = \alpha y$ represents a parabola.

More generally, we can take into account the linear terms: $ax^2 + bxy + cy^2 + dx + ey = 1$. By change of coordinates, we can convert this equation into forms such as

$$(u-u_0)^2 + (v-v_0)^2 = R^2, \quad \frac{(u-u_0)^2}{2} + \frac{(v-v_0)^2}{5} = 1, \quad \frac{(u-u_0)^2}{2} - \frac{(v-v_0)^2}{5} = 1$$

so that no cross term or linear term is involved. To do so, note that

$$ax^2 + bxy + cy^2 = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & \frac{b}{2} \\ \frac{b}{2} & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}. \tag{6.3.1}$$

Let

$$u = \begin{bmatrix} x \\ y \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} a & \frac{b}{2} \\ \frac{b}{2} & c \end{bmatrix}.$$

Since $A = A^T$, then by the spectral theorem, $A = \lambda_1 q_1 q_1^T + \lambda_2 q_2 q_2^T$, where q_1 and q_2 are orthonormal. Thus

$$\begin{aligned} ax^2 + bxy + cy^2 &= u^T(\lambda_1 q_1 q_1^T + \lambda_2 q_2 q_2^T)u \\ &= \lambda_1 (q_1^T u)^2 + \lambda_2 (q_2^T u)^2 \end{aligned}$$

Recall that $Q^T = \begin{bmatrix} q_1 & q_2 \end{bmatrix}^T$ defines a coordinate mapping that maps a vector to its coordinates with respect to the basis $\{q_1, q_2\}$. Let $v_i = q_i^T u$, then we have changed from the coordinate $u = (x,y)^T$ with respect to the basis $\{e_1, .e_2\}$ to the coordinate $v = (v_1, v_2)^T$ with respect to the basis $\{q_1, q_2\}$. Hence

$$ax^2 + bxy + cy^2 = 1 \quad \text{is equivalent to} \quad \frac{v_1^2}{1/\lambda_1} + \frac{v_2^2}{1/\lambda_2} = 1.$$

More generally, we can consider quadratic forms $x^T Ax = \sum_{i,j} a_{ij} x_i x_j$ in $\mathbb{R}^n$. Without loss of generality, we can assume that A is symmetric. (As $x^T Ax$ is a scalar, then $x^T Ax = (x^T Ax)^T = x^T A^T x$. Thus $x^T Ax = x^T(\frac{A+A^T}{2})x$, with $\frac{A+A^T}{2}$ being symmetric.) Then we have

$$x^T Ax = x^T(\sum_{i=1}^{n} \lambda_i q_i q_i^T)x = \sum_{i=1}^{n} \lambda_i (q_i^T x)^2 = \sum_{i=1}^{n} \lambda_i y_i^2$$

where $y_i = q_i^T x$. By changing the old coordinate $x = (x_1, ..., x_n)^T$ to a new one $y = (y_1, ..., y_n)^T$, we have converted the original quadratic form which involves n^2 terms, into a form such that no cross term or linear term is involved and in which there are only n terms.

6.3.2 Rank-One Approximation of a Symmetric Matrix

We now consider the problem of approximating a symmetric matrix by a rank-one matrix, which is an application of eigen-decomposition. The problem can be informally stated as follows.

Problem 6.3.1. Given a real $n \times n$ symmetric matrix A, find a unit vector $x \in \mathbb{R}^n$ and a real number λ such that λxx^T best approximates A.

Here we have made use of the fact that every real rank-one symmetric matrix can be written as λxx^T, where $x \in \mathbb{R}^n$, $\|x\| = 1$ and $\lambda \in \mathbb{R}\backslash\{0\}$.

An application of this problem is data compression. If $A \in \mathbb{R}^{n\times n}$ is symmetric, then to store A, we need to store about $\frac{1}{2}n^2 \sim O(n^2)$ entries. However, if we can find a rank-one matrix that approximates A well enough, then we can only store the rank-one matrix, thus only $(n+1) \sim O(n)$ entries (because we only need to store $x \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$). This can save a lot of storage and is therefore very meaningful.

However, to solve this problem, we have to be precise about what we mean by a "best approximation". To do so, we introduce the concept of matrix norm.

Matrix Norm

We first define the inner product of matrices.

Definition 6.3.2. The inner product of two matrices A and B in $\mathbb{R}^{m\times n}$, denoted by $\langle A,B\rangle$, is a real number defined as

$$\langle A,B\rangle = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}b_{ij}. \tag{6.3.2}$$

Proposition 6.3.3. *For all $A,B \in \mathbb{R}^{m\times n}$, $\langle A,B\rangle = \mathrm{Tr}(A^TB) = \mathrm{Tr}(BA^T)$.*

Proof.

$$\mathrm{Tr}(A^TB) = \sum_{j=1}^{n} a_j^T b_j = \sum_{j=1}^{n}\sum_{i=1}^{m} a_{ij}b_{ij} = \langle A,B\rangle.$$

$$\mathrm{Tr}(BA^T) = \sum_{i=1}^{m} \vec{b}_i(\vec{a}_i)^T = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}b_{ij} = \langle A,B\rangle.$$

□

We have actually proved the cyclic property of trace.

Proposition 6.3.4. *Suppose $A \in \mathbb{R}^{m\times n}$ and $B \in \mathbb{R}^{n\times m}$. Then* $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$.

We now define the *Frobenius norm* of a matrix.

Definition 6.3.5. The *Frobenius norm* of a matrix $A \in \mathbb{R}^{m\times n}$, denoted by $\|A\|_F$, is a real number defined as

$$\|A\|_F = \sqrt{\langle A,A\rangle} = \sqrt{\sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}^2}. \tag{6.3.3}$$

Thus for every real matrix A,

$$\|A\|_F^2 = \mathrm{Tr}(A^TA). \tag{6.3.4}$$

If A is real and symmetric, then $A = Q^T\Lambda Q$. Then

$$\|A\|_F^2 = \mathrm{Tr}(A^TA) = \mathrm{Tr}(A^2) = \mathrm{Tr}(Q^T\Lambda^2Q) = \mathrm{Tr}(QQ^T\Lambda^2) = \mathrm{Tr}(\Lambda^2) = \sum_{i=1}^{n}\lambda_i^2.$$

Therefore *the Frobenius norm of a real symmetric matrix is equal to the sum of the squares of its eigenvalues*. Thus if a real symmetric matrix has large eigenvalues, we can expect its entries to also be large. However, note that this is true only for symmetric matrices. As a counterexample for other matrices, every triangular matrix with diagonal entries all equal to zero must have its eigenvalues all equal to zero. Still, its off-diagonal entries can be large.

Formulation of the Problem

Our problem of finding the best rank-one approximation of a real symmetric $n \times n$ matrix A can now be formulated as the following optimization problem:

$$\begin{array}{ll} \min\limits_{\lambda \in \mathbb{R}\backslash\{0\},\, x \in \mathbb{R}^n} & \|A - \lambda x x^T\|_F^2 \\ \text{s.t.} & \|x\| = 1 \end{array} \tag{6.3.5}$$

(It can be shown that the best approximate matrix must be symmetric.) We can divide (6.3.5) into two subproblems (6.3.6) and (6.3.7):

$$\begin{array}{ll} \min\limits_{\lambda > 0,\, y \in \mathbb{R}^n} & \|A - \lambda y y^T\|_F^2 \\ \text{s.t.} & \|y\| = 1 \end{array} \tag{6.3.6}$$

$$\begin{array}{ll} \min\limits_{\mu < 0,\, w \in \mathbb{R}^n} & \|A - \mu w w^T\|_F^2 \\ \text{s.t.} & \|w\| = 1 \end{array} \tag{6.3.7}$$

By doing so, we can convert (6.3.6) and (6.3.7) into two equivalent problems (6.3.8) and (6.3.9) that involve only one variable, respectively:

$$\min_{x \in \mathbb{R}^n,\, x \neq 0} \quad \|A - x x^T\|_F^2 \tag{6.3.8}$$

$$\min_{z \in \mathbb{R}^n,\, z \neq 0} \quad \|A + z z^T\|_F^2 \tag{6.3.9}$$

If x^* is the optimal solution to (6.3.8), let $\lambda^* = \|x^*\|^2$ and $y^* = \dfrac{x^*}{\|x^*\|}$. Then (λ^*, y^*) is the optimal solution to (6.3.6). Similarly, if z^* is the optimal solution to (6.3.9), let $\mu^* = -\|z^*\|^2$ and $w^* = \dfrac{z^*}{\|z^*\|}$, then (μ^*, w^*) is the optimal solution to (6.3.7).

We now solve (6.3.8). Let $J(x) = \|A - x x^T\|_F^2$. By the symmetry of A,

$$\begin{aligned} J(x) &= \operatorname{Tr}\left((A - x x^T)(A - x x^T)\right) \\ &= \operatorname{Tr}\left(A^2 - x x^T A - A x x^T + x x^T x x^T\right) \\ &= \|A\|_F^2 - 2 x^T A x + \|x\|^4. \end{aligned}$$

Thus

$$\nabla J(x) = -4Ax + 4\|x\|^3 \frac{x}{\|x\|} = 4(\|x\|^2 I - A)x.$$

Setting $\nabla J(x) = 0$ gives $(\|x\|^2 I - A)x = 0$. Therefore $x = 0$ or x is an eigenvector of A corresponding to the eigenvalue $\|x\|^2 > 0$. As $x \neq 0$, then the optimal solution to (6.3.8) is an eigenvector x of A corresponding to the eigenvalue $\|x\|^2 > 0$.

In a similar manner, we find that the optimal solution to (6.3.9) is an eigenvector z of A corresponding to the eigenvalue $-\|z\|^2 < 0$. Based on the previous discussion on how we can convert (6.3.8) and (6.3.9) to (6.3.6) and (6.3.7), we conclude that the optimal solution (λ^*, x^*) to (6.3.5) consists of an eigenvalue of A and a corresponding unit eigenvector (there are two actually, but the sign will cancel off). To find the optimal solution, we make use of the spectral theorem. Suppose we reorder the eigenvalues such that $|\lambda_1| \geq |\lambda_2| \geq \cdots |\lambda_n| \geq 0$. If we choose the kth pair to approximate A, then the residual is equal to

$$r_k = \|A - \lambda_k q_k q_k^T\|_F^2 = \left\| \sum_{i=1, i\neq k}^{n} \lambda_i q_i q_i^T \right\|_F^2 = \sum_{i=1, i\neq k}^{n} \lambda_i^2. \tag{6.3.10}$$

The last equality is by the equality $\langle \lambda_i q_i q_i^T, \lambda_j q_j q_j^T \rangle = \lambda_i \lambda_j \operatorname{Tr}(q_i q_i^T q_j q_j) = 0$ whenever $i \neq j$. We conclude that the best rank-one approximation of A is given by $\lambda_1 q_1 q_1^T$, where λ_1 is the eigenvalue of A with the largest absolute value and q_1 is a corresponding unit eigenvector.

Low-rank Approximations

In a similar manner, it can be shown that the best rank-k approximation of A is given by $\sum_{i=1}^{k} \lambda_i q_i q_i^T$, where $\lambda_1, ..., \lambda_k$ are the k eigenvalues of A largest in absolute value and $q_1, ..., q_k$ are the corresponding unit eigenvectors, respectively.

In reality, the best rank-four or rank-five approximation of a matrix is usually already good enough. For example, human eye cannot tell the difference between an image and its best rank-four or rank-five approximation!

6.3.3 Spectral Theorem for Hermitian Matrices

We now discuss the complex version of the spectral theorem. We first do some review.

Definition 6.3.6. A matrix $A \in \mathbb{C}^{n\times n}$ is said to be a *Hermitian* matrix if $A^H = A$.

Example 6.3.7. The matrix $A = \begin{bmatrix} 3 & 2-i \\ 2+i & 4 \end{bmatrix}$ is Hermitian, as $A^H = A$.

If A is Hermitian, then $a_{ij} = \overline{a}_{ji}$ for all i and j. Thus the diagonal entries of a Hermitian matrix must be real numbers.

We now give a comparison between the inner products in $\mathbb{R}^n$ and $\mathbb{C}^n$.

$\mathbb{R}^n$	$\mathbb{C}^n$
$\langle x, y \rangle = x^T y$	$\langle x, y \rangle = y^H x$
$x^T y = y^T x$	$x^H y = \overline{y^H x}$
$\|x\|^2 = \sum_{i=1}^n x_i^2$	$\|x\|^2 = \sum_{i=1}^n \|x_i\|^2$
$x \perp y \Leftrightarrow x^T y = 0$	$x \perp y \Leftrightarrow y^H x = 0$
$\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$	$\langle \alpha x, y \rangle = \alpha \langle x, y \rangle$
	$\langle x, \alpha y \rangle = \overline{\alpha} \langle x, y \rangle$, $\alpha \in \mathbb{C}$

Spectral Theorem

We first introduce a useful formula.

Proposition 6.3.8. *For all $A \in \mathbb{C}^{n\times n}$ and $x, y \in \mathbb{C}^n$, $\langle Ax, y\rangle = \langle x, A^H y\rangle$.*

Proof. $y^H(Ax) = (A^H y)^H x$. □

We now give the complex version of the spectral theorem.

Theorem 6.3.9 (Spectral Theorem)**.** *Every Hermitian matrix $A \in \mathbb{C}^{n\times n}$ can be factorized as*

$$A = Q\Lambda Q^H \tag{6.3.5}$$

where $\Lambda \in \mathbb{R}^{n\times n}$ is a diagonal matrix whose diagonal entries are the eigenvalues of A and $Q \in \mathbb{C}^{n\times n}$ is a unitary matrix, i.e. $Q^H Q = I$. Therefore

$$A = \sum_{i=1}^{n} \lambda_i q_i q_i^H \tag{6.3.6}$$

where $\lambda_1, ..., \lambda_n$ are eigenvalues of A with corresponding orthonormal eigenvectors $q_1, ..., q_n$.

Eigenvalues are real-valued. We can make use of proposition 6.3.8 to prove that every eigenvalue of A is nonzero. Suppose $Ax = \lambda x$. Since

$$\langle Ax, x\rangle = \langle \lambda x, x\rangle = \lambda \langle x, x\rangle$$

and

$$\langle Ax, x\rangle = \langle x, A^H x\rangle = \langle x, Ax\rangle = \langle x, \lambda x\rangle = \overline{\lambda}\langle x, x\rangle,$$

then $\lambda\langle x, x\rangle = \overline{\lambda}\langle x, x\rangle$. As x is nonzero, then $\lambda = \overline{\lambda}$. Hence $\lambda \in \mathbb{R}$.

Eigenvectors can be orthonormal. We can also use proposition 6.3.8 to prove that two eigenvectors of A corresponding to two distinct eigenvalues are orthogonal. Suppose $Ax_1 = \lambda_1 x_1$ and $Ax_2 = \lambda_2 x_2$ with $\lambda_1 \neq \lambda_2$. As

$$\langle Ax_1, x_2\rangle = \langle \lambda_1 x_1, x_2\rangle = \lambda_1 \langle x_1, x_2\rangle$$

and

$$\langle Ax_1, x_2\rangle = \langle x_1, A^H x_2\rangle = \langle x_1, Ax_2\rangle = \langle x_1, \lambda_2 x_2\rangle = \overline{\lambda_2}\langle x_1, x_2\rangle = \lambda_2\langle x_1, x_2\rangle,$$

then $\lambda_1\langle x_1, x_2\rangle = \lambda_2\langle x_1, x_2\rangle$. As $\lambda_1 \neq \lambda_2$, then $\langle x_1, x_2\rangle = 0$. Therefore $x_1 \perp x_2$.

The proof for the third part of the theorem is basically the same as the proof for the real case, with transpose replaced by the Hermitian operator.

■

Chapter 7

Week Seven

7.1 Monday (Prof. Tom Luo)

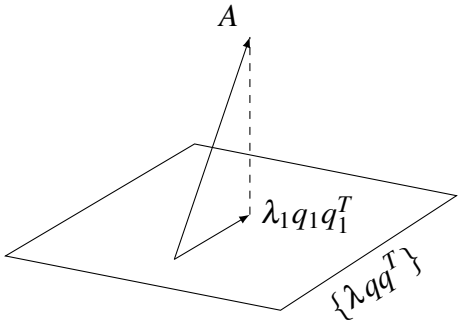
7.1.1 A Geometric Interpretation of Rank-One Approximation

Suppose A is a real symmetric matrix. Then A be can factorized as $A = Q\Lambda Q^T$, where Q is orthogonal and Λ is diagonal. As Q and Q^T are nonsingular, then $\text{rank}(A) = \text{rank}(\Lambda)$. Therefore *the rank of A is equal to the number of nonzero eigenvalues of A*. Suppose $\text{rank}(A) = r$, then $A = \sum_{i=1}^{r} \lambda_i q_i q_i^T$, where $\lambda_1, ..., \lambda_n$ are eigenvalues of A with $|\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_r| > 0 = \lambda_{r+1} = \cdots = \lambda_n$ and $q_1, ..., q_n$ are the corresponding orthonormal eigenvectors, respectively. We see that A, which is rank-r, can be written as the sum of r rank-one matrices. Note that for other types of matrices, this is not true in general.

Last lecture we showed that the best rank-one approximation of A is given by $\lambda_1 q_1 q_1^T$. Here we give a geometric interpretation. As we have mentioned above, A can be written as the sum of r rank-one matrices. These matrices are mutually orthogonal:

$$\langle \lambda_i q_i q_i^T, \lambda_j q_j q_j^T \rangle = \text{Tr}\left((\lambda_i q_i q_i^T)(\lambda_j q_j q_j^T)\right) = \lambda_i \lambda_j \,\text{Tr}\left(q_i (q_i^T q_j) q_j^T\right) = 0$$

whenever $i \neq j$. Thus $\lambda_1 q_1 q_1^T$ and $A - \lambda_1 q_1 q_1^T = \lambda_2 q_2 q_2^T + \cdots + \lambda_r q_r q_r^T$ are orthogonal. For a rank-one matrix λqq^T to best approximate A, it *should* be the projection of A onto the set of rank-one symmetric matrices. Equivalently, λqq^T and $A - \lambda qq^T$ must be orthogonal. $\lambda_1 q_1 q_1^T$ satisfies exactly this optimality condition and is thus the optimal solution. We can visualize the best rank-one approximation of A with the following graph.



Note that this analogy is not completely rigorous, as the set of all rank-one matrices is not a vector space.

7.1.2 Quadratic Forms

We go back to our discussion of quadratic forms. A *quadratic form* is an expression $x^T Ax = \sum_{i,j} a_{ij} x_i x_j$. Without loss of generality, we can assume that A is symmetric, as $x^T Ax = x^T A^T x = x^T \left(\frac{A + A^T}{2}\right) x$.

An equation in the form of $x^TAx = 1$ is called a *conic section*.

Example 7.1.1. Consider the equation $3x^2 + 2xy + 3y^2 = 1$. We can write this equation in matrix form:

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 1.$$

We now find the eigen-decomposition of A.

$$\det(\lambda I - A) = \begin{vmatrix} \lambda - 3 & -1 \\ -1 & \lambda - 3 \end{vmatrix} = (\lambda - 3)^2 - 1 = 0$$

gives $\lambda_1 = 4$ and $\lambda_2 = 2$. As

$$\lambda_1 I - A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \quad \text{and} \quad \lambda_2 I - A = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix},$$

then the unit eigenvectors corresponding to λ_1 and λ_2 are

$$q_1 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \quad \text{and} \quad q_2 = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}, \quad \text{respectively.}$$

Thus

$$\underbrace{\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}}_{A} = \underbrace{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}}_{Q} \underbrace{\begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}}_{\Lambda} \underbrace{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}}_{Q^T}.$$

Therefore

$$3x^2 + 2xy + 3y^2 = 4\left(\frac{1}{\sqrt{2}}x + \frac{1}{\sqrt{2}}y\right)^2 + 2\left(\frac{1}{\sqrt{2}}x - \frac{1}{\sqrt{2}}y\right)^2 = 4z_1^2 + 4z_2^2 = 1$$

with

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = Q^T \begin{bmatrix} x \\ y \end{bmatrix}.$$

As the matrix A in the quadratic form x^TAx is symmetric, then A can always be factorized as $A = Q\Lambda Q^T$. Thus every quadratic form x^TAx can be written as the weighted sum of squares. Now, suppose we add another quadratic form to the original one to form a new one, such as $3x^2 + 2xy + 3y^2 + x^2 + 5xy + y^2$. Originally $3x^2 + 2xy + 3y^2$ and $x^2 + 5xy + y^2$ can both be written as the weighted sum of squares. If we combine them together, we can still write the new one as the weighted sum of squares (the new one may be completely different from the original ones). Therefore every quadratic form (such as $3x^2 + 2xy + 3y^2 + x^2 + 5xy + y^2$) can always be written as the weighted sum of squares. This gives us the following theorem.

Theorem 7.1.2. *Suppose $A \in \mathbb{R}^{n\times n}$ can be written as*

$$A = \sum_{i=1}^{N} \lambda_i u_i u_i^T,$$

where $\lambda_i \in \mathbb{R}$, $u_i \in \mathbb{R}^n$, and $N > n$. Then there exist $\tilde{\lambda}_1, ..., \tilde{\lambda}_r \in \mathbb{R}$ and $\tilde{u}_1, ..., \tilde{u}_r \in \mathbb{R}^n$ such that

$$A = \sum_{i=1}^{r} \tilde{\lambda}_i \tilde{u}_i \tilde{u}_i^T$$

where $r = \operatorname{rank}(A)$.

Sketch of Proof. That $A = \sum_{i=1}^{N} \lambda_i u_i u_i^T$ implies that A is symmetric. Apply spectral theorem. Then $A = \sum_{i=1}^{r} \tilde{\lambda}_i \tilde{u}_i \tilde{u}_i^T$ with $\tilde{\lambda}_1, ..., \tilde{\lambda}_r \in \mathbb{R}$ being the eigenvalues of A and $\tilde{u}_1, ..., \tilde{u}_r \in \mathbb{R}^n$ being the corresponding orthonormal eigenvectors, respectively. □

7.1.3 Positive Definite Matrices

Motivation

Suppose $A \in \mathbb{R}^{n\times n}$ is symmetric. Let $f(x) = \frac{1}{2}x^TAx + b^Tx$, where $x \in \mathbb{R}^n$. We would like to find the minimum of f.

To find the critical points of f, set $\nabla f(x) = Ax + b = 0$. If $Ax + b = 0$ does not have a solution, then f is unbounded from below. Still, even if $Ax + b = 0$ does have a solution x^*, x^* may not be a minimizer of f. For x^* to be a minimizer of f, it must be true for all $x \in \mathbb{R}^n$ that

$$\begin{aligned} f(x) - f(x^*) &= (\nabla f(x^*))^T (x - x^*) + \frac{1}{2}(x - x^*)^T H_f (x - x^*) \\ &= \frac{1}{2}(x - x^*)^T A (x - x^*) \\ &\geq 0 \end{aligned}$$

where H_f is the Hessian matrix of f defined by $\left(H_f\right)_{ij} = \dfrac{\partial^2 f}{\partial x_i \partial x_j}$. The second equality is by the fact that $\nabla f(x^*) = 0$ and that $H_f = A$. If x^* is to be the unique minimizer, then $\frac{1}{2}(x - x^*)^T A(x - x^*) > 0$ for all $x \neq x^*$. This motivates us to study the following kinds of matrices.

Positive Definite Matrices

Definition 7.1.3. Suppose $A \in \mathbb{R}^{n\times n}$ and $A^T = A$.

(i) A is said to be *positive definite* if $x^TAx > 0$ for all $x \neq 0$. This is denoted by $A \succ 0$.

(ii) A is said to be *positive semi-definite* if $x^TAx \geq 0$ for all $x \in \mathbb{R}^n$. This is denoted by $A \succeq 0$.

(iii) A is said to be *indefinite* if there exist $x, y \in \mathbb{R}^n$ such that $x^TAx < 0 < y^TAy$.

Example 7.1.4.

(i) The matrix $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ is indefinite.

(ii) The matrix $\begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$ is positive definite because for all $x = (x_1, x_2)^T$, $x^TAx = 4x_1^2 + 2x_2^2 \geq 0$. Equality holds if and only if $x_1 = x_2 = 0$.

(iii) The matrix $\begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$ is positive definite. By example 7.1.1, for all $x \in \mathbb{R}^n$,

$$x^TAx = 4\left(\frac{1}{\sqrt{2}}x_1 + \frac{1}{\sqrt{2}}x_2\right)^2 + 2\left(\frac{1}{\sqrt{2}}x_1 - \frac{1}{\sqrt{2}}x_2\right)^2 = 4z_1^2 + 2z_2^2 \geq 0.$$

Equality holds if and only if $z_1 = z_2 = 0$. As Q^T is norm-preserving, $z = Q^Tx = 0$ if and only if $x = 0$.

Below we give some equivalent characterizations of the positive definiteness of a matrix. Statement (iii) involves the concept of leading principal submatrices, which we will define later.

Theorem 7.1.5. *Suppose A is a symmetric matrix in $\mathbb{R}^{n\times n}$. The following statements are equivalent.*

(i) A is positive definite.

(ii) All eigenvalues of A are positive.

(iii) All leading principal submatrices of A have positive determinants.

(iv) A can be factorized as $A = RR^T$, where R is a nonsingular matrix.

Proof of "(i) ⇒ (ii)". Suppose A is positive definite. If $Ax = \lambda x$ for $x \neq 0$, then $x^T Ax = \lambda \|x\|^2 > 0$ by the positive definiteness of A. As $\|x\|^2 > 0$, then $\lambda > 0$. □

Proof of "(ii) ⇒ (i)". Suppose all eigenvalues $\lambda_1, ..., \lambda_n$ of A are positive. Since A is symmetric, then $A = Q\Lambda Q^T$, with $\Lambda = \text{diag}(\lambda_1, ..., \lambda_n)$. Then for every $x \in \mathbb{R}^n$,

$$x^T Ax = x^T Q\Lambda Q^T x = (Q^T x)^T \Lambda (Q^T x) = \tilde{x}^T \Lambda \tilde{x} = \sum_{i=1}^{n} \lambda_i \tilde{x}_i^2 \geq 0$$

where $\tilde{x} = Q^T x$. Equality holds if and only if $\tilde{x} = 0$. As Q^T is norm-preserving, $\tilde{x} = 0$ is equivalent to $x = 0$. Thus equality holds if and only if $x = 0$. Therefore A is positive definite. □

Corollary 7.1.6. *If a symmetric matrix A is positive definite, then* $\det(A) > 0$*. Thus A is nonsingular.*

Proof. Suppose A has eigenvalues $\lambda_1, ..., \lambda_n$. As A is positive definite, $\lambda_i > 0$ for each $i = 1, ..., n$. Then $\det(A) = \prod_{i=1}^{n} \lambda_i > 0$. Hence A is nonsingular. □

We now introduce the concept of *principal submatrices*. Let A be an $n \times n$ matrix. Suppose α is a subset of $\{1, 2, ..., n\}$. The matrix $A_{\alpha,\alpha}$, which is obtained by deleting the rows and columns of A whose indices are not in α, is called the *principal submatrix* of A with rows and columns indexed by α. More generally, we can define $A_{\alpha,\beta}$ as the submatrix of A with rows indexed by α and columns indexed by β.

Example 7.1.7. Suppose

$$A = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}.$$

Then

$$A_{1,1} = 1, \quad A_{\{1,2\},\{1,2\}} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}$$

$$A_{\{2,3\},\{2,3\}} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \quad A_{\{1,3,4\},\{1,2\}} = \begin{bmatrix} 1 & -1 \\ 0 & -1 \\ 0 & 0 \end{bmatrix}.$$

For each $i = 1, ..., n$, we define the ith *leading principal submatrix* of A, denoted by A_i, as the principal submatrix $A_{\alpha,\alpha}$ of A with rows and columns indexed by the set $\alpha = \{1, 2, ..., i\}$.

We now go back to theorem 7.1.5.

Proof of "(i) ⇒ (iii)". Suppose A is positive definite. We first show that for each $\alpha \subseteq \{1, ..., n\}$, the matrix $A_{\alpha,\alpha}$ is positive definite. For every $x_\alpha \in \mathbb{R}^{|\alpha|}$, we can form a vector $x \in \mathbb{R}^n$ such that $x_i = 0$ for $i \notin \alpha$ and such that the other entries of x are exactly the entries of x_α in the same order. Then

$$x_\alpha^T A_{\alpha,\alpha} x_\alpha = x^T Ax \geq 0.$$

Equality holds if and only if $x = 0$, if and only if $x_\alpha = 0$. Thus each $A_{\alpha,\alpha}$ is positive definite. In particular, each leading principal submatrix A_i is positive definite. So the eigenvalues of A_i are all positive. Hence $\det(A_i)$, which is equal to the product of the eigenvalues of A_i, is positive. □

Remark. We have actually proved that all $A_{\alpha,\alpha}$'s have positive determinants. By taking $\alpha = \{i\}$ for each $i = 1,...,n$, then $A_{\alpha,\alpha} = a_{ii} > 0$. Thus the diagonal entries of a positive definite matrix must be positive.

Corollary 7.1.8. *The diagonal entries of a positive definite matrix are positive.*

Proof of "(iii) ⇒ (iv)". Suppose all leading principal submatrices of A have positive determinants. We will prove that A can be factorized as $A = LDL^T$ without row exchanges and that the pivots of A are positive. We argue by induction. Since $a_{11} > 0$, then we can use a_{11} as the first pivot to perform elimination on the first column. Suppose we have performed eliminations on columns 1 through $(k-1)$ without row exchanges and that $a_{11},...,a_{(k-1),(k-1)} > 0$. Denote the resultant matrix by $\widetilde{A}$. Then $\det(\widetilde{A}_{k-1}) = \det(A_{k-1}) > 0$ and $\det(\widetilde{A}_k) = \det(A_k) > 0$. Since $\det(\widetilde{A}_k) = \widetilde{a}_{kk}\det(\widetilde{A}_{k-1})$, then $\widetilde{a}_{kk} > 0$. Therefore we can use a_{kk} as the kth pivot to perform elimination on the kth column without any row exchange. Therefore the determinants of all principal leading submatrices remain the same. The induction is complete. Thus all pivots of A are positive and no row exchange is required during elimination. As A is symmetric, then A can be factorized as

$$\begin{aligned}
A &= LDL^T \\
&= L\begin{bmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{bmatrix} L^T \\
&= L\begin{bmatrix} \sqrt{d_1} & & \\ & \ddots & \\ & & \sqrt{d_n} \end{bmatrix}\begin{bmatrix} \sqrt{d_1} & & \\ & \ddots & \\ & & \sqrt{d_n} \end{bmatrix} L^T \\
&= \underbrace{\left(L\begin{bmatrix} \sqrt{d_1} & & \\ & \ddots & \\ & & \sqrt{d_n} \end{bmatrix}\right)}_{R}\underbrace{\left(L\begin{bmatrix} \sqrt{d_1} & & \\ & \ddots & \\ & & \sqrt{d_n} \end{bmatrix}\right)^T}_{R^T}.
\end{aligned}$$

where $d_1,...,d_n$ are the pivots of the row echelon form of A. The existence of $\sqrt{d_i}$ is by the fact that all pivots of A are positive. As $\sqrt{d_i} > 0$ for each i, then $\text{diag}(\sqrt{d_1},...,\sqrt{d_n})$ is invertible. Since both L and $\text{diag}(\sqrt{d_1},...,\sqrt{d_n})$ are invertible, then R is invertible and thus nonsingular. This completes the proof. Also, R is lower triangular. □

Proof of "(iv) ⇒ (i)". Suppose $A = RR^T$ with R nonsinuglar. For every $x \in \mathbb{R}^n$, $x^TAx = x^TRR^Tx = \|R^Tx\|^2 \geq 0$. Equality holds if and only if $R^Tx = 0$. As R is nonsingular, so is R^T. Thus $R^Tx = 0$ if and only if $x = 0$. Therefore A is positive definite. □

The proof of theorem 7.1.5 is now complete.

■

7.2 Wednesday (Prof. Tom Luo)

7.2.1 Properties of PD and PSD Matrices

Last lecture we found some equivalent characterizations of positive definite matrices. We can, by analogy, propose equivalent characterizations of positive semi-definite matrices.

Conjecture 7.2.1. Suppose A is a symmetric matrix in $\mathbb{R}^{n\times n}$. The following statements are equivalent.

(i) A is positive semi-definite.

(ii) All eigenvalues of A are non-negative.

(iii) All leading principal submatrices of A have non-negative determinants.

(iv) A can be factorized as $A = RR^T$ for some $R \in \mathbb{R}^{n\times n}$.

We first show that (i) and (ii) are indeed equivalent.

Proof of "(i) $\Leftrightarrow$ (ii)". Suppose $A \succeq 0$. Let λ be an eigenvalue of A. Then $Ax = \lambda x$ for some $x \neq 0$. As $A \succeq 0$, then $0 \leq x^T Ax = x^T(\lambda x) = \lambda \|x\|^2$. Since $\|x\|^2 > 0$, then $\lambda \geq 0$.

Suppose all eigenvalues of A are non-negative. As A is symmetric, then $A = \sum_{i=1}^n \lambda_i q_i q_i^T$ with $\lambda_i \geq 0$ for all i. For every $x \in \mathbb{R}^{n\times n}$, $x^T Ax = \sum_{i=1}^n \lambda_i (x^T q_i)^2 \geq 0$. Therefore $A \succeq 0$. □

Next we show that (i) implies (iii).

Proof of "(i) $\Rightarrow$ (iii)". Suppose A is positive semi-definite. Let $\alpha = \{1,2,...,i\}$, where $i = 1,2,...,n$. For every $x_\alpha \in \mathbb{R}^{|\alpha|}$, let

$$x = \begin{bmatrix} x_\alpha \\ O \end{bmatrix} \in \mathbb{R}^n.$$

Then $x_\alpha^T A_{\alpha,\alpha} x_\alpha = x^T Ax \geq 0$. As this this true for every $i = 1,...,n$, then every leading principle submatrix $A_{\alpha,\alpha}$ of A is positive semi-definite. Thus the eigenvalues of $A_{\alpha,\alpha}$ are non-negative. Hence the determinant of $A_{\alpha,\alpha}$, which is equal to the product of the eigenvalues of $A_{\alpha,\alpha}$, is non-negative. □

However, (iii) does not imply (i) or (ii). As a counterexample, consider the matrix

$$A = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}.$$

We have $\det(A_1) = \det(A_2) = 0$. However, A is not positive semi-definite and has -1 and 0 as its eigenvalues.

At last, (i) and (iv) are indeed equivalent.

Proof of "(i) $\Leftrightarrow$ (iv)". Suppose $A = RR^T$ for some $R \in \mathbb{R}^{n\times n}$. For every $x \in \mathbb{R}^n$, $x^T Ax = x^T RR^T x = \|R^T x\| \geq 0$. Thus A is positive semi-definite.

Suppose A is positive semi-definite. Since A is symmetric, then $A = Q\Lambda Q^T$ where the diagonal entries of $\Lambda = \text{diag}(\lambda_1,...,\lambda_n)$ are non-negative. Thus $A = \left(Q\,\text{diag}(\sqrt{\lambda_1},...,\sqrt{\lambda_n})\right)\left(Q\,\text{diag}(\sqrt{\lambda_1},...,\sqrt{\lambda_n})\right)^T$. □

We summarize our results below.

Theorem 7.2.2. *Suppose A is a symmetric matrix in $\mathbb{R}^{n\times n}$. The following statements are equivalent.*

(i) A is positive semi-definite.

(ii) All eigenvalues of A are non-negative.

(iii) A can be factorized as $A = RR^T$ for some $R \in \mathbb{R}^{n\times n}$.

Theorem 7.2.3. *If $A \in \mathbb{R}^{n\times n}$ is positive definite, then all principal submatrices of A have non-negative determinants.*

Remark. The converse of this theorem is not true.

We now give some corollaries of theorem 7.2.2 and theorem 7.2.3.

Proposition 7.2.4. *Suppose $A \succeq 0$. If $a_{kk} = 0$ for some k, then $a_{kj} = a_{jk} = 0$ for every $j = 1,...,n$.*

Proof. If $A \succeq 0$, then for every $\alpha \subseteq \{1,...,n\}$, $A_{\alpha,\alpha}$ has nonnegative determinants. For every $j = 1,...,n$, let $\alpha = \{j,k\}$. Then

$$\det(A_{\alpha,\alpha}) = \begin{vmatrix} a_{jj} & a_{jk} \\ a_{kj} & a_{kk} \end{vmatrix} \left(\text{or } \begin{vmatrix} a_{kk} & a_{kj} \\ a_{jk} & a_{jj} \end{vmatrix} \right) = a_{jj}a_{kk} - a_{jk}a_{kj} = 0 - a_{kj}^2 \geq 0.$$

Hence $a_{kj} = a_{jk} = 0$. □

Proposition 7.2.5. *If $A \succeq 0$ and $B \succeq 0$, then $\langle A,B\rangle \geq 0$.*

Proof. Since A is symmetric, then A can be factorized as $A = \sum_{i=1}^n \lambda_i q_i q_i^T$. Then

$$\begin{aligned} \langle A,B\rangle &= \operatorname{Tr}(AB) \\ &= \operatorname{Tr}\left((\sum_{i=1}^n \lambda_i q_i q_i^T)B \right) \\ &= \sum_{i=1}^n \lambda_i \operatorname{Tr}(q_i q_i^T B) \\ &= \sum_{i=1}^n \lambda_i \operatorname{Tr}(q_i^T B q_i) \\ &= \sum_{i=1}^n \lambda_i (q_i^T B q_i) \end{aligned}$$

As $A \succeq 0$, then $\lambda_i \geq 0$ for every i. As $B \succeq 0$, then $q_i^T B q_i \geq 0$ for all i. Hence $\langle A,B\rangle \geq 0$. □

Proposition 7.2.6. *Suppose $A \in \mathbb{R}^{n\times n}$ and $A \succeq 0$. Let $B \in \mathbb{R}^{n\times m}$. Then $B^T AB \succeq 0$.*

Proof. For every $x \in \mathbb{R}^m$, $x^T(B^T AB)x = (Bx)^T A(Bx) \geq 0$ since $A \succeq 0$. Thus $B^T AB \succeq 0$. □

Proposition 7.2.7. *If A and B are positive semi-definite matrices of the same size, then for all $\alpha,\beta \geq 0$, $\alpha A + \beta B$ is also positive semi-definite.*

Proposition 7.2.8. *Suppose $A \in \mathbb{R}^{n\times n}$ is symmetric. Then $(\alpha I + A) \succ 0$ for a sufficiently large number $\alpha \in \mathbb{R}$.*

Proof. As $Ax = \lambda x$ if and only if $(\alpha I + A)x = (\alpha + \lambda)x$, then $\operatorname{eig}(\alpha I + A) = \{\alpha + \lambda \mid \lambda \in \operatorname{eig}(A)\}$. Take $\alpha = \max\limits_{1\leq i\leq n} |\lambda_i| + 1$, then all the eigenvalues of $\alpha I + A$ are positive. Thus $\alpha I + A$ is positive definite. □

Remark. The eigenvalues of a matrix are also called the spectral values. Adding αI to a matrix is like shifting the spectrum of a matrix.

7.2.2 Singular Value Decomposition

We have seen that the eigen-decomposition $A = Q\Lambda Q^T$ is very powerful. As another application, once we have obtained such decomposition, we can solve linear systems very easily. However, this decomposition is limited to square symmetric matrices. Here we introduced a new decomposition which can be applied to any matrix. This decomposition, called *singular value decomposition*, takes the form

$$A = U\Sigma V^T, \tag{7.2.1}$$

which looks very similar to the eigen-decomposition $A = Q\Lambda Q^T$ of a symmetric matrix. The singular value decomposition of a matrix, or SVD for short, is probably the most important matrix decomposition.

Description of SVD

We will now describe SVD. Suppose $A \in \mathbb{R}^{m\times n}$. Then A can be factorized as $A = U\Sigma V^T$.

The matrix U. The matrix U is an $m \times m$ orthogonal matrix whose columns $u_1, ..., u_m$ are orthonormal eigenvectors of AA^T. Note that $AA^T \in \mathbb{R}^{m\times m}$ is symmetric and PSD.

The matrix V. The matrix V is an $n \times n$ orthogonal matrix whose columns $v_1, ..., v_n$ are orthonormal eigenvectors of A^TA. Note that $A^TA \in \mathbb{R}^{n\times n}$ is symmetric and PSD.

The matrix Σ. The matrix Σ is an $m \times n$ diagonal matrix. (So Σ has the same size as A.) Thus

$$\Sigma = \left[\begin{array}{ccc} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \\ \hline & O & \end{array}\right] \text{ when } m > n \quad \text{and} \quad \Sigma = \left[\begin{array}{ccc|c} \sigma_1 & & & \\ & \ddots & & O \\ & & \sigma_m & \end{array}\right] \text{ when } m > n. \tag{7.2.2}$$

Let $k = \min\{m, n\}$. The diagonal entries $\sigma_1, ..., \sigma_k$ are called the *singular values* of A. $\sigma_i \geq 0$ for every $i = 1, ..., k$. Moreover, A^TA and AA^T have the same set of nonzero eigenvalues, which are exactly $\sigma_1^2, ..., \sigma_k^2$.

With SVD, A can be written as

$$A = \sum_{i=1}^{k} \sigma_i u_i v_i^T. \tag{7.2.3}$$

We shall now prove that A can indeed be factorized into $A = U\Sigma V^T$. We will only prove the case for $m \geq n$. The proof for the other case is basically the same.

Proof of case $m \geq n$

Construction of V Suppose $\lambda_1, ..., \lambda_n$ are eigenvalues of A^TA. We will only discuss the case when the λ_i's are all positive. Thus we have

$$A^TAv_j = \lambda_j v_j \tag{7.2.4}$$

for each $j = 1, ..., n$ with $v_1, ..., v_n \in \mathbb{R}^n$ as orthonormal eigenvectors of $A^TA \in \mathbb{R}^{n\times n}$. Let

$$V = \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}. \tag{7.2.5}$$

Thus $V \in \mathbb{R}^{n\times n}$ is orthogonal.

Construction of U Define

$$u_j = \frac{1}{\sqrt{\lambda_j}} A v_j \in \mathbb{R}^m \tag{7.2.6}$$

for each $j = 1, ..., n$. We now show that $\{u_1, ..., u_n\}$ is an orthonormal set:

$$\begin{aligned}
\langle u_i, u_j \rangle &= \langle \frac{1}{\sqrt{\lambda_i}} A v_i, \frac{1}{\sqrt{\lambda_j}} A v_j \rangle \\
&= \frac{1}{\sqrt{\lambda_i \lambda_j}} \langle A v_i, A v_j \rangle \\
&= \frac{1}{\sqrt{\lambda_i \lambda_j}} \langle A^T A v_i, v_j \rangle \\
&= \frac{1}{\sqrt{\lambda_i \lambda_j}} \langle \lambda_i v_i, v_j \rangle \\
&= \sqrt{\frac{\lambda_i}{\lambda_j}} \langle v_i, v_j \rangle \\
&= \begin{cases} 0 & \text{when } i \neq j \\ 1 & \text{when } i = j \end{cases}.
\end{aligned}$$

The last step is to extend the orthonormal set $\{u_1, ..., u_n\}$ to an orthonormal basis $\{u_1, ..., u_m\}$ of $\mathbb{R}^m$. Note that this basis is not unique. Now, let

$$U = \begin{bmatrix} u_1 & \cdots & u_m \end{bmatrix}. \tag{7.2.7}$$

So U is orthogonal.

Completion of $A = U\Sigma V^T$ As $u_j = \frac{1}{\sqrt{\lambda_j}} A v_j \in \mathbb{R}^m$, then $\sqrt{\lambda_j} u_j = A v_j \in \mathbb{R}^m$. Therefore

$$\begin{bmatrix} u_1 & \cdots & u_m \end{bmatrix} \begin{bmatrix} \sqrt{\lambda_1} & & \\ & \ddots & \\ & & \sqrt{\lambda_n} \end{bmatrix} = A \begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}. \tag{7.2.8}$$

Define $\sigma_i = \sqrt{\lambda_i}$, $i = 1, ..., n$. We have

$$\underbrace{\left[\begin{array}{ccc|ccc} u_1 & \cdots & u_n & u_{n+1} & \cdots & u_m \end{array}\right]}_{U} \underbrace{\left[\begin{array}{ccc} \sigma_1 & & \\ & \ddots & \\ & & \sigma_n \\ \hline & O & \end{array}\right]}_{\Sigma} = A \underbrace{\begin{bmatrix} v_1 & \cdots & v_n \end{bmatrix}}_{V}. \tag{7.2.9}$$

Equivalently,

$$A = U\Sigma V^T. \tag{7.2.10}$$

It remains to show that u_j are eigenvectors of AA^T and that $\lambda_j \in \text{eig}(AA^T)$. Since

$$AA^T u_j = AA^T \left(\frac{1}{\sqrt{\lambda_j}} A v_j \right) = \frac{1}{\sqrt{\lambda_j}} A(A^T A v_j) = \frac{1}{\sqrt{\lambda_j}} A(\lambda_j v_j) = \sqrt{\lambda_j} A v_j = \lambda_j u_j,$$

then $\lambda_1, ..., \lambda_n$ are also eigenvalues of AA^T with corresponding orthonormal eigenvectors $u_1, ..., u_n$, respectively. This completes the proof.

□

With SVD, we can show that $\text{rank}(AA^T) = \text{rank}(A^TA)$. Since $A = U\Sigma V^T$, then $AA^T = U(\Sigma\Sigma^T)U^T$ and $A^TA = V(\Sigma^T\Sigma)V^T$. These are exactly eigen-decompositions. As $\Sigma\Sigma^T$ and $\Sigma^T\Sigma$ have the same number of non-zero diagonal entries, then $\text{rank}(AA^T) = \text{rank}(\Sigma\Sigma^T) = \text{rank}(\Sigma^T\Sigma) = \text{rank}(A^TA)$.

7.2.3 Geometric Interpretation of SVD

Suppose $A = U\Sigma V^T$. Then $A^T = V\Sigma^T U^T$. So $AV = U\Sigma$ and $A^TU = V\Sigma^T$. Thus for each $j = 1, ..., k$,

$$Av_j = \sigma_j u_j \quad \text{and} \quad A^T u_j = \sigma_j v_j. \tag{7.2.11}$$

The vectors $v_1, ..., v_n$ are called the *right singular vectors* of A and the vectors $u_1, ..., u_m$ are called the *left singular vectors* of A. Since $A = U\Sigma V^T$ with U and V orthogonal, then the rank of A is equal to the number of non-zero diagonal entries of Σ. Let $\text{rank}(A) = r$. We reorder the σ_i's such that $|\sigma_1| \geq |\sigma_2| \geq \cdots \geq |\sigma_r| > 0 = \sigma_{r+1} = \cdots = \sigma_k$. Then we have

$$A = \sum_{i=1}^{r} \sigma_i u_i v_i^T = \underbrace{\begin{bmatrix} u_1 & \cdots & u_r \end{bmatrix}}_{m\times r} \underbrace{\begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}}_{r\times r} \underbrace{\begin{bmatrix} v_1 & \cdots & v_r \end{bmatrix}^T}_{r\times n}. \tag{7.2.12}$$

This is called the *compact singular value decomposition* of A. In practice, we use compact SVD rather than SVD because compact SVD saves more storage.

By (7.2.12), for all $x \in \mathbb{R}^n$, $Ax = \sum_{i=1}^r \sigma_i (v_i^T x) u_i \in \text{Span}\{u_1, ..., u_r\}$. Thus $\text{Col}(A) \subseteq \text{Span}\{u_1, ..., u_r\}$. Moreover, as $Av_i = \sigma_i u_i$, then $u_1, ..., u_r \in \text{Col}(A)$. Therefore

$$\text{Col}(A) = \text{Span}\{u_1, ..., u_r\}. \tag{7.2.13}$$

Thus $\{u_1, ..., u_r\}$ forms a basis for $\text{Col}(A)$.

Now we turn to $N(A)$. As $Av_j = \sum_{i=1}^r \sigma_i (v_i^T v_j) u_i = 0$ for $j = r+1, ..., n$, then $v_{r+1}, ..., v_n \in N(A)$. Since $\dim(N(A)) = n - r$, then $\{v_{r+1}, ..., v_n\}$ forms a basis for $N(A)$.

Similarly, $\{v_1, ..., v_r\}$ forms a basis for $\text{Row}(A)$ and $\{u_{r+1}, ..., u_m\}$ forms a basis for $N(A^T)$.

■